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Clinical outcome measures and their evidence base in degenerative cervical myelopathy: a systematic review to inform a core measurement set (AO Spine RECODE-DCM)

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Clinical outcome measures and their evidence base in degenerative cervical myelopathy: a systematic review to inform a core measurement set (AO Spine RECODE-DCM)

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ABSTRACT

Objective: To systematically review the measurement properties of outcome measures currently used in the assessment of degenerative cervical myelopathy (DCM) for clinical research.

Methods: A systematic review of MEDLINE and EMBASE was conducted between August 2020 and January 2021. Full-text articles were included if their primary purpose was to evaluate the measurement properties of instruments used in DCM or to define clinically important differences. Psychometric properties and clinically important differences were both extracted from each study, assessed for risk of bias, and presented in accordance with the Consensus-based Standards for the selection of health Measurement Instruments (COSMIN) criteria.

Results: Twenty-nine outcome instruments were identified from 52 studies published between 1999 and 2020. They measured neuromuscular function (16 instruments), life impact (five instruments), pain (five instruments), and radiological scoring (five instruments). No instrument had evaluations for all 10 measurement properties and <50% had assessments for all three domains (i.e., reliability, validity, and responsiveness). There was a paucity of high-quality evidence. Notably, there were no studies that reported on structural validity and no high-quality evidence that discussed content validity. In this context, we identified nine instruments that are interpretable by clinicians: the arm and neck pain scores; the 12- and 36-item short form health surveys; the Japanese Orthopaedic Association (JOA) score, modified JOA (mJOA), and JOA Cervical Myelopathy Evaluation Questionnaire (JOACMEQ); the neck disability index; and the visual analogue scale for pain. These include six scores with barriers to application and one score with insufficient criterion and construct validity.

Conclusions: This review comprehensively aggregates studies evaluating outcome measures used to assess patients with DCM. Overall, there is a need for a set of agreed tools to measure outcomes in DCM. These findings will be used to inform the development of a core measurement set as part of AO Spine RECODE-DCM.

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ARTICLE SUMMARY

Strengths and limitations of this study

- Consensus-based reporting guidelines were used to evaluate the properties and clinically important differences of DCM measurement instruments.
- By focusing on measurement instruments that are currently in use, the core measurement set (CMS) can be immediately adopted and used to cross-compare historic data.
- Interpretability was used as an important characteristic to make recommendations due to the absence of category A COSMIN recommendations. This concept was based on the proposition that instruments in a CMS should be interpretable by clinicians and offer qualitative meaning to clinicians and patients.
- Developing better instruments in DCM remains a top 10 research priority and further work will be needed to close those gaps.

Keywords: degenerative cervical myelopathy; cervical spondylotic myelopathy; spinal cord compression; outcome measures; core measurement set

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LIST OF ABBREVIATIONS

30MWT	30-m Walking Test
COSMIN	Consensus-based Standards for the selection of health Measurement Instruments
AUC	Area under curve
BBS	Berg Balance Scale
BF	Bladder function
BP	Bodily pain
CMS	Core measurement set
CF	Cervical spine function
CT	computed tomography
EN	Environment health
EQ-5D	EuroQol-5 Dimension
GH	General health
JOA	Japanese Orthopaedic Association
LEF	Lower extremity function
LOA	Limits of agreement
MCID	Minimal clinically important difference
MCS	Mental component summary
MDC	Minimal detectable change
MDI	Myelopathy Disability Index
MFLE	Motor function of lower extremity
MFSE	Motor function of shoulder and elbow
MH	Mental health
mJOA	Modified Japanese Orthopaedic Association
MRI	Magnetic resonance imaging
NDI	Neck Disability Index
NMF	Neuromuscular limb function
JOACMEQ	Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire
P-mJOA	Patient-derived version of the mJOA
PCS	Physical component summary
PH	Physical health
PROM	Patient-reported outcome measure

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PS	Psychological well-being
RE	Emotional role
RP	Physical role
SCB	Substantial clinical benefit
SDC	Smallest detectable change
SF-12	12-Item Short Form Health Survey
SF-36	36-Item Short Form Health Survey
SFUE	Sensory function of upper extremity
SF	Social function
SR	Social relationships
QOL	Quality of life
QuickDASH	Quick Disabilities of the Arm, Shoulder, and Hand
UEF	Upper extremity function
VAS	Visual Analogue Scale
VT	Vitality
WHOQOL-Bref	World Health Organisation Quality of Life Instruments

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INTRODUCTION

The most common adult spinal cord disease, degenerative cervical myelopathy (DCM), is both measured and reported inconsistently across clinical research (1-4). DCM is a progressive spinal cord disease caused by degenerative changes in the cervical spine that lead to stress and injury to the cervical spinal cord. It usually initially presents as a loss of digital dexterity, subtle gait disturbances, and mild pain which, if left untreated, can potentially lead to tetraplegia and wheelchair dependence (5).

In 2019, AO Spine launched the Research Objectives and Common Data Elements for Degenerative Cervical Myelopathy (AO Spine RECODE-DCM; www.aospine.org/recode) initiative with the aim of creating a 'research toolkit' to help accelerate knowledge discovery and improve outcomes in DCM (3, 6). The initiative identified the need to improve consistency in measurement and reporting across DCM research to enable studies to be compared and/or aggregated, and to ensure the most meaningful aspects of the disease are captured (7, 8). This process started by creating a list of essential outcomes (i.e., core outcome set) and baseline characteristics (i.e., core data elements). To truly enable consistent reporting, however, these data sets should be partnered with a core measurement set (CMS): a set of agreed tools that are used to measure the outcomes in DCM and other data elements (9-17).

Several approaches have been employed to form a CMS, ranging from the development of novel measurement instruments to adopting the use of existing ones (18-20). For AO Spine RECODE DCM, it was decided to recommend existing instruments and, preferably, preferentially those already used in DCM. This was to a more rapid introduction, cognisant that many new tools are in development and the CMS can be updated in the future.

Consequently, we sought to examine the tools used in DCM research and assess their quality (21) using objective criteria. In recognition of variable quality amongst reported outcome measures, the COSMIN initiative has developed clinimetric tools to assess instrument quality (22). We searched the literature for studies evaluating one or more psychometric properties defined by the Consensus-based Standards for the selection of health Measurement Instruments (COSMIN) guidelines, as well as studies

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that defined clinically important differences such as the minimally clinical important difference (MCID) and substantial clinical benefits (SCB). Data were rated, aggregated, and assessed for methodology bias using the COSMIN manual for systematic reviews of patient-reported outcome measures (PROM) (23-25). This work builds on the protocol for the AO Spine RECODE-DCM initiative (3, 6) and complements two earlier reviews of outcome measures in DCM (2, 21).

METHODS

Search

A search string was developed to identify original research assessing the psychometric properties of instruments currently used in the clinical research of DCM (26). This comprised synonyms of 'psychometric' and 'DCM' (**Supplementary Table 1**). The search was developed with oversight of a medical librarian (IK) and informed by previously developed search filters for DCM (26-28). The search was applied to MEDLINE and EMBASE, from inception until 4 August 2020, using OVID (Wolters Kluwer, Netherlands). The search also focused on DCM tools identified in previous scoping reviews (2, 21, 29).

Study selection

All titles and abstracts were screened independently against a set of pre-defined eligibility criteria by four reviewers (AYT, AB, ED, and FB). A full list of inclusion and exclusion criteria of studies are stated in **Table 1**.

Table 1. Inclusion and exclusion criteria

Inclusion	Exclusion
Publication type	
• Article written in English • Primary clinical research articles	• Article not written in English • Conference abstracts or posters • Editorials, commentaries, opinion papers or letters • Book chapters or theses
Study type	
• Study includes primary clinical data	• Study uses only secondary data • Case reports

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	• Narrative reviews • Systematic reviews • Meta-analyses
Populations	
• Human studies	• Non-human studies
Indications	
• Exclusively DCM (CSM, OPLL, cervical stenosis, spondylosis, spinal cord compression, cervical myelopathy)	• Populations with DCM and at least one other condition (e.g., radiculopathy)
Comparator	
• At least one assessment tool from (2, 21, 29)	
Outcomes	
• At least one psychometric property • At least one MCID or SCB	

Potentially eligible studies were selected for full-text analysis. In the event of multiple publications analysing the same cohort for the same purpose, the most recent paper was used for evaluation. At each stage, two reviewers independently (AYT, AB, ED, FB) reviewed all the screened studies for inclusion to ensure reliability of study selection. Disagreements were resolved by consensus or appeal to a third senior reviewer (BMD).

Quality assessment

The quality of included studies was assessed using the COSMIN risk of bias checklist (23-25). Briefly, the COSMIN risk of bias tool assesses 10 measurement properties, including nine psychometric properties (i.e., content validity, structural validity, internal consistency, cross-cultural validity/measurement invariance, reliability, measurement error, criterion validity, hypotheses testing for construct validity, and responsiveness) and clinically important differences. A list of definitions is presented in **Table 2**. Interpretability and feasibility and were also evaluated using criteria adapted by the authors from the COSMIN methodology (**Supplementary Tables 2** and **3**, respectively). The methodological quality of each study was scored as 'very good', 'adequate', 'doubtful', 'inadequate', or 'not applicable'. Overall ratings were then made

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for each property using a modified Grading of Recommendations Assessment, Development, and Evaluation (mGRADE) approach.

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Table 2. Definitions of domains, measurement properties, and aspects of measurement properties, adapted from the COSMIN guidelines (23-25, 30) and studies of clinically important differences (31, 32)

Domain	Measurement property	Aspect	Definition
Reliability			The degree to which the measurement is free from measurement error. The extent to which scores for patients who have not changed are the same for repeated measurement under several conditions; over time; by different persons on the same occasion; or by the same persons on different occasions.
	Internal consistency		The degree of interrelatedness among the items included in a measurement instrument.
	Reliability		The proportion of the total variance in the measurements which is due to 'true'* differences between patients.
	Measurement error		The systematic and random error of a patient's score that is not attributed to true changes in the construct to be measured.
Validity			The degree to which a measurement instrument measures the construct(s) it purports to measure.
	Content validity		The degree to which the content of a measurement tool is an adequate reflection of all facets of a given construct.
	Construct validity		The degree to which the scores of a measurement instrument are consistent with hypotheses (for instance with regard to internal relationships, relationships to scores of other instruments, or differences between relevant groups) based on the assumption that the instrument validly measures the construct to be measured.
		Structural validity	The degree to which the scores of a measurement instrument are an adequate reflection of the dimensionality of the construct to be measured.

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		Hypotheses testing	Idem construct validity
		Cross-cultural validity	The degree to which the performance of the items on a translated or culturally adapted measurement instrument are an adequate reflection of the performance of the items of the original version of the instrument.
	Criterion validity		The degree to which the scores of a measurement instrument are an adequate reflection of a 'gold standard'.
Responsiveness			The ability of a measurement instrument to detect change over time in the construct to be measured.
	Responsiveness		Idem responsiveness
Interpretability†			The ability of a measurement instrument to detect change over time in the construct to be measured.
	Clinically important differences		
		Minimal clinically important difference	The smallest measured change score that patients perceive to be important, also known as the MCID or MID (Minimally Important Difference)
		Substantial clinical benefit	The change in outcome associated with patient perception of a large meaningful improvement

*The word 'true' must be seen in the context of the classical test theory, which states that any observation is composed of two components – a true score and error associated with the observation. 'True' is the average score that would be obtained if the scale were applied an infinite number of times. It refers only to the consistency of the score, and not to its accuracy (33).

†Interpretability is not considered a measurement property, but an important characteristic of a measurement instrument.

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Data extraction

A proforma adapted from COSMIN was employed by one reviewer (AYT) to extract the following: study details, sample size, patient demographics, measurement properties, and qualitative and/or quantitative results for each property. Examples of qualitative and quantitative results included observations (e.g., narrative syntheses) and statistics (e.g., correlation coefficients). These result types are specific for each measurement property and are listed in the COSMIN guidelines (23-25).

Data analysis

Each result was rated as 'sufficient', 'indeterminate', or 'insufficient'. All results were qualitatively summarised and given an overall rating as 'sufficient', 'indeterminate', 'inconsistent', or 'insufficient'. Measurement instruments were categorised into three recommendation groups:

A. Instruments with evidence of sufficient content validity and at least low-quality evidence of sufficient internal consistency.
B. Instruments categorised not in A or C.
C. Instruments with high-quality evidence of an insufficient measurement property (23-25).

Recommendations for each instrument were presented in tandem with interpretability and feasibility assessments and reported as a narrative synthesis (34). We used the PRISMA checklist when writing our report (35).

Patient and public involvement

This project forms part of a larger, international multi-stakeholder co-production initiative called AO Spine RECODE-DCM, which aims to develop a framework to accelerate knowledge discovery that can improve outcomes in DCM. Patients and the public were therefore involved in its overall design, conduct, management, and dissemination, and are recognised amongst the authors of this article. For further information, please refer to aospine.org/recode.

RESULTS

Literature search

The primary literature search identified a total of 3,239 unduplicated studies (MEDLINE: 2,389, EMBASE: 1,550). Abstract and full-text screening excluded 3,187

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studies. Therefore, this review included a total of 52 studies (**Figure 1** and **Supplementary Table 2**).

Figure 1. PRISMA flow chart. A systematic review of MEDLINE and EMBASE was conducted between August 2020 and January 2021 to identify original research on the measurement properties of instruments currently used in DCM research.

Study properties

The 52 included studies assessed a total of 7,395 patients worldwide (female: 3,217, male: 4,178) with 29 instruments (**Table 3**). These were classified into four domains based on the DCM core outcome set (36): neuromuscular function, life impact, pain, and radiological scoring.

Table 3. Study properties

Property	Number	%
Total studies, included	52	100
Prospective	31	60
Retrospective	21	40
Total patient sample	7,395	100
Male	4,178	56
Female	3,217	44
Measurement instruments by domain*	29	100
Neuromuscular function	16	55
Life impact	5	17
Pain	5	17
Radiological scoring	5	17
Publication year		
Maximum year of publication	2020	–
Median year of publication	2014	–
Mean year of publication	2012	–
Minimum year of publication	1999	–
Countries, by no. of patients	7,395	100
Japan	2,014	27
United States of America	1,802	24
Canada	1,361	18
South Korea	726	10

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Global/Multicentre	601	8
China	255	3
India	121	2
Iran	87	1
Brazil	85	1
Italy	75	1
Hong Kong	72	1
Thailand	70	1
Taiwan	45	1
United Kingdom	41	1
France	40	1

*Instrument counts per domain do not add up to total due to the one-to-many relationship between certain instruments and domains (e.g., JOACMEQ is used both for life impact and neuromuscular function; see **Table 4**).

Measurement properties

The measurement properties of the 29 instruments were evaluated using the COSMIN methodology for systematic reviews (23-25). A summary of findings is presented in **Table 4**: (1) the overall feasibility rating, (2) the overall interpretability rating, and (3) the overall recommendation category based on existing evidence. Included studies reported on at least one of the 10 COSMIN properties for all instruments. No instrument had evidence for all 10 properties and <50% (13/29) of instruments had evidence for at least one property per measurement domain (**Figure 2**).

Figure 2. Number of studies for each outcome measure and property [normalised]. Included studies reported on at least one of the 10 COSMIN properties for all instruments. No instrument had evidence for all 10 properties and <50% (13/29) of instruments had evidence for at least one property per measurement domain (see **Table 2** for definitions). Notably, no instruments were evaluated for structural validity, attained sufficient evidence for content validity, or obtained a Category A recommendation based on COSMIN criteria.

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Table 4. Summary of findings

Domain	Instrument	Feasibility	Interpretability	Recommendation category	Recommendation justification
Life Impact					
	EQ-5D	+	+	C	High quality evidence for insufficient construct validity
	SF-12	–	+	B	Indeterminate result rating for internal consistency
	SF-36	–	+	B	Indeterminate result rating for internal consistency
	WHOQOL-Bref	+	–	B	Indeterminate result rating for internal consistency
Life Impact and Neuromuscular function					
	JOACMEQ	+	+	B	
Neuromuscular function					
	10-s step test	+	–	C	High quality evidence for insufficient criterion validity
	30MWT	+	–	C	High quality evidence for insufficient responsiveness
	9-Hole peg test	++	–	B	
	BBS	++	–	B	
	European Myelopathy Scale	+	–	B	
	Foot tapping test	+	–	C	High quality evidence for insufficient criterion validity

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	Grip-and-release test	+	–	B	
	JOA	–	+	B	
	MDI	+	–	B	
	mJOA	–	+	C	High quality evidence for insufficient criterion and construct validity
	Nurick scale	+	–	B	
	P-mJOA	+	–	B	
	Ranawat classification of disease severity	–	–	B	
	Triangle step test	+	–	B	
Pain and Neuromuscular function					
	QuickDASH	–	–	B	
Pain					
	NDI	+	+	B	
	Arm pain score	–	+	B	
	Neck pain score	+	+	B	
	VAS for pain	+	+	B	
Radiology					
	Cobb's method	+	–	B	
	CT (Tsuyama's classification, 2D & 3D)	+	–	B	
	CT (Tsuyama's classification, lateral + axial)	+	–	B	
	Isihara's Cervical Curvature Index	+	–	B	

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MRI (Depiction of intramedullary hyperintensity at eight cervical disc levels, T2W, 1.5-T or 3-T)	+	–	B
MRI (Kang's classification, 1.5-T or 3-T)	+	–	B
MRI (Muhle's classification, 1.5-T)	+	–	B
MRI (Vaccaro's classification, 1.5-T)	+	–	B
X-rays (Computer-assisted measurement of length & thickness)	+	–	B

Feasibility

++ = No barriers

+ = Minimal barriers

– = Barriers

Interpretability

+ = Interpretable

– = Uninterpretable, due to absence of MCID (23-25)

Recommendation category

A = Measurement instruments with evidence for sufficient content validity (any level) AND at least low-quality evidence for sufficient internal consistency

B = Measurement instruments categorized not in A or C

C = Measurement instruments with high quality evidence for an insufficient measurement property

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Content validity

Only three measurement instruments were evaluated for content validity: the JOA Cervical Myelopathy Evaluation Questionnaire (JOACMEQ), the modified JOA (mJOA) score, and the Berg Balance Scale (BBS) (**Supplementary Table 5**). The overall ratings for content validity, however, were indeterminate due to the uncertainty of the methods used to assess comprehensibility, and the very low quality of the evidence.

Structural validity

No instruments were assessed for structural validity.

Internal consistency

Ten measurement instruments were evaluated for internal consistency, including the JOACMEQ, JOA, mJOA, 12-Item Short Form Health Survey (SF-12), and SF-36 (**Supplementary Table 6**). Since structural is required for the interpretation of internal consistency, the overall ratings for internal consistency were indeterminate given the absence of studies on structural validity.

Cross-cultural validity

Only three measurement instruments were evaluated for cross-cultural validity: JOACMEQ, JOA, and mJOA (**Supplementary Table 7**). The overall ratings were indeterminate due to the absence of multiple group factors analyses and differential item functioning analyses. The quality of evidence was also very low due to the uncertainty of the approaches used to analyse the data.

Reliability

Seventeen measurement instruments were evaluated for reliability, including JOACMEQ, JOA, and mJOA (**Supplementary Table 8**). The reported measures of reliability were test–retest reliability, intra-observer reliability, and inter-observer reliability. No instrument attained high-quality evidence for sufficient or insufficient reliability due to (a) imprecision (sample sizes < 100), (b) serious inconsistency, and/or (c) serious risk of bias.

Measurement error

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Nine instruments were evaluated for measurement error, including JOACMEQ, JOA, mJOA, NDI, SF-36, and Visual Analogue Scale (VAS) for pain (**Supplementary Table 9**). The measures of error reported were minimal detectable change (MDC) and distribution-based MCID (23-25, 37). The mJOA was the only score to attain high-quality evidence for sufficiency (distribution-based MCID range: 1.2–1.4, total sample size: 868). Due to the inconsistency of results, the quality of the evidence of most other instruments could not to be rated.

Criterion validity

Twelve measurement instruments were evaluated for criterion validity, including the JOACMEQ, JOA, mJOA, NDI, and SF-36 (**Supplementary Table 10**). Both the mJOA and the patient-derived version of the mJOA (P-mJOA) attained high-quality evidence for sufficient criterion validity as whole scales. However, three of four items of the mJOA, along with the 10-second step test and foot tapping test, attained high-quality evidence for insufficient criterion validity (i.e., these subdomains lack criterion validity for their use as separate measures). The quality of the evidence of most of the remaining instruments was not high due to (a) imprecision (i.e., sample sizes < 100) or (b) important methodological flaws in the design or statistical methods.

Construct validity

Sixteen measurement instruments were evaluated for construct validity, including JOACMEQ, JOA, mJOA, NDI, arm and neck pain scores, and SF-12 (**Supplementary Table 11**). From these, eight of 16 attained high-quality evidence for sufficient construct validity; these included the NDI, arm and neck pain scores, and SF-12. Two instruments achieved high-quality evidence for insufficient construct validity. Notably, the mJOA had both high-quality sufficiency and insufficiency depending on the comparator tool (e.g., sufficiency with respect to the NDI and SF-36 and insufficiency with respect to the 30-m walking test [30MWT] and EuroQol-5 Dimension [EQ-5D]). While the designs and statistical methods applied were adequate for the research questions posed, the quality of the evidence of most of the remaining tools ranged from ‘low’ to ‘moderate’ due to imprecision (i.e., sample sizes < 100). Importantly, only one study formulated an a priori hypothesis (38).

Responsiveness

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Sixteen measurement instruments were evaluated for responsiveness, including the JOACMEQ, JOA, mJOA, NDI, SF-12, and SF-36 (**Supplementary Table 12**). The mJOA was the only score to attain high-quality evidence for sufficient responsiveness (effect size range: 0.87–1.0, total sample size: 352). The 30MWT, on the other hand, was the only score to attain high-quality evidence for insufficient responsiveness (standardised response mean: 0.3, total sample size: 484). The quality of the evidence of most of the remaining tools ranged from 'very low' to 'moderate' due to (a) imprecision (i.e., sample sizes < 100) and (b) uncertainty of the statistical methods.

Clinically important differences

Ten measurement instruments were evaluated for clinically important differences, including the JOACMEQ, JOA, mJOA, NDI, arm and neck pain scores, SF-12, SF-36, and VAS for pain (**Supplementary Table 3**). From these, seven of 10 attained a sufficient rating, including the JOACMEQ, JOA, mJOA, NDI, and SF-36. Only anchor-based measures were accepted for assessment of the MCID (23-25, 39-42).

Interpretability and feasibility

Interpretability and feasibility and were described using criteria adapted by the authors from the COSMIN methodology (**Supplementary Tables 2** and **3**, respectively). Interpretability was summarised in terms of the degree to which clinicians may assign qualitative meaning to the scores or change in scores (i.e., the clinically important differences) while feasibility was described in terms of the ease of application of the measurement instrument while. No or minimal application barriers were identified for most outcome measures (**Table 4**). Nine instruments were, however, deemed uninterpretable due to the absence of MCIDs.

Recommendations

No category A recommendations were made as no measurement instrument had sufficient evidence for content validity (**Table 4** and **Figure 2**). Next, due to the availability of high-quality evidence for insufficient criterion validity, construct validity, and/or responsiveness, five instruments were recommended for category C. However, most instruments were categorised into category B due to the notable absence of high-quality evidence in most measurement properties.

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In light of these results, and given both (a) the very strict quality standards of the COSMIN framework and (b) the absence of evidence is not the same as poor-quality evidence, we propose that instruments most suitable for use should be interpretable by clinicians and offer qualitative meaning to either clinicians or patients. To this end, the measurement properties of the nine interpretable instruments are presented in **Table 5**: the arm and neck pain scores; SF-12 and SF-36; JOA, mJOA, and JOACMEQ; NDI; and VAS for pain. These include one score with insufficient criterion and construct validity (i.e., mJOA) and six scores with barriers to application.

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Table 5. Interpretable measurement instruments

Domain	Instrument	Psychometric properties*	Feasibility	Recommendation category
Life Impact				
	SF-12	Cronbach's α coefficient (0.77)	–	B
	MCS	SCB (51.5)		
	PCS	SCB (30.1) Responsiveness: SF-12 PCS (Mean change score: 8.17)		
	SF-36	Cronbach's α coefficient (0.79–0.93) Responsiveness: SF-36 (Normalised change: 0.32)	–	B
	MCS	MDC or SDC (Distribution: 3.3–5.7) MCID (Distribution: 3.4–6.8, anchor: 3.0–7.4) Construct validity: Arm pain score (Pearson's correlation: –0.23) mJOA scale (Pearson's correlation: 0.19) NDI (Spearman's rank correlation: –0.17) Neck pain score (Pearson's correlation: –0.28) SF-12 PCS (Pearson's correlation: 0.01) Responsiveness: SF-36 MCS (Effect size range: 0.81, sensitivity: 0.67)		
	PCS	MDC or SDC (Distribution: 5.2–5.7, anchor: 4.9) MCID (Distribution: 2.9–5.5, distribution: 10, anchor: 3.9–9.6)		

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	SCB (16) Criterion validity (Likert scale) AUC: 0.67–0.69 Construct validity: Arm pain score (Pearson's correlation: –0.44) mJOA scale (Pearson's correlation: 0.43) NDI (Spearman's rank correlation: –0.49) Neck pain score (Pearson's correlation: –0.41) SF-12 PCS (Pearson's correlation: –0.29) Responsiveness: SF-36 PCS (Effect size range: 0.84, sensitivity: 0.85)		
Life Impact and Neuromuscular function			
JOACMEQ	Patient comprehensibility: "No questions elicited no answer or “I am not sure” in more than 5% of patients" Test–retest stability: Cronbach's α coefficient (0.91) Forward-backward translation [Persian and Thai]: n/a	+	B
Bladder function	Cronbach's α coefficient (0.32–0.74) Test–retest stability: ICC (0.62) MDC or SDC (Distribution: 7.7) MCID (Anchor: 6.0)		

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	Responsiveness: JOACMEQ Bladder function (AUC: 0.82, Effect size: 0.33, Mean change score: 18.0)
Cervical spine function	Cronbach's α coefficient (0.77–0.78) Test–retest stability: ICC (0.63) MDC or SDC (Distribution: 12.9, anchor: 12.5) MCID (Anchor: 2.5) Criterion validity (Likert scale) AUC: 0.58 Responsiveness: JOACMEQ Cervical spine function (AUC: 0.72, Effect size: 0.28, Mean change score: 25.8)
Lower extremity function	Cronbach's α coefficient (0.80–0.86) Test–retest stability: ICC (0.83) MDC or SDC (Distribution: 6.6, anchor: 8.5) MCID (Anchor: 8.5–9.5) Criterion validity (Likert scale) AUC: 0.66–0.70 Construct validity: NDI (Pearson's correlation: –0.66) SF-12 MCS (Spearman's rank correlation: 0.40) SF-12 PCS (Spearman's rank correlation: 0.29)

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		Responsiveness: JOACMEQ Quality of life (AUC: 0.83, Effect size: 0.46, Mean change score: 23.7)
	QOL	Cronbach's α coefficient (0.80–0.86) Test–retest stability: ICC (0.83) MDC or SDC (Distribution: 6.6, anchor: 8.5) MCID (Anchor: 8.5–9.5) Criterion validity (Likert scale) AUC: 0.66–0.70 Construct validity: NDI (Pearson's correlation: –0.66) SF-12 MCS (Spearman's rank correlation: 0.40) SF-12 PCS (Spearman's rank correlation: 0.29) Responsiveness: JOACMEQ Quality of life (AUC: 0.83, Effect size: 0.46, Mean change score: 23.7)
	Upper extremity function	Cronbach's α coefficient (0.72–0.74) Test–retest stability: ICC (0.93) MDC or SDC (Distribution: 9.5, anchor: 6.1) MCID (Anchor: 2.5–13.0) Responsiveness:

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	JOACMEQ Upper extremity function (AUC: 0.74, Effect size: 0.17, Mean change score: 10.7)		
Neuromuscular function			
JOA	Cronbach's α coefficient (0.72) Forward-backward translation [Brazilian Portuguese]: Comprehension rate (>81.2%) Inter-observer reliability: ICC (0.81) MDC or SDC (Distribution: 1.0, anchor: 2.5) LOA (1.2 [–1.2, 3.6]) MCID (Anchor: 2.5) Criterion validity (Likert scale) AUC: 0.59–0.62 Construct validity: JOACMEQ QOL (Spearman's rank correlation: 0.41) mJOA (Spearman's rank correlation: 0.87) NDI (Spearman's rank correlation: –0.50 to –0.76) SF-12 MCS (Spearman's rank correlation: –0.05) SF-12 PCS (Spearman's rank correlation: 0.50) Responsiveness: JOA (Mean change score: 4.6, normalised change: 0.21) JOA Motor function of lower extremity (Mean change score: 0.60) mJOA (Spearman's rank correlation: 0.75)	–	B

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	Bladder function	Intra-observer reliability (κ = 0.64) Inter-observer reliability (κ = 0.47)		
	Motor function of fingers	Intra-observer reliability (κ = 0.68) Inter-observer reliability (κ = 0.53)		
	Motor function of shoulder and elbow	Intra-observer reliability (κ = 0.50) Inter-observer reliability (κ = 0.31)		
	Motor function of lower extremity	Intra-observer reliability (κ = 0.55) Inter-observer reliability (κ = 0.49)		
	Sensory function of lower extremity	Intra-observer reliability (κ = 0.54) Inter-observer reliability (κ = 0.58)		
	Sensory function of upper extremity	Intra-observer reliability (κ = 0.51) Inter-observer reliability (κ = 0.42)		
mJOA		Cronbach's α coefficient (0.60–0.63) Forward-backward translation [Brazilian Portuguese and Italian]: n/a Test–retest stability (Spearman's rank correlation: 0.91) Intra-observer reliability (ICC: 0.87) Inter-observer reliability (ICC: 0.97, κ = 0.80) MDC or SDC (Distribution: 2.1) MCID (Distribution: 1.2–1.4, anchor: 1.3–3.1) SCB (14) Criterion validity (Nurick scale) Spearman's rank correlation: –0.41, Pearson's correlation: –0.62 to –0.63	–	C

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		Construct validity: 30MWT (Pearson's correlation: –0.38) EQ-5D (Spearman's rank correlation: 0.42) JOACMEQ QOL (Spearman's rank correlation: 0.41) NDI (Spearman's rank correlation: –0.51, Pearson's correlation: –0.33 to –0.34) SF-12 MCS (Pearson's correlation: 0.03) SF-12 PCS (Pearson's correlation: 0.42) SF-36 MCS (Pearson's correlation: 0.25) SF-36 PCS (Pearson's correlation: 0.30) Responsiveness: mJOA (Effect size: 0.87–1.0, normalised change: 1.47)
	Motor dysfunction of lower extremities	Inter-observer reliability (ICC: 0.73) Criterion validity (Nurick scale) Pearson's correlation: –0.65 to –0.68 Construct validity: 30MWT (Pearson's correlation: –0.43) NDI (Pearson's correlation: –0.31) SF-36 MCS (Pearson's correlation: 0.21) SF-36 PCS (Pearson's correlation: 0.31–0.50)
	Motor dysfunction of upper extremities	Inter-observer reliability (ICC: 0.77) Criterion validity (Nurick scale) Pearson's correlation: –0.42 Construct validity:

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		30MWT (Pearson's correlation: –0.21) NDI (Pearson's correlation: –0.24) SF-36 MCS (Pearson's correlation: 0.20) SF-36 PCS (Pearson's correlation: 0.22)		
	Sensory dysfunction of sphincter dysfunction	Inter-observer reliability (ICC: 0.78) Criterion validity (Nurick scale) Pearson's correlation: –0.25 Construct validity: 30MWT (Pearson's correlation: –0.23) NDI (Pearson's correlation: –0.16) SF-36 MCS (Pearson's correlation: 0.08) SF-36 PCS (Pearson's correlation: 0.06)		
	Sensory dysfunction of upper extremities	Inter-observer reliability (ICC: 0.93) Criterion validity (Nurick scale) Pearson's correlation: –0.23 Construct validity: 30MWT (Pearson's correlation: –0.05) NDI (Pearson's correlation: –0.23) SF-36 MCS (Pearson's correlation: 0.19) SF-36 PCS (Pearson's correlation: 0.19)		
Pain				
NDI		MDC or SDC (Distribution: 6.2%, anchor: 5.2%) MCID (Anchor: 5–13) SCB (Anchor: 9.5–36)	+	B

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		Criterion validity (Likert scale) AUC: 0.66–0.75 Construct validity: Arm pain score (Pearson's correlation: 0.68) mJOA (Pearson's correlation: –0.36) Neck pain score (Pearson's correlation: 0.64) SF-12 MCS (Pearson's correlation: –0.40) SF-12 PCS (Pearson's correlation: –0.54) Responsiveness: Anchor (AUC: 0.66) NDI (Mean change score: –15.8)		
	Pain, "Numeric rating scale" (Arm pain score)	MCID (Anchor: 2.5) SCB (3.5) Construct validity: mJOA (Pearson's correlation: –0.19) Neck pain score (Pearson's correlation: 0.72)	–	B
	Pain, "Numeric rating scale" (Neck pain scores)	MCID (Anchor: 2.5) SCB (3.5) Construct validity: mJOA (Pearson's correlation: –0.07)	–	B
	VAS for pain	MDC or SDC (Distribution: 3.1) MCID (Distribution: 24.0–30.0, anchor: 0.4–2.7) SCB (1.1)	+	B

n/a = No info available

LOA: Limits of agreement

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Feasibility

++ = No barriers

+ = Minimal barriers

– = Barriers

Interpretability

+ = Interpretable

– = Uninterpretable, due to absence of MCID

Recommendation category

A = Measurement instruments with evidence for sufficient content validity (any level) AND at least low-quality evidence for sufficient internal consistency

B = Measurement instruments categorized not in A or C

C = Measurement instruments with high quality evidence for an insufficient measurement property

*Comparators shown as indented tools

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DISCUSSION

DCM is measured and reported inconsistently across clinical trials (1-4). In light of these inconsistencies, AO Spine launched RECODE-DCM (www.aospine.org/recode) with the aim of creating a 'research toolkit' that helps to accelerate knowledge discovery and improve outcomes in DCM. One of the objectives of the RECODE-DCM initiative was to develop a CMS (3, 6). This systematic review consists of an initial step towards building this CMS by identifying tools that have been used in DCM research and examining their quality, in accordance with the COSMIN standards (23-25).

Overall, we identified 29 instruments with at least one in 10 measurement properties evaluated (**Figure 2**); none, however, had evaluations for all 10 properties and <50% had more than one property evaluated per measurement domain (i.e., reliability, validity, and responsiveness) (**Table 2**). We also noted a paucity in the quantity and quality of studies evaluating DCM instruments; this is visible by the absence of category A recommendations and the classification of most tools in category B (**Table 4**). Acknowledging both the stringency of the COSMIN standards and that absence of evidence is not equivalent to poor-quality evidence, we proposed nine instruments that seem interpretable to clinicians and appear to offer qualitative meaning to clinicians and people with lived experience of DCM. These instruments are the SF-12 and SF-36; JOA, mJOA, and JOACMEQ; NDI; and VAS for pain (**Table 5**).

The fact that most outcomes received B-category recommendations due to absence of high-quality evidence is not unexpected. In this review, the most common reasons for low-quality evidence, as per the COSMIN guidelines, were: important methodological flaws in study design or statistical methods, uncertainty of approaches used to analyse the data, and imprecision due to sample size below the recommended power and significance levels. The rigor (or stringency) of the COSMIN standards may have emphasised these quality limitations due to the highly specific nature of some standards and the expectation of psychometric expertise relative to the DCM context. For example, results for internal consistency must be rated 'indeterminate' if there is not at least low-quality evidence for structural validity; no such studies were available in this review, possibly because this is a more recent and complex criterion or because of the search or selection criteria. Similarly, studies on content validity cannot score higher than 'inadequate' if there are no recordings/verbatim transcriptions of patient

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focus groups or interviews. Likewise, analyses of reliability cannot score higher than 'doubtful' if statistics other than the Pearson or Spearman correlation coefficients are used. These thresholds of acceptability may account for some of the lacking information and are an important entry challenge for instruments into DCM research—a field where the routine involvement of stakeholders with lived experience is at an early stage (3, 8), inconsistent study reporting is prevalent (2, 4), and few studies have involved >100 patients. From the application of these COSMIN criteria in other research fields, however, it appears that these methodological deficiencies are not exclusive to DCM instruments, including those in current use (43-45). The lack of high-quality assessments, thus, should not necessarily imply that the identified outcome measures are generally inadequate, and should take into consideration an availability bias in the literature (i.e., that some tools, such as the SF-12, are used because they are the only tools available and, therefore, have available literature due to their routine use).

Measurement rigor is universally important and, in DCM, particularly relevant as the development of new instruments is a top 10 research priority. This rank reinforces the decision of the steering committee to make the initial CMS recommendations based existing on tools, rather than on tools under development. This decision was taken recognising that the success of a CMS requires widespread adoption, and that the adoption of clinical recommendations can be challenging without stakeholder awareness, familiarity, and/or confidence (46-49). We hypothesised that asking the global field to align with new innovations would be more challenging, and premature, at this stage. Thus, for this first iteration of the DCM CMS, there is a focus on current instruments in academic usage. Whilst, currently, few have met the bar set by the COSMIN methodology, there are nine reasonable candidates using our post-hoc thresholds (**Table 5**). Ultimately, the CMS process will need to lean significantly on the expertise of those involved in the consensus phase in order to make final recommendations that are methodologically rigorous and representative of those with lived experience.

Despite its conscientious design, this systematic review has limitations. In searching for existing instruments, we have neither identified nor assessed tools under development, or those currently being translated into clinical or research settings or

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published in languages other than English. To the extent that DCM instruments are currently in use, however, this review only identified tools in four of the six core domains from RECODE-DCM's minimum data set (36) and did not considered the construct of the disease as a factor in evaluating the outcomes. For those missing outcomes, focused scoping reviews (informed by a gap analysis that will be published separately) will be conducted in the future. Next, clinician-reported outcome measures and performance-based outcome measures were analysed with the exact same methods as PROMs. Whilst COSMIN explicitly allows this (23-25), methods may be differentially adapted to tailor to these distinct instrument types; we chose not to do so out of prudence and consistency, and results across these instrument groups should be interpreted accordingly. Feasibility and interpretability were also evaluated using bespoke criteria which, despite being adapted from the COSMIN methodology, may not weigh all criteria accurately. Importantly, our decision to shortlist the clinically interpretable instruments was made a posteriori due to the unexpected absence of category A recommendations. This decision was informed by our judgement that instruments in a CMS should be interpretable by clinicians and offer qualitative meaning to clinicians and people with lived experience. While the COSMIN taxonomy does indeed class interpretability as an important and stand-alone characteristic (23-25), this shortlist may inevitably represent a placement bias. Notably, some nuances of different versions of measurement instruments (e.g., mJOA) were not extensively evaluated (50). Lastly, and as is frequently the case in this body of reviews (43-45), none of the authors is specifically trained in measurement theory and, therefore, this work represents our best attempt to implement the guidelines and standards set forward by the COSMIN methodology in the context of DCM.

CONCLUSIONS

Currently, none of the measurement instruments used in DCM holds sufficient evidence to meet the COSMIN criteria for a strong recommendation for use. However, there are leading contenders; namely, the SF-12 and SF-36; JOA, mJOA, and JOACMEQ; NDI; and VAS for pain. The findings of this review will inform a consensus process to form a CMS for DCM. As the development of new assessments for DCM is an active research priority, greater awareness of the COSMIN framework is pertinent to DCM researchers.

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AUTHOR CONTRIBUTIONS

BMD conceived the article. AYT, KSL, AB, ED, and FB conducted the literature search. AYT and BMD analysed the data and wrote the manuscript. All authors reviewed and critically revised the manuscript prior to submission.

ETHICS APPROVAL

Ethics approval for this study was not applicable because this study did not involve human participants.

DATA AVAILABILITY STATEMENT

All data relevant to the study are included in the article or uploaded as supplementary information.

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COMPETING INTERESTS

None declared.

REFERENCES

1. Fehlings MG, Tetreault LA, Riew KD, Middleton JW, Aarabi B, Arnold PM, et al. A Clinical Practice Guideline for the Management of Patients With Degenerative Cervical Myelopathy: Recommendations for Patients With Mild, Moderate, and Severe Disease and Nonmyelopathic Patients With Evidence of Cord Compression. Global Spine Journal. 2017;7(3_suppl):70S-83S.
2. Davies BM, McHugh M, Elgheriani A, Kolias AG, Tetreault LA, Hutchinson PJA, et al. Reported Outcome Measures in Degenerative Cervical Myelopathy: A Systematic Review. PLOS ONE. 2016;11(8):e0157263.

For peer review only

3. Davies BM, Khan DZ, Mowforth OD, McNair AGK, Gronlund T, Kolias AG, et al. RE-CODE DCM (REsearch Objectives and Common Data Elements for Degenerative Cervical Myelopathy): A Consensus Process to Improve Research Efficiency in DCM, Through Establishment of a Standardized Dataset for Clinical Research and the Definition of the Research Priorities. Global Spine Journal. 2019;9(1_suppl):65S-76S.
4. Davies BM, McHugh M, Elgheriani A, Kolias AG, Tetreault L, Hutchinson PJA, et al. The reporting of study and population characteristics in degenerative cervical myelopathy: A systematic review. PloS one. 2017;12(3):e0172564-e.
5. Davies BM, Mowforth OD, Smith EK, Kotter MR. Degenerative cervical myelopathy. BMJ. 2018;360:k186.
6. AO Spine. AO Spine RECODE-DCM: REsearch Objectives and COmmon Data Elements for Degenerative Cervical Myelopathy 2021 [Available from: www.aospine.org/recode.
7. Davies B, Mowforth O, Sadler I, Aarabi B, Kwon B, Kurpad S, et al. Recovery priorities in degenerative cervical myelopathy: a cross-sectional survey of an international, online community of patients. BMJ Open. 2019;9(10):e031486.
8. Boerger TF, Davies BM, Sadler I, Sarewitz E, Kotter MRN. Patient, sufferer, victim, casualty or person with cervical myelopathy: let us decide our identifier. Integrated Healthcare Journal. 2020;2(1):e000023.
9. Clarke M. Standardising outcomes for clinical trials and systematic reviews. Trials. 2007;8(1):39.
10. Kirkham JJ, Dwan KM, Altman DG, Gamble C, Dodd S, Smyth R, et al. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. BMJ. 2010;340:c365.
11. Kirkham JJ, Gargon E, Clarke M, Williamson PR. Can a core outcome set improve the quality of systematic reviews? – a survey of the Co-ordinating Editors of Cochrane review groups. Trials. 2013;14(1):21.
12. Prinsen CAC, Vohra S, Rose MR, Boers M, Tugwell P, Clarke M, et al. How to select outcome measurement instruments for outcomes included in a "Core Outcome Set" – a practical guideline. Trials. 2016;17(1):449.
13. Williamson PR, Altman DG, Bagley H, Barnes KL, Blazeby JM, Brookes ST, et al. The COMET Handbook: version 1.0. Trials. 2017;18(3):280.
14. Boers M, Kirwan JR, Wells G, Beaton D, Gossec L, d'Agostino M-A, et al. Developing Core Outcome Measurement Sets for Clinical Trials: OMERACT Filter 2.0. Journal of Clinical Epidemiology. 2014;67(7):745-53.
15. Schmitt J, Apfelbacher C, Spuls PI, Thomas KS, Simpson EL, Furue M, et al. The Harmonizing Outcome Measures for Eczema (HOME) roadmap: a methodological framework to develop core sets of outcome measurements in dermatology. J Invest Dermatol. 2015;135(1):24-30.
16. Kirkham JJ, Davis K, Altman DG, Blazeby JM, Clarke M, Tunis S, et al. Core Outcome Set-STAndards for Development: The COS-STAD recommendations. PLOS Medicine. 2017;14(11):e1002447.
17. Kirkham JJ, Gorst S, Altman DG, Blazeby JM, Clarke M, Devane D, et al. Core Outcome Set-STAndards for Reporting: The COS-STAR Statement. PLoS Med. 2016;13(10):e1002148.
18. Potter S, Davies C, Holcombe C, Weiler-Mithoff E, Skillman J, Vidya R, et al. International development and implementation of a core measurement set for research and audit studies in implant-based breast reconstruction: a study protocol. BMJ Open. 2020;10(1):e035505.

For peer review only

19. Grieve S, Perez R, Birklein F, Brunner F, Bruehl S, Harden RN, et al. Recommendations for a first Core Outcome Measurement set for complex regional PAin syndrome Clinical sTudies (COMPACT). Pain. 2017;158(6):1083-90.
20. Davies CF, Macefield R, Avery K, Blazeby JM, Potter S. Patient-Reported Outcome Measures for Post-mastectomy Breast Reconstruction: A Systematic Review of Development and Measurement Properties. Ann Surg Oncol. 2021;28(1):386-404.
21. Kalsi-Ryan S, Singh A, Massicotte EM, Arnold PM, Brodke DS, Norvell DC, et al. Ancillary outcome measures for assessment of individuals with cervical spondylotic myelopathy. Spine (Phila Pa 1976). 2013;38(22 Suppl 1):S111-22.
22. Mokkink LB, Terwee CB, Knol DL, Stratford PW, Alonso J, Patrick DL, et al. Protocol of the COSMIN study: COnsensus-based Standards for the selection of health Measurement INstruments. BMC Medical Research Methodology. 2006;6(1):2.
23. Prinsen CAC, Mokkink LB, Bouter LM, Alonso J, Patrick DL, de Vet HCW, et al. COSMIN guideline for systematic reviews of patient-reported outcome measures. Qual Life Res. 2018;27(5):1147-57.
24. Mokkink LB, de Vet HCW, Prinsen CAC, Patrick DL, Alonso J, Bouter LM, et al. COSMIN Risk of Bias checklist for systematic reviews of Patient-Reported Outcome Measures. Quality of Life Research. 2018;27(5):1171-9.
25. Terwee CB, Prinsen CAC, Chiarotto A, Westerman MJ, Patrick DL, Alonso J, et al. COSMIN methodology for evaluating the content validity of patient-reported outcome measures: a Delphi study. Qual Life Res. 2018;27(5):1159-70.
26. Davies BM, Goh S, Yi K, Kuhn I, Kotter MRN. Development and validation of a MEDLINE search filter/hedge for degenerative cervical myelopathy. BMC Medical Research Methodology. 2018;18(1):73.
27. Khan DZ, Khan MS, Kotter MR, Davies BM. Tackling Research Inefficiency in Degenerative Cervical Myelopathy: Illustrative Review. JMIR Res Protoc [Internet]. 2020 2020/06//; 9(6):[e15922 p.]. Available from: http://europepmc.org/abstract/MED/32525490 https://doi.org/10.2196/15922 https://europepmc.org/articles/PMC7317636.
28. Khan MA, Mowforth OD, Kuhn I, Kotter MRN, Davies BM. Development of a validated search filter for Ovid Embase for degenerative cervical myelopathy. Health Information & Libraries Journal.n/a(n/a).
29. Singh A, Tetreault L, Casey A, Laing R, Statham P, Fehlings MG. A summary of assessment tools for patients suffering from cervical spondylotic myelopathy: a systematic review on validity, reliability and responsiveness. Eur Spine J. 2015;24 Suppl 2:209-28.
30. Mokkink LB, Terwee CB, Patrick DL, Alonso J, Stratford PW, Knol DL, et al. The COSMIN study reached international consensus on taxonomy, terminology, and definitions of measurement properties for health-related patient-reported outcomes. J Clin Epidemiol. 2010;63(7):737-45.
31. van Kampen DA, Willems WJ, van Beers LWAH, Castelein RM, Scholtes VAB, Terwee CB. Determination and comparison of the smallest detectable change (SDC) and the minimal important change (MIC) of four-shoulder patient-reported outcome measures (PROMs). Journal of Orthopaedic Surgery and Research. 2013;8(1):40.
32. Nwachukwu BU, Beck EC, Kunze KN, Chahla J, Rasio J, Nho SJ. Defining the Clinically Meaningful Outcomes for Arthroscopic Treatment of Femoroacetabular Impingement Syndrome at Minimum 5-Year Follow-up. The American Journal of Sports Medicine. 2020;48(4):901-7.

For peer review only

33. Streiner DL, Norman GR, Cairney J. Health Measurement ScalesA practical guide to their development and use: A practical guide to their development and use: Oxford University Press; 2015 2015-01.
34. Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. BMJ. 2020;368:l6890.
35. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 2021;372:n71.
36. AO Spine. Minimum Dataset [Internet] 2021 [Available from: https://aospine.aofoundation.org/research/recode-dcm/minimum--dataset.
37. de Vet HCW, Terwee CB, Mokkink LB, Knol DL. Measurement in Medicine: A Practical Guide. Cambridge: Cambridge University Press; 2011.
38. Longo UG, Berton A, Denaro L, Salvatore G, Denaro V. Development of the Italian version of the modified Japanese orthopaedic association score (mJOA-IT): cross-cultural adaptation, reliability, validity and responsiveness. European Spine Journal. 2016;25(9):2952-7.
39. de Vet HC, Terwee CB. The minimal detectable change should not replace the minimal important difference. J Clin Epidemiol. 2010;63(7):804-5; author reply 6.
40. de Vet HC, Ostelo RW, Terwee CB, van der Roer N, Knol DL, Beckerman H, et al. Minimally important change determined by a visual method integrating an anchor-based and a distribution-based approach. Qual Life Res. 2007;16(1):131-42.
41. de Vet HC, Terwee CB, Ostelo RW, Beckerman H, Knol DL, Bouter LM. Minimal changes in health status questionnaires: distinction between minimally detectable change and minimally important change. Health Qual Life Outcomes. 2006;4:54-.
42. Crosby RD, Kolotkin RL, Williams GR. Defining clinically meaningful change in health-related quality of life. J Clin Epidemiol. 2003;56(5):395-407.
43. Francis DO, McPheeters ML, Noud M, Penson DF, Feurer ID. Checklist to operationalize measurement characteristics of patient-reported outcome measures. Syst Rev. 2016;5(1):129.
44. Francis DO, Daniero JJ, Hovis KL, Sathe N, Jacobson B, Penson DF, et al. Voice-Related Patient-Reported Outcome Measures: A Systematic Review of Instrument Development and Validation. Journal of Speech, Language, and Hearing Research. 2017;60(1):62-88.
45. Terwee CB, de Vet HC, Prinsen CAC, Mokkink LB. Comment on "Checklist to operationalize measurement characteristics of patient-reported outcome measures" [Internet] [Available from: https://www.cosmin.nl/wp-content/uploads/Letter-comment-on-Francis.pdf.
46. Kirkham JJ, Clarke M, Williamson PR. A methodological approach for assessing the uptake of core outcome sets using ClinicalTrials.gov: findings from a review of randomised controlled trials of rheumatoid arthritis. BMJ. 2017;357:j2262.
47. Bauer MS, Kirchner J. Implementation science: What is it and why should I care? Psychiatry Research. 2020;283:112376.
48. Gupta DM, Boland RJ, Aron DC. The physician's experience of changing clinical practice: a struggle to unlearn. Implementation Science. 2017;12(1):28.
49. Braithwaite J, Churruca K, Long JC, Ellis LA, Herkes J. When complexity science meets implementation science: a theoretical and empirical analysis of systems change. BMC Medicine. 2018;16(1):63.

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50. Furlan JC, Catharine Craven B. Psychometric analysis and critical appraisal of the original, revised, and modified versions of the Japanese Orthopaedic Association score in the assessment of patients with cervical spondylotic myelopathy. Neurosurg Focus. 2016;40(6):E6.

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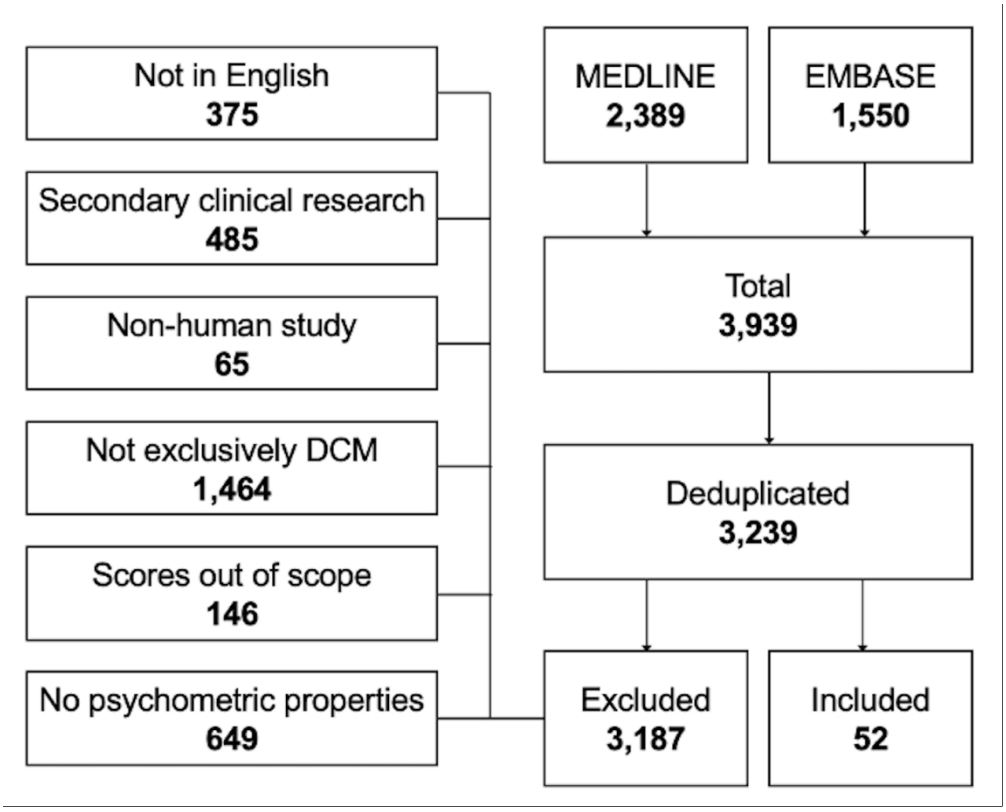


Figure 1. PRISMA flow chart. A systematic review of MEDLINE and EMBASE was conducted between August 2020 and January 2021 to identify original research on the measurement properties of instruments currently used in DCM research.

119x96mm (300 x 300 DPI)

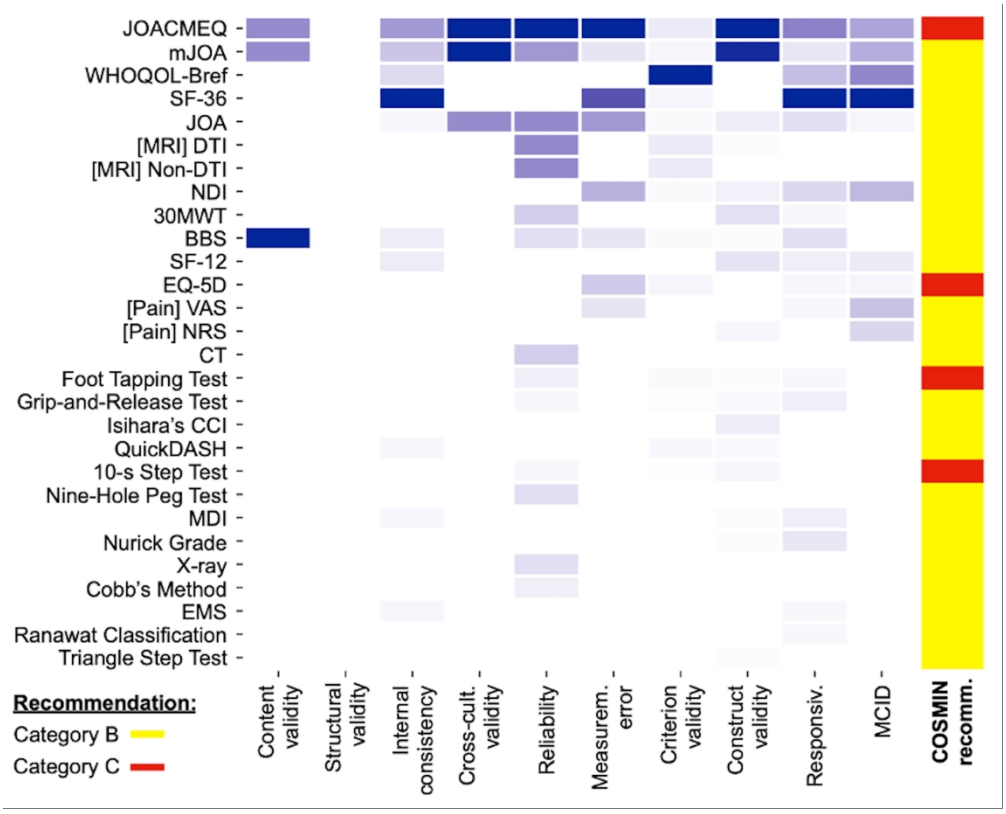


Figure 2. Number of studies for each outcome measure and property [normalised]. Included studies reported on at least one of the 10 COSMIN properties for all instruments. No instrument had evidence for all 10 properties and <50% (13/29) of instruments had evidence for at least one property per measurement domain (see Table 2 for definitions). Notably, no instruments were evaluated for structural validity, attained sufficient evidence for content validity, or obtained a Category A recommendation based on COSMIN criteria.

159x128mm (300 x 300 DPI)

SUPPLEMENTARY INFORMATION

Supplementary Table 1. Full search phrases used for MEDLINE and EMBASE on 4 August 2020

Ovid MEDLINE		**2389 articles**
Cervical spine concept		
1	exp Cervical Vertebrae/ or exp Cervical Cord/ or cervical.tw	
DCM concept		
2	Exp Spinal Cord Diseases/ or Exp Spinal Diseases/	
3	degenerat*.tw	
4	2 and 3	
5	Myelopath*.tw	
6	Myeloradiculopath*.tw	
7	Radiculopath*.tw	
8	Exp Spinal Cord Compression/	
9	Exp “Ossification of the Posterior Longitudinal Ligament”/	
10	Ossification of the Posterior Longitudinal Ligament.tw	
11	OPLL.tw	
12	Exp Spinal Stenosis/	
13	Cervical stenosis.tw	
14	Exp Spondylosis/	
15	Spondylosis.tw	
16	Spondylotic.tw	
17	Degenerative cervical myelopathy.tw	
18	DCM.tw	
19	Cervical spondylotic myelopathy.tw	
20	CSM.tw	
21	4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20	
22	1 and 21	
Tools for function concept (neurological/gait/mobility centric) concept		
23	Japanese Orthopaedic Association score.tw	
24	JOA.tw	
25	modified JOA.tw	
26	mJOA.tw	
27	Graded Redefined Assessment of Sensation Strength and Prehension.tw	
28	GRASSP.tw	
29	Quick Disability of the Arm Shoulder and Hand.tw	
30	QuickDASH.tw	

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31 Myelopathy Disability Index.tw
32 MDI.tw
33 Nurick score.tw
34 Neck functional disability scale.tw
35 NFDS.tw
36 Neck Disability Index.tw
37 NDI.tw
38 Cooper myelopathy scale.tw
39 CMS.tw
40 European myelopathy score.tw
41 EMS.tw
42 Bournemouth questionnaire.tw
43 BQ.tw
44 Cervical spine outcomes questionnaire.tw
45 CSOQ.tw
46 Patient specific functional scale.tw
47 PSFS.tw
48 World Health Organization Quality of Life Instruments.tw
49 WHOQOL.tw
50 Grip and release test.tw
51 GRT.tw
52 Grip Dynamometer.tw
53 Triangle step test.tw
54 Foot tapping test.tw
55 30 m walking test.tw
56 30MWT.tw
57 10 m walking test.tw
58 10MWT.tw
59 Berg Balance Scale.tw
60 BBS.tw
61 GAITRite.tw
62 10 second step test.tw
63 9 hole peg test.tw
64 Prolo.tw
65 Mental component score.tw
66 MCS.tw
67 Physical component score.tw
68 PCS.tw

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69	Hospital anxiety depression scale.tw
70	HADS.tw
71	Global rating of change.tw
72	GROC.tw
73	23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67 or 68 or 69 or 70 or 71 or 72
Tools for QOL concept (including pain) concept	
74	Exp “Quality of Life”/ or exp “Surveys and Questionnaires”/
75	Short Form Health Survey.tw
76	SF-36.tw
77	SF-12.tw
78	EQ-5D.tw
79	Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire.tw
80	JOACMEQ.tw
81	Visual Analogue Scale.tw
82	VAS.tw
83	Likert scale.tw
84	Numeric pain rating scale.tw
85	NPRS.tw
86	North American Spine Satisfaction.tw
87	NASS.tw
88	74 or 75 or 76 or 77 or 78 or 79 or 80 or 81 or 82 or 83 or 84 or 85 or 86 or 87
Psychometric concept	
89	Exp Psychometrics/
90	Pyschometr*.tw
91	(clinimetr* or clinometr*).tw.
92	Outcome assessment*.tw
93	exp Health Status Indicators/
94	Exp “Reproducibility of Results”/
95	Reproducib*.tw
96	Exp Validation Study/
97	exp Discriminant Analysis/
98	(reliab* or unreliab* or valid* or coefficient or homogeneity or homogeneous or internal consistency).tw
99	(cronbach* and (alpha or alphas)).tw.
100	(item and (correlation* or selection* or reduction*)).tw

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101 (agreement or precision or imprecision or precise values or test–retest).tw
102 (reliab* and (test or retest)).tw

103 (stability or interrater or inter-rater or intrarater or intra-rater or intertester or inter-tester or intratester or intra-tester or interobserver or inter-observer or intraobserver or intraobserver or intertechnician or inter-technician or intratechnician or intra-technician or interexaminer or inter-examiner or intraexaminer or intra-examiner or interassay or interassay or intraassay or intra-assay or interindividual or inter-individual or intraindividual or intra-individual or interparticipant or inter-participant or intraparticipant or intra-participant or kappa or kappas or repeatab*).tw
104 ((replicab* or repeated) and (measure or measures or findings or result or results or test or tests)).tw
105 (generaliza* or generalisa* or concordance).tw
106 (intraclass and correlation*).tw
107 Exp Observer Variation/
108 Observer variation.tw
109 (multitrait and scaling and (analysis or analyses)).tw
110 Measurement error*.tw
111 (item discriminant or interscale correlation* or error or errors or individual variability).tw
112 (variability and (analysis or values)).tw
113 (uncertainty and (measurement or measuring)).tw
114 Exp Diagnostic Errors/
115 Exp Data accuracy/
116 Exp Dimensional Measurement Accuracy/
117 Accuracy.tw
118 ((minimal or minimally or clinical or clinically) and (important or significant or detectable) and (change or difference)).tw
119 Minimally clinically important difference*.tw
120 MCID.tw
121 (small* and (real or detectable) and (change or difference)).tw
122 (meaningful change or ceiling effect or floor effect or Item response model or IRT or Rasch or Differential item functioning or DIF or computer adaptive testing or item bank or cross-cultural equivalence).tw
123 Exp Bias/ or exp Selection Bias/
124 Bias.tw
125 Exp “Predictive Value of Test”/

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126	89 or 90 or 91 or 92 or 93 or 94 or 95 or 96 or 97 or 98 or 99 or 100 or 101 or 102 or 103 or 104 or 105 or 106 or 107 or 108 or 109 or 110 or 111 or 112 or 113 or 114 or 115 or 116 or 117 or 118 or 119 or 120 or 121 or 122 or 123 or 124 or 125
Combined concepts	
127	73 or 88
128	22 and 126 and 127

EMBASE **1550 articles**

Cervical spine concept	
1	exp Cervical Vertebra/ or cervical spine/ or exp Cervical spinal cord/ or cervical.tw
DCM concept	
2	Exp Spinal Cord Disease/ or Exp Spine Disease/
3	Exp degeneration/
4	degenerat*.tw
5	3 or 4
6	2 and 5
7	Myelopath*.tw
8	Myeloradiculopath*.tw
9	Exp radiculopathy/
10	Radiculopath*.tw
11	Exp Spinal Cord Compression/
12	Exp Posterior Longitudinal Ligament/ and exp ossification/
13	Ossification of the Posterior Longitudinal Ligament.tw
14	OPLL.tw
15	Exp vertebral canal stenosis/
16	Cervical stenosis.tw
17	Exp Cervical Spondylosis/
18	Exp Spondylosis/
19	Spondylosis.tw
20	Spondylotic.tw
21	Exp Cervical myelopathy/
22	Degenerative cervical myelopathy.tw
23	DCM.tw
24	Exp Cervical spondylotic myelopathy/
25	Cervical spondylotic myelopathy.tw
26	CSM.tw
27	6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26

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28	1 and 27
Tools for function concept (neurological/gait/mobility centric) concept	
29	Exp Japanese Orthopaedic Association score/
30	Japanese Orthopaedic Association score.tw
31	JOA.tw
32	modified JOA.tw
33	mJOA.tw
34	Exp “Disabilities of the Arm, Shoulder and Hand (score)”/
35	Graded Redefined Assessment of Sensation Strength and Prehension.tw
36	GRASSP.tw
37	Quick Disability of the Arm Shoulder and Hand.tw
38	QuickDASH.tw
39	Myelopathy Disability Index.tw
40	MDI.tw
41	Exp “Nurick (grade)”/
42	Nurick score.tw
43	Neck functional disability scale.tw
44	NFDS.tw
45	Exp Neck Disability Index/
46	Neck Disability Index.tw
47	NDI.tw
48	Cooper myelopathy scale.tw
49	CMS.tw
50	European myelopathy score.tw
51	EMS.tw
52	Bournemouth questionnaire.tw
53	BQ.tw
54	Cervical spine outcomes questionnaire.tw
55	CSOQ.tw
56	Patient specific functional scale.tw
57	PSFS.tw
58	World Health Organization Quality of Life Instruments.tw
59	WHOQOL.tw
60	Grip and release test.tw
61	GRT.tw
62	Grip Dynamometer.tw
63	Triangle step test.tw
64	Foot tapping test.tw

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65	30 m walking test.tw
66	30MWT.tw
67	10 m walking test.tw
68	10MWT.tw
69	Berg Balance Scale.tw
70	BBS.tw
71	GAITRite.tw
72	10 second step test.tw
73	9 hole peg test.tw
74	Prolo.tw
75	Mental component score.tw
76	MCS.tw
77	Physical component score.tw
78	PCS.tw
79	Hospital anxiety depression scale.tw
80	HADS.tw
81	Global rating of change.tw
82	GROC.tw
83	29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67 or 68 or 69 or 70 or 71 or 72 or 73 or 74 or 75 or 76 or 77 or 78 or 79 or 80 or 81 or 82
Tools for QOL concept (including pain) concept	
84	Short Form Health Survey.tw
85	Exp Short Form 36/
86	SF-36.tw
87	Exp Short Form 12/
88	SF-12.tw
89	Exp “European Quality of Life 5 Dimensions questionnaire”/
90	EQ-5D.tw
91	Exp Japanese Orthopaedic Association Cervical Myelopathy Evaluation/
92	Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire.tw
93	JOACMEQ.tw
94	Exp visual analog scale/
95	Visual Analogue Scale.tw
96	VAS.tw
97	Exp Likert scale/
98	Likert scale.tw

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99	Numeric pain rating scale.tw
100	NPRS.tw
101	North American Spine Satisfaction.tw
102	NASS.tw
103	84 or 85 or 86 or 87 or 88 or 89 or 90 or 91 or 92 or 93 or 94 or 95 or 96 or 97 or 98 or 99 or 100 or 101 or 102
Psychometric concept	
104	Exp Psychometry/
105	Pyschometr*.tw
106	(clinimetr* or clinometr*).tw.
107	Outcome assessment*.tw
108	exp Health Status Indicator/
109	Exp Reproducibility/
110	Reproducib*.tw
111	Exp Validation Study/
112	exp Discriminant Analysis/
113	(reliab* or unreliab* or valid* or coefficient or homogeneity or homogeneous or internal consistency).tw
114	(cronbach* and (alpha or alphas)).tw.
115	(item and (correlation* or selection* or reduction*)).tw
116	(agreement or precision or imprecision or precise values or test–retest).tw
117	(reliab* and (test or retest)).tw
118	(stability or interrater or inter-rater or intrarater or intra-rater or intertester or inter-tester or intratester or intra-tester or interobserver or inter-observer or intraobserver or intraobserver or intertechnician or inter-technician or intratechnician or intra-technician or interexaminer or inter-examiner or intraexaminer or intra-examiner or interassay or interassay or intraassay or intra-assay or interindividual or inter-individual or intraindividual or intra-individual or interparticipant or inter-participant or intraparticipant or intra-participant or kappa or kappas or repeatab*).tw
119	((replicab* or repeated) and (measure or measures or findings or result or results or test or tests)).tw
120	(generaliza* or generalisa* or concordance).tw
121	(intraclass and correlation*).tw
122	Exp Observer Variation/
123	Observer variation.tw
124	(multitrait and scaling and (analysis or analyses)).tw
125	Measurement error*.tw
126	(item discriminant or interscale correlation* or error or errors or individual variability).tw

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127	(variability and (analysis or values)).tw
128	(uncertainty and (measurement or measuring)).tw
129	Exp Diagnostic Error/
130	Exp Data accuracy/
131	Exp Dimensional Measurement Accuracy/
132	((minimal or minimally or clinical or clinically) and (important or significant or detectable) and (change or difference)).tw
133	Minimally clinically important difference*.tw
134	MCID.tw
135	(small* and (real or detectable) and (change or difference)).tw
136	(meaningful change or ceiling effect or floor effect or Item response model or IRT or Rasch or Differential item functioning or DIF or computer adaptive testing or item bank or cross-cultural equivalence).tw
137	Exp Bias/ or exp Selection Bias/
138	Bias.tw
139	104 or 105 or 106 or 107 or 108 or 109 or 110 or 111 or 112 or 113 or 114 or 115 or 116 or 117 or 118 or 119 or 120 or 121 or 122 or 123 or 124 or 125 or 126 or 127 or 128 or 129 or 130 or 131 or 132 or 133 or 134 or 135 or 136 or 137 or 138
Combined concepts	
140	83 or 103
141	28 and 139 and 140

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Supplementary Table 2. Study characteristics

Study	Country	Sample size	Psychometric properties	Outcome measures
Auffinger, Lall (1)	United States	30	MCID/SCB, Measurement error	NDI, VAS for pain, SF-36
Augusto, Diniz (2)	Brazil	30	Cross-cultural validity/Measurement invariance, Reliability, Responsiveness	JOA, NDI
Azimi, Rezaei (3)	Iran	87	Cross-cultural validity/Measurement invariance, Responsiveness	JOACMEQ
Badhiwala, Witiw (4)	Canada	606	MCID/SCB	SF-36, mJOA, Nurick Scale, NDI
Bohm, Fehlings (5)	Multicenter/ Global	601	Reliability, Hypotheses testing for construct validity, Responsiveness	Walking tests (timed or steps), mJOA, Nurick Scale, NDI, SF-36
Carreon, Glassman (6)	United States	505	MCID/SCB	NDI, SF-36, "Numeric rating scale" for pain
Chang, Kong (7)	Korea	108	Reliability	CT / CTM
Chiba, Kato (8)	Japan		Reliability	X-rays
Chien, Lai (9)	Taiwan	45	Responsiveness, MCID/SCB	JOACMEQ, NDI
Chiu and Pang (10)	Hong Kong	72	Internal consistency, Reliability, Content validity, Hypotheses testing for construct validity, Criterion validity, Measurement error, Responsiveness	BBS, mJOA
Fukui, Chiba (11)	Japan	368	Content validity	JOACMEQ
Fukui, Chiba (12)	Japan	201	Reliability	JOACMEQ
Goyal, Murphy (13)	United States	118	Responsiveness	NDI, SF-12
Gwinn, Iannotti (14)	United States	20	Reliability	X-rays, Cobb's method
Hosono, Sakaura (15)	Japan	30	Reliability, Criterion validity	Grip-and-release test, JOA
Hosono, Takenaka (16)	Japan	48	Responsiveness	Grip-and-release test, JOA
Kang, Lee (17)	Korea	82	Reliability	MRI (not DTI)
Kato, Oshima (18)	Japan	92	Measurement error, Hypotheses testing for construct validity, Responsiveness	JOA, mJOA, JOACMEQ, NDI, SF-12
Kato, Oshima (19)	Japan	101	Measurement error, Criterion validity, MCID/SCB	JOA, Likert scale

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Kato, Oshima (20)	Japan	101	Measurement error, Criterion validity, MCID/SCB	JOACMEQ, NDI, EQ-5D, SF-36, Likert scale
King and Roberts (21)	United States	88	Internal consistency	SF-36
Ko, Choi (22)	Korea	357	Reliability	MRI (not DTI)
Kopjar, Tetreault (23)	USA	277	Responsiveness, Hypotheses testing for construct validity, Internal consistency	mJOA, Nurick Scale, NDI, SF-36, Walking tests (timed or steps)
Latimer, Haden (24)	England	70	Responsiveness	SF-36, NDI, VAS for pain, MDI
Longo, Berton (25)	Italy	75	Cross-cultural validity/Measurement invariance, Reliability, Internal consistency, Hypotheses testing for construct validity, Responsiveness, Criterion validity	mJOA, Nurick Scale, NDI, SF-36
Lubelski, Alvin (26)	United States	119	Hypotheses testing for construct validity, Responsiveness, Criterion validity	mJOA, Nurick Scale, EQ-5D
Mihara, Kondo (27)	Japan	270	Hypotheses testing for construct validity	Grip-and-release test, Triangle step test
Nakamoto, Oshima (28)	Japan	94	Internal consistency, Hypotheses testing for construct validity, Criterion validity	QuickDASH, JOA, NDI, SF-36, "Numeric rating scale" for pain
Nakashima, Yukawa (29)	Japan	101	Hypotheses testing for construct validity	
Nicholson, Millhouse (30)	United States	235	Hypotheses testing for construct validity	MRI (not DTI), mJOA, NDI, SF-12, "Numeric rating scale" for pain, Isihara's Cervical Curvature Index
Nikaido, Kikuchi (31)	Japan	87	Hypotheses testing for construct validity)	JOACMEQ, SF-36
Numasawa, Ono (32)	Japan	126	Hypotheses testing for construct validity, Responsiveness, Reliability	JOA, Foot tapping test, Grip-and-release test
Olindo, Signate (33)	France	40	Reliability	9-Hole peg test, MRI (not DTI), Nurick Scale, mJOA, Walking tests (timed or steps)
Park, Kim (34)	Korea	100	Reliability	MRI (not DTI)
Pratali, Smith (35)	Brazil		Cross-cultural validity	mJOA
Pratali, Smith (36)	Brazil	55	Reliability	mJOA

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Rhee, Shi (37)	United States	100	Criterion validity, Reliability, Content validity	mJOA
Sato, Horikoshi (38)	Japan	66	Hypotheses testing for construct validity	MRI (DTI), JOA
Shim, Lee (39)	Korea	79	Reliability, Criterion validity	MRI (not DTI)
Singh and Crockard (40)	England	100	Internal consistency, Responsiveness	Odom's Criteria, Nurick Scale, Ranawat classification of disease severity, MDI, JOA, EMS, SF-36
Singh and Crockard (41)	United Kingdom	41	Hypotheses testing for construct validity	Walking tests (timed or steps), MDI, Nurick Scale
Singh, Gnanalingham (42)	England	105	Internal consistency, Criterion validity, Responsiveness	SF-12, SF-36
Spurgas, Abbas (43)	USA	35	MCID/SCB	VAS for pain, NDI, SF-12, mJOA
Tetreault, Nouri (44)	Canada	755	MCID/SCB	mJOA, NDI
Thakar and Rajshekhar (45)	India	51	MCID/SCB, Responsiveness	VAS for pain, Nurick Scale, SF-36
Thakar, Christopher (46)	India	70	Internal consistency, Criterion validity, Responsiveness, MCID/SCB	WHOQOL-Bref, SF-36, Nurick Scale
Wada, Fukui (47)	Japan	137	Responsiveness	JOACMEQ, JOA, 10-s step test
Witayakom, Paholpak (48)	Thailand	70	Cross-cultural validity/Measurement invariance, Reliability, Internal consistency, Hypotheses testing for construct validity	JOACMEQ, SF-36
Yonenobu, Abumi (49)	Japan	29	Reliability	JOA
Yukawa, Kato (50)	Japan	163	Hypotheses testing for construct validity, Reliability, Criterion validity	10-s step test, JOA, Grip-and-release test
Zhang, Zhou (51)	China	142	Internal consistency, Responsiveness, MCID/SCB	SF-36, mJOA
Zhou, Zhang (52)	China	113	MCID/SCB, Measurement error	mJOA, SF-36

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Supplementary Table 3. Interpretability (i.e., MCID and SCB).

Instrument		Result summary	Overall rating
EQ-5D		MCID: 0.05; total sample size: 101	Sufficient
JOA		MCID: 2.5; total sample size: 101	Sufficient
JOACMEQ			
	Bladder function	MCID: 6.0; total sample size: 78	Sufficient
	Cervical spine function	MCID: 2.5; total sample size: 179	Sufficient
	Lower extremity function	MCID range 2.5–9.4; total sample size: 179	Sufficient
	QOL	MCID range 8.5–9.5; total sample size: 179	Sufficient
	Upper extremity function	MCID range 2.5–13.0; total sample size: 179	Sufficient
mJOA		MCID range 1.3–3.1; total sample size: 868	Sufficient
		SCB: 14; total sample size: 35	Indeterminate
NDI		MCID range 5–13; total sample size: 108	Sufficient
		SCB range 9.5–36; total sample size: 65	Indeterminate
Pain, "Numeric rating scale" (Arm pain)		MCID: 2.5; total sample size: 30 SCB: 3.5; total sample size: 30	Indeterminate
Pain, "Numeric rating scale" (Neck pain)		MCID: 2.5; total sample size: 30 SCB: 3.5; total sample size: 30	Indeterminate
SF-12			

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	MCS	SCB: 51.5; total sample size: 35	Indeterminate
	PCS	SCB: 30.1; total sample size: 35	Indeterminate
SF-36			
	MCS	MCID range 3.0–7.4; total sample size: 749	Sufficient
	PCS	"MCID range 3.9–9.6; total sample size: 890 SCB: 16; total sample size: 30"	Sufficient
VAS for pain		MCID range 0.4–2.7; total sample size: 30	Sufficient
		SCB: 1.1; total sample size: 30	Indeterminate

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Supplementary Table 4. Feasibility assessment.

Tool	Time (min)	Equipment	Training	License	Money	Ease of administration	Overall assessment
10-s step test	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
30MWT	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
9-Hole peg test	5	Yes	No	No	No	Barriers	Barriers
Berg Balance Scale	>15	Yes	Yes	No	No	Barriers	Barriers
Cobb's method (C2-C7)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
CT (Tsuyama's classification, 2D & 3D)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
CT (Tsuyama's classification, lateral + axial)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
EQ-5D	5	Minimal	No	Yes	Yes	Minimal barriers	Minimal barriers
European Myelopathy Scale	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
Foot tapping test	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
Grip-and-release test	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
Isihara's Cervical Curvature Index	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
JOA	5	No	No	No	No	No barriers	No barriers

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JOACMEQ	5–15	Minimal	No	No	No	Minimal barriers	Minimal barriers
MDI	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
mJOA	5	No	No	No	No	No barriers	No barriers
MRI (Depiction of intramedullary hyperintensity at eight cervical disc levels, T2W, 1.5-T or 3-T)	5–15	Minimal	No	No	No	Minimal barriers	Minimal barriers
MRI (Kang's classification, 1.5-T or 3-T)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
MRI (Muhle's classification, 1.5-T)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
MRI (Vaccaro's classification, 1.5-T)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
NDI	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
Nurick scale	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
P-mJOA	5	Minimal	No	No	No	Minimal barriers	Minimal barriers

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Arm pain score	5	No	No	No	No	No barriers	No barriers
Neck pain score	5	No	No	No	No	No barriers	No barriers
QuickDASH	5	Minimal	No	Yes	Yes	Minimal barriers	Barriers
Ranawat classification of disease severity	5	No	No	No	No	No barriers	No barriers
SF-12	5	Minimal	No	Yes	Yes	Minimal barriers	Barriers
SF-36	5–15	Minimal	No	Yes	Yes	Minimal barriers	Barriers
Triangle step test	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
VAS for pain	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
WHOQOL-Bref	5–15	Minimal	No	No	No	Minimal barriers	Minimal barriers
X-rays (Computer-assisted measurement of length & thickness)	5–15	Minimal	No	No	No	Minimal barriers	Minimal barriers

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Supplementary Table 5. Content validity.

Instrument	Result summary	Overall rating	Quality of evidence
BBS	Patient comprehensibility: Item discrimination index >0.589	Indeterminate	Very low
JOACMEQ	Patient comprehensibility: "No questions elicited no answer or “I am not sure” in more than 5% of patients"	Indeterminate	Very low
P-mJOA	Patient comprehensibility: "In patients preferring to complete the mJOA them- selves, the most popular answers were: “ease of answering the questions” (n = 33), “understanding of the questions” (n = 17)"	Indeterminate	Very low

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Supplementary Table 6. Internal consistency.

Instrument	Result summary	Overall rating	Quality of evidence
BBS	Cronbach's alpha range 0.95–0.98; consistent; total sample size: 72	Indeterminate	Moderate
European Myelopathy Scale	Cronbach's alpha: 0.68; consistent; total sample size: 100	Indeterminate	Low
JOA	Cronbach's alpha: 0.72; consistent; total sample size: 100	Indeterminate	Low
JOACMEQ	Cronbach's alpha: 0.91; total sample size: 70	Indeterminate	Moderate
Bladder function	Cronbach's alpha range 0.77–0.78; consistent; total sample size: 157	Indeterminate	High
Cervical spine function	Cronbach's alpha range 0.75–0.88; consistent; total sample size: 157	Indeterminate	High
QOL	Cronbach's alpha range 0.80–0.86; consistent; total sample size: 157	Indeterminate	High
Upper extremity function	Cronbach's alpha range 0.72–0.74; consistent; total sample size: 157	Indeterminate	High
MDI	Cronbach's alpha: 0.92; consistent; total sample size: 100	Indeterminate	Low
mJOA	Cronbach's alpha range 0.60–0.63; consistent; total sample size: 352	Indeterminate	High
QuickDASH	Cronbach's alpha: 0.94; consistent; total sample size: 94	Indeterminate	Very low
SF-12	Cronbach's alpha: 0.77; consistent; total sample size: 105	Indeterminate	n/a

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SF-36	Cronbach's alpha range 0.79–0.93; consistent; total sample size: 473	Indeterminate	n/a
WHOQOL-Bref	Cronbach's alpha range 0.86–0.87; consistent; total sample size: 38	Indeterminate	n/a

n/a = No info available

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Supplementary Table 7. Cross-cultural validity.

Instrument	Result summary	Overall rating	Quality of evidence
JOA	Forward-backward translation [Brazilian Portuguese] Comprehension rate: >81.2%	Indeterminate	Very low
JOACMEQ	Forward-backward translation [Persian and Thai] No info available	Indeterminate	Very low
mJOA	Forward-backward translation [Brazilian Portuguese and Italian] No info available	Indeterminate	Very low

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Supplementary Table 8. Reliability.

Instrument	Result summary	Overall rating	Quality of evidence
10-s step test	Test–retest stability: Spearman's rank correlation: 0.89; total sample size: 163	Indeterminate	Low
30MWT	Test–retest stability: Pearson's correlation range 0.89–1.00; total sample size: 16	Indeterminate	Very low
9-Hole peg test	Intra-observer reliability: ICC range 0.97–0.98; consistent; total sample size: 41 Inter-observer reliability: ICC range 0.97–0.99; consistent; total sample size: 41	Sufficient	Very low
BBS*	Test–retest stability: ICC: 0.99; total sample size: 32 Inter-observer reliability: ICC: 0.99; total sample size: 32	Sufficient	Very low
	Test–retest stability: Kappa: 0.67; total sample size: 32 Inter-observer reliability: Kappa: 0.43; total sample size: 32	Insufficient	Very low
Cobb's method	Intra-observer reliability: ICC: 0.84; total sample size: 20 Inter-observer reliability: ICC: 0.77; total sample size: 20	Sufficient	Very low
CT	Intra-observer reliability:	Sufficient	Moderate

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(Tsuyama's classification, 2D & 3D)	Kappa range 0.85–0.86; consistent; total sample size: 108 Inter-observer reliability: Kappa range 0.71–0.76; consistent; total sample size: 108		
CT (Tsuyama's classification, lateral + axial)	Intra-observer reliability: Kappa: 0.67; total sample size: 108 Inter-observer reliability: Kappa: 0.51; total sample size: 108	Insufficient	Moderate
Foot tapping test	Test–retest stability: Pearson's correlation range 0.90–0.93; total sample size: 126	Indeterminate	Low
Grip-and-release test	Inter-observer reliability: ICC: 0.99; total sample size: 30	Sufficient	Very low
JOA	Inter-observer reliability: ICC: 0.81; total sample size: 29	Sufficient	Very low
Bladder function	Intra-observer reliability: Kappa: 0.64; total sample size: 29 Inter-observer reliability: Kappa: 0.47; total sample size: 29	Insufficient	Very low
Motor function of fingers	Intra-observer reliability: Kappa: 0.68; total sample size: 29 Inter-observer reliability: Kappa: 0.53; total sample size: 29	Insufficient	Very low
Motor function of	Intra-observer reliability: Kappa: 0.50; total sample size: 29	Insufficient	Very low

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	shoulder and elbow	Inter-observer reliability: Kappa: 0.31; total sample size: 29		
	Motor function of lower extremity	Intra-observer reliability: Kappa: 0.55; total sample size: 29 Inter-observer reliability: Kappa: 0.49; total sample size: 29	Insufficient	Very low
	Sensory function of lower extremity	Intra-observer reliability: Kappa: 0.44; total sample size: 29 Inter-observer reliability: Kappa: 0.34; total sample size: 29	Insufficient	Very low
	Sensory function of trunk	Intra-observer reliability: Kappa: 0.54; total sample size: 29 Inter-observer reliability: Kappa: 0.58; total sample size: 29	Insufficient	Very low
	Sensory function of upper extremity	Intra-observer reliability: Kappa: 0.51; total sample size: 29 Inter-observer reliability: Kappa: 0.42; total sample size: 29	Insufficient	Very low
JOACMEQ				
	Bladder function	Test–retest stability: ICC: 0.62; total sample size: 70	Insufficient	Very low
	Cervical spine function	Test–retest stability: ICC: 0.63; total sample size: 70	Insufficient	Very low
	Lower extremity function	Test–retest stability: ICC: 0.93; total sample size: 70	Sufficient	Very low
	QOL	Test–retest stability:	Sufficient	Very low

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		ICC: 0.83; total sample size: 70		
	Upper extremity function	Test–retest stability: ICC: 0.93; total sample size: 70	Sufficient	Very low
mJOA		Test–retest stability: Spearman's rank correlation: 0.91; total sample size: 75	Indeterminate	Very low
		Intra-observer reliability: ICC: 0.87; total sample size: 55	Sufficient	Very low
		Inter-observer reliability: ICC: 0.97; total sample size: 55 Kappa: 0.80; total sample size: 75	Sufficient	Low
	Motor dysfunction of lower extremities	Inter-observer reliability: ICC: 0.73; total sample size: 75	Sufficient	Low
	Motor dysfunction of upper extremities	Inter-observer reliability: ICC: 0.77; total sample size: 75	Sufficient	Low
	Sensory dysfunction of sphincter dysfunction	Inter-observer reliability: ICC: 0.78; total sample size: 75	Sufficient	Low
	Sensory dysfunction of upper extremities	Inter-observer reliability: ICC: 0.93; total sample size: 75	Sufficient	Low
MRI (Depiction of intramedullary hyperintensity at eight cervical disc		Inter-observer reliability: Kendall's W range 0.72–0.78; total sample size: 79	Indeterminate	Very low

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levels, T2W, 1.5-T or 3-T)			
MRI (Kang's classification, 1.5-T or 3-T)	Intra-observer reliability: Kappa: 0.67; total sample size: 439 ICC: 0.77, total sample size: 82 Inter-observer reliability: Kappa range 0.60–0.93; total sample size: 539 ICC range 0.74–0.75; total sample size: 82	Inconsistent	n/a
MRI (Muhle's classification, 1.5-T)	Intra-observer reliability: Kappa: 0.72; total sample size: 357 Inter-observer reliability: Kappa range 0.61; total sample size: 357	Inconsistent	n/a
MRI (Vaccaro's classification, 1.5-T)	Intra-observer reliability: Kappa: 0.71; total sample size: 357 Inter-observer reliability: Kappa range 0.69; total sample size: 357	Sufficient	Moderate
P-mJOA			
Motor dysfunction of lower extremities	Inter-observer reliability: Kappa: 0.61; total sample size: 755	Insufficient	Moderate
Motor dysfunction of upper extremities	Inter-observer reliability: Kappa: 0.66; total sample size: 755	Insufficient	Moderate
Sensory dysfunction of sphincter dysfunction	Inter-observer reliability: Kappa: 0.55; total sample size: 755	Insufficient	Moderate

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Sensory dysfunction of upper extremities	Inter-observer reliability: Kappa: 0.55; total sample size: 755	Insufficient	Moderate
X-rays (Computer-assisted measurement of length)	Intra-observer reliability: ICC: 0.94; total sample size: 9 Inter-observer reliability: ICC: 0.93; total sample size: 9	Sufficient	Very low
X-rays (Computer-assisted measurement of thickness)	Intra-observer reliability: ICC: 0.96; total sample size: 9 Inter-observer reliability: ICC: 0.97; total sample size: 9	Sufficient	Very low

*Result ratings for BBS were split by statistic used due to their associated differences in sufficiency.

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Supplementary Table 9. Measurement error.

Instrument	Result summary	Overall rating	Quality of evidence
BBS	MDC or SDC Distribution: 1.5; total sample size: 32	Indeterminate	n/a
EQ-5D	MDC or SDC Distribution: 0.13; total sample size: 101 Anchor: 0.04; total sample size: 101	Inconsistent	n/a
JOA	MDC or SDC Distribution: 1.0; total sample size: 101 Anchor: 2.5; total sample size: 101 LOA 1.2 (–1.2, 3.6); total sample size: 92	Sufficient	Very low
JOACMEQ			
Bladder function	MDC or SDC Distribution: 7.7; total sample size: 101	Insufficient	Very low
Cervical spine function	MDC or SDC Distribution: 12.9; total sample size: 101 Anchor: 12.5; total sample size: 101	Insufficient	Very low
Lower extremity function	MDC or SDC Distribution: 7.3; total sample size: 101 Anchor: 9.4; total sample size: 101	Inconsistent	n/a
QOL	MDC or SDC Distribution: 6.6; total sample size: 101 Anchor: 8.5; total sample size: 101	Sufficient	Very low

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	Upper extremity function	MDC or SDC Distribution: 9.5; total sample size: 101 Anchor: 6.1; total sample size: 101	Sufficient	Very low
mJOA		MDC or SDC Distribution: 2.1; total sample size: 113	Inconsistent	Very low
		MCID range; total sample size: 868 Distribution: 1.2–1.4	Sufficient	High
NDI		MDC or SDC Distribution: 6.2%; total sample size: 101 Anchor: 5.2%; total sample size: 101	Insufficient	Very low
SF-36				
	MCS	MDC or SDC Distribution: 3.3–5.7; total sample size: 244	Inconsistent	n/a
		MCID; total sample size: 748 Distribution: 3.4–6.8	Inconsistent	n/a
	PCS	MDC or SDC Distribution: 5.2–5.7; total sample size: 214 Anchor: 4.9; total sample size: 101	Inconsistent	n/a
		MCID range; total sample size: 861 Distribution: 2.9–5.5	Inconsistent	n/a
		MCID; total sample size: 51 Distribution: 10		
VAS for pain		MDC or SDC Distribution: 3.1; total sample size: 30	Insufficient	Very low

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	MCID range 24.0–30.0; total sample size: 51	Insufficient	Very low
WHOQOL-Bref			
PH	MCID Distribution: 8.2; total sample size: 38	Indeterminate	n/a
PS	MCID Distribution: 7.9; total sample size: 38	Indeterminate	n/a
SR	MCID Distribution: 8.0; total sample size: 38	Indeterminate	n/a
EN	MCID Distribution: 5.6; total sample size: 38	Indeterminate	n/a
PF	MCID Distribution: 10.5; total sample size: 38	Indeterminate	n/a
RP	MCID Distribution: 17.2; total sample size: 38	Indeterminate	n/a
BP	MCID Distribution: 13.2; total sample size: 38	Indeterminate	n/a
GH	MCID Distribution: 12.3; total sample size: 38	Indeterminate	n/a
VT	MCID Distribution: 10.8; total sample size: 38	Indeterminate	n/a
SF	MCID Distribution: 13.6; total sample size: 38	Indeterminate	n/a
RE	MCID Distribution: 18.0; total sample size: 38	Indeterminate	n/a
MH	MCID	Indeterminate	n/a

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	Distribution: 11.2; total sample size: 38

n/a = No info available

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Supplementary Table 10. Criterion validity.

Instrument	Result summary*	Overall rating	Quality of evidence
10-s step test	JOA Spearman's rank correlation: 0.66; total sample size: 163	Insufficient	High
BBS	mJOA AUC range 0.88–0.94; total sample size: 31	Sufficient	Low
Foot tapping test	JOA Pearson's correlation: 0.66; total sample size: 126 JOA MFLE Pearson's correlation: 0.70; total sample size: 126	Insufficient	High
Grip-and-release test	JOA Pearson's correlation: 0.72; total sample size: 30	Sufficient	Low
JOA	Likert scale, "Health transition question" AUC: 0.59; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.62; total sample size: 101	Insufficient	Very low
JOACMEQ			
Cervical spine function	Likert scale, "Health transition question" AUC: 0.58; total sample size: 101 Likert scale, "Patient satisfaction question"	Insufficient	Very low

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		AUC: 0.58; total sample size: 101		
	Upper extremity function	Likert scale, "Health transition question" AUC: 0.66; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.65; total sample size: 101	Insufficient	Very low
	Lower extremity function	Likert scale, "Health transition question" AUC: 0.61; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.66; total sample size: 101	Insufficient	Very low
	QOL	Likert scale, "Health transition question" AUC: 0.70; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.66; total sample size: 101	Insufficient	Very low
mJOA		Nurick scale [convergent] Spearman's rank correlation: –0.41; total sample size: 119 Pearson's correlation range: –0.62 to –0.63; total sample size: 352	Sufficient	High
	Motor dysfunction of upper extremities	Nurick scale [convergent] Pearson's correlation range –0.42 to –0.42; total sample size: 352	Insufficient	High

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	Motor dysfunction of lower extremities	Nurick scale [convergent] Pearson's correlation: –0.65 to –0.68; total sample size: 352	Sufficient	High
	Sensory dysfunction of upper extremities	Nurick scale [convergent] Pearson's correlation: –0.23; total sample size: 277	Insufficient	High
	Sensory dysfunction of sphincter dysfunction	Nurick scale [convergent] Pearson's correlation: –0.25; total sample size: 277	Insufficient	High
NDI		Likert scale, "Health transition question" AUC: 0.66; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.75; total sample size: 101	Inconsistent	n/a
P-mJOA		mJOA Spearman's rank correlation: 0.83; total sample size: 755	Sufficient	High
QuickDASH		JOA MFSE Spearman's rank correlation: –0.50; total sample size: 94 JOA SFUE Spearman's rank correlation: –0.32; total sample size: 94	Insufficient	Moderate
SF-36				
	PCS	Likert scale, "Health transition question" AUC: 0.67; total sample size: 101	Insufficient	Very low

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	Likert scale, "Patient satisfaction question" AUC: 0.69; total sample size: 101		
WHOQOL-Bref			
PH	SF-36 PCS Pearson's correlation: 0.51; total sample size: 38 SF-36 MCS Pearson's correlation: 0.30; total sample size: 38	Inconsistent	n/a
PS	SF-36 PCS Pearson's correlation: 0.34; total sample size: 38 SF-36 MCS Pearson's correlation: 0.23; total sample size: 38	Insufficient	Low
SR	SF-36 PCS Pearson's correlation: 0.35; total sample size: 38 SF-36 MCS Pearson's correlation: 0.28; total sample size: 38	Insufficient	Low
EN	SF-36 PCS Pearson's correlation: 0.05; total sample size: 38 SF-36 MCS Pearson's correlation: 0.03; total sample size: 38	Insufficient	Low

n/a = No info available

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*Instruments listed are comparators

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Supplementary Table 11. Construct validity.

Instrument	Result summary*	Overall rating	Quality of evidence
10-s step test	Grip-and-release test [convergent] Spearman's rank correlation: 0.53; total sample size: 163	Sufficient	Moderate
30MWT	mJOA [convergent] Pearson's correlation: –0.44; total sample size: 16 MDI [convergent] Spearman's rank correlation: 0.65; total sample size: 41 Nurick scale [convergent] Pearson's correlation: 0.50; total sample size: 16 Spearman's rank correlation: 0.61; total sample size: 41	Sufficient	Moderate
	NDI Pearson's correlation: 0.21; total sample size: 16	Sufficient	Low
	SF-36 PCS Pearson's correlation: –0.35; total sample size: 16	Sufficient	Low
	SF-36 MCS Pearson's correlation: –0.20; total sample size: 16	Sufficient	Low
BBS	mJOA [convergent] Spearman's rank correlation: 0.81; total sample size: 72	Sufficient	Moderate

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EQ-5D	mJOA AUC: 0.68; total sample size: 119 Nurick scale AUC: 0.61; total sample size: 119	Insufficient	High
Foot tapping test	Grip-and-release test [convergent] Pearson's correlation: 0.58; total sample size: 126	Sufficient	High
Isihara's Cervical Curvature Index	mJOA Pearson's correlation: 0.04; total sample size: 235	Sufficient	High
	SF-12 PCS Pearson's correlation: 0.06; total sample size: 235 SF-12 MCS Pearson's correlation: 0.11; total sample size: 235	Sufficient	High
	Pain, "Numeric rating scale" (Arm pain score) Pearson's correlation: –0.28; total sample size: 235 Pain, "Numeric rating scale" (Neck pain scores) Pearson's correlation: –0.27; total sample size: 235 NDI Pearson's correlation: –0.10; total sample size: 235	Sufficient	High
JOA	mJOA [convergent]	Sufficient	Low

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		Spearman's rank correlation: 0.87; total sample size: 92		
		JOACMEQ QOL [convergent] Spearman's rank correlation: 0.41; total sample size: 92	Sufficient	Low
		SF-12 PCS Spearman's rank correlation: 0.50; total sample size: 92	Sufficient	Low
		SF-12 MCS Spearman's rank correlation: –0.05; total sample size: 92	Sufficient	Low
		NDI Spearman's rank correlation range –0.50 to –0.76; total sample size: 122	Sufficient	Moderate
JOACMEQ				
	QOL	NDI Spearman's rank correlation: –0.66; total sample size: 92	Sufficient	Low
	QOL	SF-12 PCS Spearman's rank correlation: 0.29; total sample size: 92 SF-12 MCS Spearman's rank correlation: 0.40; total sample size: 92	Insufficient	Low
MDI		Nurick scale [convergent] Spearman's rank correlation: 0.66; total sample size: 41	Sufficient	Low

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mJOA		30MWT [convergent] Pearson's correlation: –0.38; total sample size: 193	Insufficient	High
		JOACMEQ QOL [convergent] Spearman's rank correlation: 0.41; total sample size: 92	Insufficient	Low
		EQ-5D Spearman's rank correlation: 0.42; total sample size: 119	Insufficient	High
		SF-36 PCS Pearson's correlation range: 0.30–0.30; total sample size: 338 SF-12 PCS Spearman's rank correlation: 0.47; total sample size: 92	Sufficient	High
		SF-36 MCS Pearson's correlation: 0.25–0.25; total sample size: 338 SF-12 MCS Spearman's rank correlation: 0.03; total sample size: 92	Sufficient	High
		NDI Spearman's rank correlation: –0.51; total sample size: 92 Pearson's correlation rage –0.33 to –0.34; total sample size: 336	Sufficient	High
	Motor dysfunction	30MWT [convergent]	Insufficient	High

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	of lower extremities	Pearson's correlation: –0.43; total sample size: 193		
		SF-36 PCS Pearson's correlation range: 0.31–0.50; total sample size: 338	Sufficient	High
		SF-36 MCS Pearson's correlation: 0.21; total sample size: 268	Sufficient	High
		NDI Pearson's correlation: –0.31; total sample size: 261	Sufficient	High
	Motor dysfunction of upper extremities	30MWT [convergent] Pearson's correlation: –0.21; total sample size: 193	Insufficient	High
		SF-36 PCS Pearson's correlation: 0.22; total sample size: 268	Insufficient	High
		SF-36 MCS Pearson's correlation: 0.20; total sample size: 268	Sufficient	High
		NDI Pearson's correlation: –0.24; total sample size: 261	Sufficient	High
	Sensory dysfunction of sphincter dysfunction	30MWT [convergent] Pearson's correlation: –0.23; total sample size: 193	Insufficient	High
		SF-36 PCS	Sufficient	High

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		Pearson's correlation: 0.06; total sample size: 268 SF-36 MCS Pearson's correlation: 0.08; total sample size: 268		
		NDI Pearson's correlation: –0.16; total sample size: 261	Sufficient	High
	Sensory dysfunction of upper extremities	30MWT [convergent] Pearson's correlation: –0.05; total sample size: 193	Insufficient	High
		SF-36 PCS Pearson's correlation: 0.19; total sample size: 268 SF-36 MCS Pearson's correlation: 0.19; total sample size: 268	Sufficient	High
		NDI Pearson's correlation: –0.23; total sample size: 261	Insufficient	High
NDI		mJOA Pearson's correlation: –0.36; total sample size: 235	Sufficient	High
		SF-12 PCS Pearson's correlation: –0.54; total sample size: 235 SF-12 MCS	Sufficient	High

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	Pearson's correlation: –0.40; total sample size: 235		
	Pain, "Numeric rating scale" (Arm pain score) Pearson's correlation: 0.68; total sample size: 235 Pain, "Numeric rating scale" (Neck pain scores) Pearson's correlation: 0.64; total sample size: 235	Sufficient	High
Nurick scale	EQ-5D Spearman's rank correlation: –0.28; total sample size: 119	Sufficient	High
Pain, "Numeric rating scale" (Arm pain scores)	mJOA Pearson's correlation: –0.19; total sample size: 235	Sufficient	High
	Pain, “Numeric rating scale” (Neck pain score) [convergent] Pearson’s correlation: 0.72; total sample size: 235	Sufficient	High
Pain, "Numeric rating scale" (Neck pain scores)	mJOA Pearson's correlation: –0.07; total sample size: 235	Sufficient	High
QuickDASH	SF-36 Spearman's rank correlation: –0.75; total sample size: 94	Sufficient	Moderate
	NDI and Pain, "Numeric rating scale" [convergent]	Sufficient	Moderate

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		Spearman's rank correlation range 0.69–0.83; total sample size: 94		
SF-12				
	MCS	mJOA Pearson's correlation: 0.19; total sample size: 235	Sufficient	High
		Pain, "Numeric rating scale" (Arm pain score) Pearson's correlation: –0.23; total sample size: 235 Pain, "Numeric rating scale" (Neck pain score) Pearson's correlation: –0.28; total sample size: 235	Sufficient	High
		NDI Spearman's rank correlation: –0.17; total sample size: 92	Sufficient	Moderate
		SF-12 PCS Pearson's correlation: 0.01; total sample size: 235	Sufficient	High
	PCS	mJOA Pearson's correlation: 0.43; total sample size: 235	Sufficient	High
		Pain, "Numeric rating scale" (Arm pain score) Pearson's correlation: –0.44; total sample size: 235 Pain, "Numeric rating scale" (Neck pain score)	Sufficient	High

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	Pearson's correlation: –0.41; total sample size: 235		
	NDI Spearman's rank correlation: –0.49; total sample size: 92	Sufficient	Moderate
	SF-12 MCS Spearman's rank correlation: –0.29; total sample size: 92	Sufficient	Low
Triangle step test	Grip-and-release test [convergent] Spearman's rank correlation: 0.55; total sample size: 270	Sufficient	High

*Instruments listed are comparators

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Supplementary Table 12. Responsiveness.

Instrument	Result summary*	Overall rating	Quality of evidence
30MWT	30MWT SRM: 0.3; total sample size: 484	Insufficient	High
BBS	mJOA Sensitivity range 77.4-80.0; total sample size: 31 Specificity range 87.8–92.9; total sample size: 31	Sufficient	Low
EQ-5D	EQ-5D Mean change score: 0.06; total sample size: 108	Indeterminate	High
European Myelopathy Scale	EMS Normalised change: 0.18; total sample size: 99	Indeterminate	Very low
Foot tapping test	Foot tapping test Mean change score: 6; total sample size: 6	Indeterminate	Very low
Grip-and-release test	Grip-and-release test Spearman's rank correlation: 0.69; total sample size: 48	Sufficient	Very low
	JOA Spearman's rank correlation: 0.32; total sample size: 48	Insufficient	Low
JOA	mJOA Spearman's rank correlation: 0.75; total sample size: 92	Sufficient	Very low
	JOA	Indeterminate	Very low

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		Mean change score range 4.6; total sample size: 126 Normalised change: 0.21; total sample size: 99 JOA MFLE Mean change score range 0.6; total sample size: 126		
JOACMEQ				
	Bladder function	JOACMEQ BF AUC: 0.82; total sample size: 78	Sufficient	Moderate
		JOACMEQ BF Mean change score: 18.0; total sample size: 87	Indeterminate	Very low
		JOACMEQ BF Effect size: 0.33; total sample size: 78	Insufficient	Moderate
	Cervical spine function	JOACMEQ CF AUC: 0.72; total sample size: 78	Sufficient	Moderate
		JOACMEQ CF Mean change score: 25.8; total sample size: 87	Indeterminate	Very low
		JOACMEQ CF Effect size: 0.28; total sample size: 78	Insufficient	Moderate
	Lower extremity function	JOACMEQ LEF AUC: 0.75; total sample size: 78	Sufficient	Moderate
		JOACMEQ LEF Mean change score: 28.4; total sample size: 87	Indeterminate	Very low

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		JOACMEQ LEF Effect size: 0.02; total sample size: 78	Insufficient	Moderate
	Upper extremity function	JOACMEQ UEF AUC: 0.74; total sample size: 78	Sufficient	Moderate
		JOACMEQ UEF Mean change score: 10.7; total sample size: 87	Indeterminate	Very low
		JOACMEQ UEF Effect size: 0.17; total sample size: 78	Insufficient	Moderate
	QOL	JOACMEQ QOL AUC: 0.83; total sample size: 78	Sufficient	Moderate
		JOACMEQ QOL Mean change score: 23.7; total sample size: 87	Indeterminate	Very low
		JOACMEQ QOL Effect size: 0.46; total sample size: 78	Insufficient	Moderate
MDI		MDI Normalised change: 0.52; total sample size: 99	Indeterminate	Very low
mJOA		mJOA Effect size range 0.87–1.0; total sample size: 352	Sufficient	High
		mJOA Normalised change: 1.47; total sample size: 42	Indeterminate	Very low
NDI		Anchor-based approach AUC: 0.66; total sample size: 78	Insufficient	Moderate

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	Effect size: 0.44; total sample size: 78		
	NDI Mean change score: –15.8; total sample size: 118	Indeterminate	Very low
Nurick scale	Nurick scale Normalised change: 0.42; total sample size: 99 Mean change score range –0.76 to –1.3; total sample size: 93	Indeterminate	Very low
Ranawat classification of disease severity	Ranawat classification of disease severity Normalised change: 0.34; total sample size: 99	Indeterminate	Very low
SF-12			
PCS	SF-12 PCS Mean change score: 8.17; total sample size: 118	Indeterminate	Very low
SF-36	SF-36 Normalised change: 0.32; total sample size: 99	Indeterminate	Very low
PCS	SF-36 PCS Effect size range: 0.84; total sample size: 142	Sufficient	Low
	SF-36 PCS Sensitivity: 0.85; total sample size: 105	Sufficient	Moderate
MCS	SF-36 MCS Effect size range: 0.81; total sample size: 142	Sufficient	Low
	SF-36 MCS	Sufficient	Moderate

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	Sensitivity: 0.67; total sample size: 105		
WHOQOL-Bref			
PH	WHOQOL-Bref PH Effect size: 0.68; total sample size: 38	Insufficient	Low
PS	WHOQOL-Bref PS Effect size: 0.39; total sample size: 38	Insufficient	Low
SR	WHOQOL-Bref SR Effect size: 0.03; total sample size: 38	Insufficient	Low
EN	WHOQOL-Bref EN Effect size: 0.45; total sample size: 38	Insufficient	Low

*Instruments listed are comparators

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REFERENCES

1. Auffinger BM, Lall RR, Dahdaleh NS, Wong AP, Lam SK, Koski T, et al. Measuring surgical outcomes in cervical spondylotic myelopathy patients undergoing anterior cervical discectomy and fusion: assessment of minimum clinically important difference. PLoS One. 2013;8(6):e67408.
2. Augusto MT, Diniz JM, Dantas FLR, Oliveira MF, Rotta JM, Botelho RV. Development of the Portuguese Version of the Modified Japanese Orthopaedic Association Score: Cross-Cultural Adaptation, Reliability, Validity, and Responsiveness. World Neurosurg. 2018;116:e1092-e7.
3. Azimi P, Rezaei O, Montazeri A. An outcome measure of functionality and quality of life in patients with cervical myelopathy. Iran Red Crescent Med J. 2014;16(6):e8102.
4. Badhiwala JH, Witiw CD, Nassiri F, Akbar MA, Jaja B, Wilson JR, et al. Minimum Clinically Important Difference in SF-36 Scores for Use in Degenerative Cervical Myelopathy. Spine (Phila Pa 1976). 2018;43(21):E1260-e6.
5. Bohm PE, Fehlings MG, Kopjar B, Tetreault LA, Vaccaro AR, Anderson KK, et al. Psychometric properties of the 30-m walking test in patients with degenerative cervical myelopathy: results from two prospective multicenter cohort studies. Spine J. 2017;17(2):211-7.
6. Carreon LY, Glassman SD, Campbell MJ, Anderson PA. Neck Disability Index, short form-36 physical component summary, and pain scales for neck and arm pain: the minimum clinically important difference and substantial clinical benefit after cervical spine fusion. Spine J. 2010;10(6):469-74.
7. Chang H, Kong CG, Won HY, Kim JH, Park JB. Inter- and intra-observer variability of a cervical OPLL classification using reconstructed CT images. Clin Orthop Surg. 2010;2(1):8-12.
8. Chiba K, Kato Y, Tsuzuki N, Nagata K, Toyama Y, Iwasaki M, et al. Computer-assisted measurement of the size of ossification in patients with ossification of the posterior longitudinal ligament in the cervical spine. J Orthop Sci. 2005;10(5):451-6.
9. Chien A, Lai DM, Cheng CH, Wang SF, Hsu WL, Wang JL. Responsiveness of the Chinese versions of the Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire and Neck Disability Index in postoperative patients with cervical spondylotic myelopathy. Spine (Phila Pa 1976). 2015;40(17):1315-21.
10. Chiu AY, Pang MY. Assessment of psychometric properties of various balance assessment tools in persons with cervical spondylotic myelopathy. journal of orthopaedic & sports physical therapy. 2017;47(9):673-82.
11. Fukui M, Chiba K, Kawakami M, Kikuchi S, Konno S, Miyamoto M, et al. Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire (JOACMEQ): Part 2. Endorsement of the alternative item. J Orthop Sci. 2007;12(3):241-8.
12. Fukui M, Chiba K, Kawakami M, Kikuchi S, Konno S, Miyamoto M, et al. Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire: part 3. Determination of reliability. J Orthop Sci. 2007;12(4):321-6.
13. Goyal DKC, Murphy HA, Hollern DA, Divi SN, Nicholson K, Stawicki C, et al. Is the Neck Disability Index an Appropriate Measure for Changes in Physical Function After Surgery for Cervical Spondylotic Myelopathy? Int J Spine Surg. 2020;14(1):53-8.
14. Gwinn DE, Iannotti CA, Benzel EC, Steinmetz MP. Effective lordosis: analysis of sagittal spinal canal alignment in cervical spondylotic myelopathy. J Neurosurg Spine. 2009;11(6):667-72.

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15. Hosono N, Sakaura H, Mukai Y, Kaito T, Makino T, Yoshikawa H. A simple performance test for quantifying the severity of cervical myelopathy. J Bone Joint Surg Br. 2008;90(9):1210-3.
16. Hosono N, Takenaka S, Mukai Y, Makino T, Sakaura H, Miwa T, et al. Postoperative 24-hour result of 15-second grip-and-release test correlates with surgical outcome of cervical compression myelopathy. Spine (Phila Pa 1976). 2012;37(15):1283-7.
17. Kang Y, Lee JW, Koh YH, Hur S, Kim SJ, Chai JW, et al. New MRI grading system for the cervical canal stenosis. AJR Am J Roentgenol. 2011;197(1):W134-40.
18. Kato S, Oshima Y, Oka H, Chikuda H, Takeshita Y, Miyoshi K, et al. Comparison of the Japanese Orthopaedic Association (JOA) score and modified JOA (mJOA) score for the assessment of cervical myelopathy: a multicenter observational study. PloS one. 2015;10(4):e0123022.
19. Kato S, Oshima Y, Matsubayashi Y, Taniguchi Y, Tanaka S, Takeshita K. Minimum Clinically Important Difference and Patient Acceptable Symptom State of Japanese Orthopaedic Association Score in Degenerative Cervical Myelopathy Patients. Spine (Phila Pa 1976). 2019;44(10):691-7.
20. Kato S, Oshima Y, Matsubayashi Y, Taniguchi Y, Tanaka S, Takeshita K. Minimum clinically important difference in outcome scores among patients undergoing cervical laminoplasty. Eur Spine J. 2019;28(5):1234-41.
21. King JT, Jr., Roberts MS. Validity and reliability of the Short Form-36 in cervical spondylotic myelopathy. J Neurosurg. 2002;97(2 Suppl):180-5.
22. Ko S, Choi W, Chae S. Comparison of inter- and intra-observer reliability among the three classification systems for cervical spinal canal stenosis. Eur Spine J. 2017;26(9):2290-6.
23. Kopjar B, Tetreault L, Kalsi-Ryan S, Fehlings M. Psychometric properties of the modified Japanese Orthopaedic Association scale in patients with cervical spondylotic myelopathy. Spine (Phila Pa 1976). 2015;40(1):E23-8.
24. Latimer M, Haden N, Seeley HM, Laing RJ. Measurement of outcome in patients with cervical spondylotic myelopathy treated surgically. Br J Neurosurg. 2002;16(6):545-9.
25. Longo UG, Berton A, Denaro L, Salvatore G, Denaro V. Development of the Italian version of the modified Japanese orthopaedic association score (mJOA-IT): cross-cultural adaptation, reliability, validity and responsiveness. Eur Spine J. 2016;25(9):2952-7.
26. Lubelski D, Alvin MD, Nesterenko S, Sundar SJ, Thompson NR, Benzel EC, et al. Correlation of quality of life and functional outcome measures for cervical spondylotic myelopathy. Journal of Neurosurgery: Spine. 2016;24(3):483-9.
27. Mihara H, Kondo S, Murata A, Ishida K, Niimura T, Hachiya M. A new performance test for cervical myelopathy: the triangle step test. Spine (Phila Pa 1976). 2010;35(1):32-5.
28. Nakamoto H, Oshima Y, Takeshita K, Chikuda H, Ono T, Taniguchi Y, et al. Usefulness of QuickDASH in patients with cervical laminoplasty. J Orthop Sci. 2014;19(2):218-22.
29. Nakashima H, Yukawa Y, Ito K, Machino M, Kanbara S, Morita D, et al. Validity of the 10-s step test: prospective study comparing it with the 10-s grip and release test and the 30-m walking test. Eur Spine J. 2011;20(8):1318-22.
30. Nicholson KJ, Millhouse PW, Pflug E, Woods B, Schroeder GD, Anderson DG, et al. Cervical Sagittal Range of Motion as a Predictor of Symptom Severity in Cervical Spondylotic Myelopathy. Spine (Phila Pa 1976). 2018;43(13):883-9.
31. Nikaido T, Kikuchi S, Yabuki S, Otani K, Konno S. Surgical treatment assessment using the Japanese orthopedic association cervical myelopathy evaluation questionnaire in patients

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with cervical myelopathy: a new outcome measure for cervical myelopathy. Spine (Phila Pa 1976). 2009;34(23):2568-72.
32. Numasawa T, Ono A, Wada K, Yamasaki Y, Yokoyama T, Aburakawa S, et al. Simple foot tapping test as a quantitative objective assessment of cervical myelopathy. Spine (Phila Pa 1976). 2012;37(2):108-13.
33. Olindo S, Signate A, Richech A, Cabre P, Catonne Y, Smadja D, et al. Quantitative assessment of hand disability by the Nine-Hole-Peg test (9-HPT) in cervical spondylotic myelopathy. Journal of Neurology, Neurosurgery & Psychiatry. 2008;79(8):965-7.
34. Park H-J, Kim SS, Chung E-C, Lee S-Y, Park N-H, Rho M-H, et al. Clinical correlation of a new practical MRI method for assessing cervical spinal canal compression. American Journal of Roentgenology. 2012;199(2):W197-W201.
35. Pratali RR, Smith JS, Motta RL, Martins SM, Motta MM, Rocha RD, et al. A Brazilian Portuguese cross-cultural adaptation of the modified JOA scale for myelopathy. Clinics. 2017;72:103-5.
36. Pratali RR, Smith JS, Rocha RD, Matos TD, Defino HLA, Herrero C. RELIABILITY OF A BRAZILIAN PORTUGUESE TRANSLATED AND CROSS-CULTURALLY ADAPTED VERSION OF THE MJOA SCALE. Acta Ortop Bras. 2018;26(5):335-7.
37. Rhee JM, Shi WJ, Cyriac M, Kim JY, Zhou F, Easley KA, et al. The P-mJOA: A Patient-derived, Self-reported Outcome Instrument for Evaluating Cervical Myelopathy: Comparison with the mJOA. Clin Spine Surg. 2018;31(2):E115-e20.
38. Sato T, Horikoshi T, Watanabe A, Uchida M, Ishigame K, Araki T, et al. Evaluation of cervical myelopathy using apparent diffusion coefficient measured by diffusion-weighted imaging. AJNR Am J Neuroradiol. 2012;33(2):388-92.
39. Shim E, Lee JW, Lee E, Kang Y, Kang HS, Kang WY, et al. Cervical spondylotic myelopathy: diagnostic performance of radiologists with varying levels of experience in comparing MR images acquired using field strengths of 1.5 and 3 Tesla. Acta Radiol. 2019;60(10):1314-20.
40. Singh A, Crockard HA. Comparison of seven different scales used to quantify severity of cervical spondylotic myelopathy and post-operative improvement. J Outcome Meas. 2001;5(1):798-818.
41. Singh A, Crockard HA. Quantitative assessment of cervical spondylotic myelopathy by a simple walking test. Lancet. 1999;354(9176):370-3.
42. Singh A, Gnanalingham K, Casey A, Crockard A. Quality of life assessment using the Short Form-12 (SF-12) questionnaire in patients with cervical spondylotic myelopathy: comparison with SF-36. Spine (Phila Pa 1976). 2006;31(6):639-43.
43. Spurgas MP, Abbas SF, Szewczyk BS, Yim B, Ata A, German JW. The effect of length of follow-up on substantial clinical benefit thresholds in patients undergoing surgery for cervical degenerative myelopathy. Journal of Clinical Neuroscience. 2019;62:88-93.
44. Tetreault L, Nouri A, Kopjar B, Côté P, Fehlings MG. The Minimum Clinically Important Difference of the Modified Japanese Orthopaedic Association Scale in Patients with Degenerative Cervical Myelopathy. Spine (Phila Pa 1976). 2015;40(21):1653-9.
45. Thakar S, Rajshekhar V. Evaluation of pain as a preference-based health status measure in patients with cervical spondylotic myelopathy undergoing central corpectomy. Acta Neurochir (Wien). 2012;154(2):335-40.
46. Thakar S, Christopher S, Rajshekhar V. Quality of life assessment after central corpectomy for cervical spondylotic myelopathy: comparative evaluation of the 36-Item Short

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Form Health Survey and the World Health Organization Quality of Life-Bref. J Neurosurg Spine. 2009;11(4):402-12.
47. Wada E, Fukui M, Takahashi K, Takeuchi D, Hashizume H, Kanamori M, et al. Japanese orthopaedic association cervical myelopathy evaluation questionnaire (JOACMEQ): Part 5. Determination of responsiveness. Journal of Orthopaedic Science. 2019;24(1):57-61.
48. Witayakom W, Paholpak P, Jirarattanaphochai K, Kosuwon W, Sirichativapee W, Wisanuyotin T, et al. Validation of the reliability of the Thai version of the Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire (JOACMEQ). Journal of Orthopaedic Science. 2016;21(2):124-7.
49. Yonenobu K, Abumi K, Nagata K, Taketomi E, Ueyama K. Interobserver and intraobserver reliability of the japanese orthopaedic association scoring system for evaluation of cervical compression myelopathy. Spine (Phila Pa 1976). 2001;26(17):1890-4; discussion 5.
50. Yukawa Y, Kato F, Ito K, Horie Y, Nakashima H, Masaaki M, et al. "Ten second step test" as a new quantifiable parameter of cervical myelopathy. Spine (Phila Pa 1976). 2009;34(1):82-6.
51. Zhang Y, Zhou F, Sun Y. Assessment of health-related quality of life using the SF-36 in Chinese cervical spondylotic myelopathy patients after surgery and its consistency with neurological function assessment: a cohort study. Health Qual Life Outcomes. 2015;13(1):1-7.
52. Zhou F, Zhang Y, Sun Y, Zhang F, Pan S, Liu Z. Assessment of the minimum clinically important difference in neurological function and quality of life after surgery in cervical spondylotic myelopathy patients: a prospective cohort study. Eur Spine J. 2015;24(12):2918-23.

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Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			
Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	3
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	7
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	7-8
Methods			

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Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	8-9
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	8
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	8, S1-S9
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	9
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	9-10
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	9-13
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	9-13
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	9-13
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	9-13
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention	13

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		characteristics and comparing against the planned groups for each synthesis (item #5))	
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	13
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	13
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	13
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	NA
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	NA
Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	9-13
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	NA
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	13
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	13-14
Study characteristics	#17	Cite each included study and present its characteristics	15, S10-S12
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	S13-S50
Results of individual	#19	For all outcomes, present for each study (a) summary statistics for	S13-S50

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studies		each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	16-18, 23-32
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	16-18, 23-32
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	NA
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	NA
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	NA
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	NA
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	33-34
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	34-35
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	34-35
Implications	#23d	Discuss implications of the results for practice, policy, and future research	34
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	2
Registration and	#24b	Indicate where the review protocol can be accessed, or state that a	2

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protocol		protocol was not prepared	
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	2
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	36
Competing interests	#26	Declare any competing interests of review authors	36
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	

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BMJ Open

Clinical outcome measures and their evidence base in degenerative cervical myelopathy: a systematic review to inform a core measurement set (AO Spine RECODE-DCM)

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Clinical outcome measures and their evidence base in degenerative cervical myelopathy: a systematic review to inform a core measurement set (AO Spine RECODE-DCM)

Alvaro Yanez Touzet[1], Aniqah Bhatti[2], Esmee Dohle[2], Faheem Bhatti[2], Keng Siang Lee[3], Julio C. Furlan[4–6], Michael G. Fehlings[7], James Harrop[8], Carl M. Zipser[9], Ricardo Rodrigues-Pinto[10–11], James Milligan[12], Ellen Sarewitz[13], Armin Curt[9], Vafa Rahimi-Movaghar[14], Bizhan Aarabi[15], Timothy F. Boerger[16], Lindsay Tetreault[17], Robert Chen[17–18], James D. Guest[19], Sukhvinder Kalsi-Ryan[6], Angus G. K. McNair[20], Mark R. N. Kotter[21–22], Benjamin M. Davies[23] On behalf of the AO Spine RECODE-DCM Steering Committee

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Registration: This systematic review was not registered.

Word count: 3,532/4,000

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ABSTRACT

Objectives: To evaluate the measurement properties of outcome measures currently used in the assessment of degenerative cervical myelopathy (DCM) for clinical research.

Design: Systematic review

Data sources: MEDLINE and EMBASE were searched through 4 August 2020.

Eligibility criteria: Primary clinical research published in English and whose primary purpose was to evaluate the measurement properties or clinically important differences of instruments used in DCM.

Data extraction and synthesis: Psychometric properties and clinically important differences were both extracted from each study, assessed for risk of bias, and presented in accordance with the Consensus-based Standards for the selection of health Measurement Instruments (COSMIN) criteria.

Results: Twenty-nine outcome instruments were identified from 52 studies published between 1999 and 2020. They measured neuromuscular function (16 instruments), life impact (five instruments), pain (five instruments), and radiological scoring (five instruments). No instrument had evaluations for all 10 measurement properties and <50% had assessments for all three domains (i.e., reliability, validity, and responsiveness). There was a paucity of high-quality evidence. Notably, there were no studies that reported on structural validity and no high-quality evidence that discussed content validity. In this context, we identified nine instruments that are interpretable by clinicians: the arm and neck pain scores; the 12- and 36-item short form health surveys; the Japanese Orthopaedic Association (JOA) score, modified JOA (mJOA), and JOA Cervical Myelopathy Evaluation Questionnaire (JOACMEQ); the neck disability index; and the visual analogue scale for pain. These include six scores with barriers to application and one score with insufficient criterion and construct validity.

Conclusions: This review aggregates studies evaluating outcome measures used to assess patients with DCM. Overall, there is a need for a set of agreed tools to measure outcomes in DCM. These findings will be used to inform the development of a core measurement set as part of AO Spine RECODE-DCM.

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ARTICLE SUMMARY

Strengths and limitations of this study

- Consensus-based reporting guidelines were used to evaluate the properties and clinically important differences of DCM measurement instruments.
- Only instruments that are currently in use were evaluated in this study.
- Interpretability was used as an important characteristic to make recommendations, a posteriori, due to the absence of category A COSMIN recommendations.
- Interpretability and feasibility were evaluated using bespoke criteria adapted, a priori, from the COSMIN methodology.

Keywords: degenerative cervical myelopathy; cervical spondylotic myelopathy; spinal cord compression; outcome measures; core measurement set

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LIST OF ABBREVIATIONS

30MWT	30-m Walking Test
COSMIN	Consensus-based Standards for the selection of health Measurement Instruments
AUC	Area under curve
BBS	Berg Balance Scale
BF	Bladder function
BP	Bodily pain
CMS	Core measurement set
CF	Cervical spine function
CT	computed tomography
EN	Environment health
EQ-5D	EuroQol-5 Dimension
GH	General health
JOA	Japanese Orthopaedic Association
LEF	Lower extremity function
LOA	Limits of agreement
MCID	Minimal clinically important difference
MCS	Mental component summary
MDC	Minimal detectable change
MDI	Myelopathy Disability Index
MFLE	Motor function of lower extremity
MFSE	Motor function of shoulder and elbow
MH	Mental health
mJOA	Modified Japanese Orthopaedic Association
MRI	Magnetic resonance imaging
NDI	Neck Disability Index
NMF	Neuromuscular limb function
JOACMEQ	Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire
P-mJOA	Patient-derived version of the mJOA
PCS	Physical component summary
PH	Physical health
PROM	Patient-reported outcome measure

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PS	Psychological well-being
RE	Emotional role
RP	Physical role
SCB	Substantial clinical benefit
SDC	Smallest detectable change
SF-12	12-Item Short Form Health Survey
SF-36	36-Item Short Form Health Survey
SFUE	Sensory function of upper extremity
SF	Social function
SR	Social relationships
QOL	Quality of life
QuickDASH	Quick Disabilities of the Arm, Shoulder, and Hand
UEF	Upper extremity function
VAS	Visual Analogue Scale
VT	Vitality
WHOQOL-Bref	World Health Organisation Quality of Life Instruments

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INTRODUCTION

The most common adult spinal cord disease, degenerative cervical myelopathy (DCM), is both measured and reported inconsistently across clinical research (1-4). DCM is a progressive spinal cord disease caused by degenerative changes in the cervical spine that lead to stress and injury to the cervical spinal cord. It usually initially presents as a loss of digital dexterity, subtle gait disturbances, and mild pain which, if left untreated, can potentially lead to tetraplegia and wheelchair dependence (5).

In 2019, AO Spine launched the Research Objectives and Common Data Elements for Degenerative Cervical Myelopathy (AO Spine RECODE-DCM; www.aospine.org/recode) initiative with the aim of creating a 'research toolkit' to help accelerate knowledge discovery and improve outcomes in DCM (3, 6). The initiative identified the need to improve consistency in measurement and reporting across DCM research to enable studies to be compared and/or aggregated, and to ensure the most meaningful aspects of the disease are captured (7, 8). This process started by creating a list of essential outcomes (i.e., core outcome set) and baseline characteristics (i.e., core data elements). To truly enable consistent reporting, however, these data sets should be partnered with a core measurement set (CMS): a set of agreed tools that are used to measure the outcomes and data elements of DCM (9-17).

Several approaches have been employed to form a CMS, ranging from the development of novel measurement instruments to adopting the use of existing ones (18-20). For AO Spine RECODE DCM, it was decided to recommend existing instruments and, preferably, those already used in DCM. This was to allow a more rapid introduction of the CMS, cognisant that many new tools are in development and the CMS can be updated in the future.

Consequently, we sought to examine the tools used in DCM research and assess their quality (21) using objective criteria. In recognition of variable quality amongst reported outcome measures, the Consensus-based Standards for the selection of health Measurement Instruments (COSMIN) initiative has developed clinimetric tools to assess instrument quality (22). We searched the literature for studies evaluating one or more psychometric properties defined by the COSMIN guidelines, as well as

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studies that defined clinically important differences such as the minimally clinical important difference (MCID) and substantial clinical benefits (SCB). Data were rated, aggregated, and assessed for methodology bias using the COSMIN manual for systematic reviews of patient-reported outcome measures (PROM) (23-25). This work builds on the protocol for the AO Spine RECODE-DCM initiative (3, 6) and complements two earlier reviews of outcome measures in DCM (2, 21).

METHODS

Search

A search string was developed to identify original research assessing the psychometric properties of instruments currently used in the clinical research of DCM (26). This comprised synonyms of 'psychometric' and 'DCM' (**Supplementary Table 1**). The search was developed with oversight of a medical librarian (IK) and informed by previously developed search filters for DCM (26-28). The search was applied to MEDLINE and EMBASE, from inception until 4 August 2020, using OVID (Wolters Kluwer, Netherlands). The search also focused on DCM tools identified in previous scoping reviews (2, 21, 29).

Study selection

All titles and abstracts were screened independently against a set of pre-defined eligibility criteria by four reviewers (AYT, AB, ED, and FB). A full list of inclusion and exclusion criteria of studies are stated in **Table 1**.

Table 1. Inclusion and exclusion criteria

Inclusion	Exclusion
Publication type	
• Article written in English • Primary clinical research articles	• Article not written in English • Conference abstracts or posters • Editorials, commentaries, opinion papers or letters • Book chapters or theses
Study type	
• Study includes primary clinical data	• Study uses only secondary data • Case reports

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	• Narrative reviews • Systematic reviews • Meta-analyses
Populations	
• Human studies	• Non-human studies
Indications	
• Exclusively DCM (CSM, OPLL, cervical stenosis, spondylosis, spinal cord compression, cervical myelopathy)	• Populations with DCM and at least one other condition (e.g., radiculopathy)
Comparator	
• At least one assessment tool from (2, 21, 29)	
Outcomes	
• At least one psychometric property • At least one MCID or SCB	

Potentially eligible studies were selected for full-text analysis. In the event of multiple publications analysing the same cohort for the same purpose, the most recent paper was used for evaluation. At each stage, two reviewers independently (AYT, AB, ED, FB) reviewed all the screened studies for inclusion to ensure reliability of study selection (**Supplementary Table 2**). Disagreements were resolved by consensus or appeal to a third senior reviewer (BMD).

Quality assessment

The quality of included studies was assessed using the COSMIN risk of bias checklist (23-25). Briefly, the COSMIN risk of bias tool assesses 10 measurement properties, including nine psychometric properties (i.e., content validity, structural validity, internal consistency, cross-cultural validity/measurement invariance, reliability, measurement error, criterion validity, hypotheses testing for construct validity, and responsiveness) and clinically important differences. A list of definitions is presented in **Table 2**. Interpretability and feasibility were also evaluated using criteria adapted a priori from the COSMIN methodology (**Supplementary Tables 3** and **4**, respectively). Namely, interpretability was evaluated for each measurement instrument through the

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availability of anchor-based MCIDs (23-25), while feasibility was assessed with respect to the ease of application of the instrument.

The methodological quality of each study was scored as ‘very good’, ‘adequate’, ‘doubtful’, ‘inadequate’, or ‘not applicable’. Overall ratings were then made for each property using the modified Grading of Recommendations Assessment, Development, and Evaluation (mGRADE) approach from the COSMIN risk of bias checklist (23-25). For each study, one review author (AYT) assessed the quality, feasibility, and interpretability from included studies and a second (BMD) checked the assessments. Disagreements were resolved by consensus.

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Table 2. Definitions of domains, measurement properties, and aspects of measurement properties, adapted from the COSMIN guidelines (23-25, 30) and studies of clinically important differences (31, 32)

Domain	Measurement property	Aspect	Definition
Reliability			The degree to which the measurement is free from measurement error. The extent to which scores for patients who have not changed are the same for repeated measurement under several conditions; over time; by different persons on the same occasion; or by the same persons on different occasions.
	Internal consistency		The degree of interrelatedness among the items included in a measurement instrument.
	Reliability		The proportion of the total variance in the measurements which is due to 'true'* differences between patients.
	Measurement error		The systematic and random error of a patient's score that is not attributed to true changes in the construct to be measured.
Validity			The degree to which a measurement instrument measures the construct(s) it purports to measure.
	Content validity		The degree to which the content of a measurement tool is an adequate reflection of all facets of a given construct.
	Construct validity		The degree to which the scores of a measurement instrument are consistent with hypotheses (for instance, with regard to internal relationships, relationships to scores of other instruments, or differences between relevant groups) based on the assumption that the instrument validly measures the construct to be measured.
		Structural validity	The degree to which the scores of a measurement instrument are an adequate reflection of the dimensionality of the construct to be measured.

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		Hypotheses testing	Idem construct validity
		Cross-cultural validity	The degree to which the performance of the items on a translated or culturally adapted measurement instrument are an adequate reflection of the performance of the items of the original version of the instrument.
	Criterion validity		The degree to which the scores of a measurement instrument are an adequate reflection of a 'gold standard'.
Responsiveness			The ability of a measurement instrument to detect change over time in the construct to be measured.
	Responsiveness		Idem responsiveness
Interpretability†			Interpretability is the degree to which one can assign qualitative meaning - that is, clinical or commonly understood connotations – to a PROM's quantitative scores or change in scores.
	Clinically important differences		
		Minimal clinically important difference	The smallest measured change score that patients perceive to be important, also known as the MCID or MID (Minimally Important Difference)
		Substantial clinical benefit	The change in outcome associated with patient perception of a large meaningful improvement

*The word 'true' must be seen in the context of the classical test theory, which states that any observation is composed of two components – a true score and error associated with the observation. 'True' is the average score that would be obtained if the scale were applied an infinite number of times. It refers only to the consistency of the score, and not to its accuracy (33).

†Interpretability is not considered a measurement property, but an important characteristic of a measurement instrument.

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Data extraction

A proforma adapted from COSMIN was employed by one reviewer (AYT) to extract the following: study details, sample size, patient demographics, measurement properties, and qualitative and/or quantitative results for each property. This was checked by a second reviewer (BMD) and any disagreements were resolved by consensus. Examples of qualitative and quantitative results included observations (e.g., narrative syntheses) and statistics (e.g., correlation coefficients). These result types are specific for each measurement property and are listed in the COSMIN guidelines (23-25).

Data analysis

Each result was rated as 'sufficient', 'indeterminate', or 'insufficient'. All results were qualitatively summarised and given an overall rating as 'sufficient', 'indeterminate', 'inconsistent', or 'insufficient'. The definitions of these ratings are available in the COSMIN guidelines (23-25). Measurement instruments were categorised into three recommendation groups:

A. Instruments with evidence of sufficient content validity and at least low-quality evidence of sufficient internal consistency.
B. Instruments categorised not in A or C.
C. Instruments with high-quality evidence of an insufficient measurement property (23-25).

Recommendations for each instrument were presented in tandem with interpretability and feasibility assessments and reported as a narrative synthesis (34). We used the PRISMA checklist when writing our report (35).

Patient and public involvement

This project forms part of a larger, international multi-stakeholder co-production initiative called AO Spine RECODE-DCM, which aims to develop a framework to accelerate knowledge discovery that can improve outcomes in DCM. Patients and the public were therefore involved in its overall design, conduct, management, and dissemination, and are recognised amongst the authors of this article. For further information, please refer to aospine.org/recode.

RESULTS

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Literature search

The primary literature search identified a total of 3,239 unduplicated studies (MEDLINE: 2,389, EMBASE: 1,550). Abstract and full-text screening excluded 3,187 studies. Therefore, this review included a total of 52 studies (**Figure 1** and **Supplementary Table 2**).

Figure 1. PRISMA flow chart. A systematic review of MEDLINE and EMBASE was conducted through 4 August 2020 to identify original research on the measurement properties of instruments currently used in DCM research.

Study properties

The 52 included studies assessed a total of 7,395 patients worldwide (female: 3,217, male: 4,178) with 29 instruments (**Table 3**). These were classified into four domains based on the DCM core outcome set (36): neuromuscular function, life impact, pain, and radiological scoring.

Table 3. Study properties

Property	Number	%
Total studies, included	52	100
Prospective	31	60
Retrospective	21	40
Total patient sample	7,395	100
Male	4,178	56
Female	3,217	44
Measurement instruments by domain*	29	100
Neuromuscular function	16	55
Life impact	5	17
Pain	5	17
Radiological scoring	5	17
Publication year		
Maximum year of publication	2020	–
Median year of publication	2014	–
Mean year of publication	2012	–
Minimum year of publication	1999	–
Countries, by no. of patients	7,395	100

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Japan	2,014	27
United States of America	1,802	24
Canada	1,361	18
South Korea	726	10
Global/Multicentre	601	8
China	255	3
India	121	2
Iran	87	1
Brazil	85	1
Italy	75	1
Hong Kong	72	1
Thailand	70	1
Taiwan	45	1
United Kingdom	41	1
France	40	1

*Instrument counts per domain do not add up to the total due to the one-to-many relationship between certain instruments and domains (e.g., JOACMEQ is used both for life impact and neuromuscular function; see **Table 4**).

Measurement properties

The measurement properties of the 29 instruments were evaluated using the COSMIN methodology for systematic reviews (23-25). A summary of findings is presented in **Table 4**: (1) the overall feasibility rating, (2) the overall interpretability rating, and (3) the overall recommendation category based on existing evidence. Included studies reported on at least one of the 10 COSMIN properties for all instruments. No instrument had evidence for all 10 properties and <50% (13/29) of instruments had evidence for at least one property per measurement domain (**Figure 2**).

Figure 2. Number of studies for each outcome measure and property [normalised]. Included studies reported on at least one of the 10 COSMIN properties for all instruments. No instrument had evidence for all 10 properties and <50% (13/29) of instruments had evidence for at least one property per measurement domain (see **Table 2** for definitions). Notably, no instruments were evaluated for structural validity, attained sufficient evidence for content validity, or obtained a Category A recommendation based on COSMIN criteria.

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Table 4. Summary of findings

Domain	Instrument	Feasibility	Interpretability	Recommendation category	Recommendation justification
Life Impact					
	EQ-5D	+	+	C	High quality evidence for insufficient construct validity
	SF-12	–	+	B	Indeterminate result rating for internal consistency
	SF-36	–	+	B	Indeterminate result rating for internal consistency
	WHOQOL-Bref	+	–	B	Indeterminate result rating for internal consistency
Life Impact and Neuromuscular function					
	JOACMEQ	+	+	B	
Neuromuscular function					
	10-s step test	+	–	C	High quality evidence for insufficient criterion validity
	30MWT	+	–	C	High quality evidence for insufficient responsiveness
	9-Hole peg test	++	–	B	
	BBS	++	–	B	
	European Myelopathy Scale	+	–	B	
	Foot tapping test	+	–	C	High quality evidence for insufficient criterion validity

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	Grip-and-release test	+	–	B	
	JOA	–	+	B	
	MDI	+	–	B	
	mJOA	–	+	C	High quality evidence for insufficient criterion and construct validity
	Nurick scale	+	–	B	
	P-mJOA	+	–	B	
	Ranawat classification of disease severity	–	–	B	
	Triangle step test	+	–	B	
Pain and Neuromuscular function					
	QuickDASH	–	–	B	
Pain					
	NDI	+	+	B	
	Arm pain score	–	+	B	
	Neck pain score	+	+	B	
	VAS for pain	+	+	B	
Radiology					
	Cobb's method	+	–	B	
	CT (Tsuyama's classification, 2D & 3D)	+	–	B	
	CT (Tsuyama's classification, lateral + axial)	+	–	B	
	Isihara's Cervical Curvature Index	+	–	B	

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MRI (Depiction of intramedullary hyperintensity at eight cervical disc levels, T2W, 1.5-T or 3-T)	+	–	B
MRI (Kang's classification, 1.5-T or 3-T)	+	–	B
MRI (Muhle's classification, 1.5-T)	+	–	B
MRI (Vaccaro's classification, 1.5-T)	+	–	B
X-rays (Computer-assisted measurement of length & thickness)	+	–	B

Feasibility

++ = No barriers

+ = Minimal barriers

– = Barriers

Interpretability

+ = Interpretable

– = Uninterpretable, due to absence of anchor-based MCIDs (23-25)

Recommendation category

A = Measurement instruments with evidence for sufficient content validity (any level) AND at least low-quality evidence for sufficient internal consistency

B = Measurement instruments categorized not in A or C

C = Measurement instruments with high quality evidence for an insufficient measurement property

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Content validity

Only three measurement instruments were evaluated for content validity: the JOA Cervical Myelopathy Evaluation Questionnaire (JOACMEQ), the modified JOA (mJOA) score, and the Berg Balance Scale (BBS) (**Supplementary Table 5**). The overall ratings for content validity, however, were indeterminate due to the uncertainty of the methods used to assess comprehensibility, and the very low quality of the evidence.

Structural validity

No instruments were assessed for structural validity.

Internal consistency

Ten measurement instruments were evaluated for internal consistency, including the JOACMEQ, JOA, mJOA, 12-Item Short Form Health Survey (SF-12), and SF-36 (**Supplementary Table 6**). Since structural is required for the interpretation of internal consistency, the overall ratings for internal consistency were indeterminate, given the aforementioned absence of studies on structural validity.

Cross-cultural validity

Only three measurement instruments were evaluated for cross-cultural validity: JOACMEQ, JOA, and mJOA (**Supplementary Table 7**). The overall ratings were indeterminate due to the absence of multiple group factors analyses and differential item functioning analyses. The quality of evidence was also very low due to the uncertainty of the approaches used to analyse the data.

Reliability

Seventeen measurement instruments were evaluated for reliability, including JOACMEQ, JOA, and mJOA (**Supplementary Table 8**). The reported measures of reliability were test–retest reliability, intra-observer reliability, and inter-observer reliability. No instrument attained high-quality evidence for sufficient or insufficient reliability due to (a) imprecision (sample sizes < 100), (b) serious inconsistency, and/or (c) serious risk of bias.

Measurement error

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Nine instruments were evaluated for measurement error, including JOACMEQ, JOA, mJOA, NDI, SF-36, and Visual Analogue Scale (VAS) for pain (**Supplementary Table 9**). The measures of error reported were minimal detectable change (MDC) and distribution-based MCID (23-25, 37). The mJOA was the only score to attain high-quality evidence for sufficiency (distribution-based MCID range: 1.2–1.4, total sample size: 868). Due to the inconsistency of results, the quality of the evidence of most other instruments could not to be rated.

Criterion validity

Twelve measurement instruments were evaluated for criterion validity, including the JOACMEQ, JOA, mJOA, NDI, and SF-36 (**Supplementary Table 10**). Both the mJOA and the patient-derived version of the mJOA (P-mJOA) attained high-quality evidence for sufficient criterion validity as whole scales. However, three of four items of the mJOA, along with the 10-second step test and foot tapping test, attained high-quality evidence for insufficient criterion validity (i.e., these subdomains lack criterion validity for their use as separate measures). The quality of the evidence of most of the remaining instruments was not high due to (a) imprecision (i.e., sample sizes < 100) or (b) important methodological flaws in the design or statistical methods.

Construct validity

Sixteen measurement instruments were evaluated for construct validity, including JOACMEQ, JOA, mJOA, NDI, arm and neck pain scores, and SF-12 (**Supplementary Table 11**). From these, eight of 16 attained high-quality evidence for sufficient construct validity; these included the NDI, arm and neck pain scores, and SF-12. Two instruments achieved high-quality evidence for insufficient construct validity. Notably, the mJOA had both high-quality sufficiency and insufficiency depending on the comparator tool (e.g., sufficiency with respect to the NDI and SF-36 and insufficiency with respect to the 30-m walking test [30MWT] and EuroQol-5 Dimension [EQ-5D]). While the designs and statistical methods applied were adequate for the research questions posed, the quality of the evidence of most of the remaining tools ranged from ‘low’ to ‘moderate’ due to imprecision (i.e., sample sizes < 100). Importantly, only one study formulated a hypothesis a priori (38).

Responsiveness

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Sixteen measurement instruments were evaluated for responsiveness, including the JOACMEQ, JOA, mJOA, NDI, SF-12, and SF-36 (**Supplementary Table 12**). The mJOA was the only score to attain high-quality evidence for sufficient responsiveness (effect size range: 0.87–1.0, total sample size: 352). The 30MWT, on the other hand, was the only score to attain high-quality evidence for insufficient responsiveness (standardised response mean: 0.3, total sample size: 484). The quality of the evidence of most of the remaining tools ranged from 'very low' to 'moderate' due to (a) imprecision (i.e., sample sizes < 100) and (b) uncertainty of the statistical methods.

Clinically important differences

Ten measurement instruments were evaluated for clinically important differences, including the JOACMEQ, JOA, mJOA, NDI, arm and neck pain scores, SF-12, SF-36, and VAS for pain (**Supplementary Table 3**). From these, seven of 10 attained a sufficient rating, including the JOACMEQ, JOA, mJOA, NDI, and SF-36. Only anchor-based measures were accepted for assessment of the MCID (23-25, 39-42).

Interpretability and feasibility

Interpretability and feasibility were described using criteria adapted from the COSMIN methodology (**Supplementary Tables 3** and **4**, respectively). Interpretability was summarised in terms of the degree to which clinicians may assign qualitative meaning to the scores or change in scores (i.e., the clinically important differences) while feasibility was described in terms of the ease of application of the measurement instrument. No or minimal application barriers were identified for most outcome measures (**Table 4**). Nine instruments were, however, deemed uninterpretable due to the absence of anchor-based MCIDs (23-25).

Recommendations

No category A recommendations were made as no measurement instrument had sufficient evidence for content validity (**Table 4** and **Figure 2**). Furthermore, five instruments were recommended for category C due to the availability of high-quality evidence for insufficient criterion validity, construct validity, and/or responsiveness. Most instruments were classed into category B due to the notable absence of high-quality evidence for most measurement properties.

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In light of these results, and given both (a) the very strict quality standards of the COSMIN framework and (b) that the absence of category A evidence is not the same as presence of poor-quality evidence, we propose that instruments most suitable for use should be interpretable by clinicians and offer qualitative meaning to either clinicians or people with lived experience of DCM (i.e., they should have an available assessment of clinically important differences). To this end, the measurement properties of the nine interpretable instruments are presented in **Table 5**: the arm and neck pain scores; SF-12 and SF-36; JOA, mJOA, and JOACMEQ; NDI; and VAS for pain. These include one score with insufficient criterion and construct validity (i.e., mJOA) and six scores with barriers to application.

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Table 5. Interpretable measurement instruments

Domain	Instrument	Psychometric properties*	Feasibility	Recommendation category
Life Impact				
	SF-12	Cronbach's α coefficient (0.77)	–	B
	MCS	SCB (51.5)		
	PCS	SCB (30.1) Responsiveness: SF-12 PCS (Mean change score: 8.17)		
	SF-36	Cronbach's α coefficient (0.79–0.93) Responsiveness: SF-36 (Normalised change: 0.32)	–	B
	MCS	MDC or SDC (Distribution: 3.3–5.7) MCID (Distribution: 3.4–6.8, anchor: 3.0–7.4) Construct validity: Arm pain score (Pearson's correlation: –0.23) mJOA scale (Pearson's correlation: 0.19) NDI (Spearman's rank correlation: –0.17) Neck pain score (Pearson's correlation: –0.28) SF-12 PCS (Pearson's correlation: 0.01) Responsiveness: SF-36 MCS (Effect size range: 0.81, sensitivity: 0.67)		
	PCS	MDC or SDC (Distribution: 5.2–5.7, anchor: 4.9) MCID (Distribution: 2.9–5.5, distribution: 10, anchor: 3.9–9.6)		

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	SCB (16) Criterion validity (Likert scale) AUC: 0.67–0.69 Construct validity: Arm pain score (Pearson's correlation: –0.44) mJOA scale (Pearson's correlation: 0.43) NDI (Spearman's rank correlation: –0.49) Neck pain score (Pearson's correlation: –0.41) SF-12 PCS (Pearson's correlation: –0.29) Responsiveness: SF-36 PCS (Effect size range: 0.84, sensitivity: 0.85)		
Life Impact and Neuromuscular function			
JOACMEQ	Patient comprehensibility: "No questions elicited no answer or “I am not sure” in more than 5% of patients" Test–retest stability: Cronbach's α coefficient (0.91) Forward-backward translation [Persian and Thai]: n/a	+	B
Bladder function	Cronbach's α coefficient (0.32–0.74) Test–retest stability: ICC (0.62) MDC or SDC (Distribution: 7.7) MCID (Anchor: 6.0)		

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		Responsiveness: JOACMEQ Bladder function (AUC: 0.82, Effect size: 0.33, Mean change score: 18.0)
	Cervical spine function	Cronbach's α coefficient (0.77–0.78) Test–retest stability: ICC (0.63) MDC or SDC (Distribution: 12.9, anchor: 12.5) MCID (Anchor: 2.5) Criterion validity (Likert scale) AUC: 0.58 Responsiveness: JOACMEQ Cervical spine function (AUC: 0.72, Effect size: 0.28, Mean change score: 25.8)
	Lower extremity function	Cronbach's α coefficient (0.80–0.86) Test–retest stability: ICC (0.83) MDC or SDC (Distribution: 6.6, anchor: 8.5) MCID (Anchor: 8.5–9.5) Criterion validity (Likert scale) AUC: 0.66–0.70 Construct validity: NDI (Pearson's correlation: –0.66) SF-12 MCS (Spearman's rank correlation: 0.40) SF-12 PCS (Spearman's rank correlation: 0.29)

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		Responsiveness: JOACMEQ Quality of life (AUC: 0.83, Effect size: 0.46, Mean change score: 23.7)
	QOL	Cronbach's α coefficient (0.80–0.86) Test–retest stability: ICC (0.83) MDC or SDC (Distribution: 6.6, anchor: 8.5) MCID (Anchor: 8.5–9.5) Criterion validity (Likert scale) AUC: 0.66–0.70 Construct validity: NDI (Pearson's correlation: –0.66) SF-12 MCS (Spearman's rank correlation: 0.40) SF-12 PCS (Spearman's rank correlation: 0.29) Responsiveness: JOACMEQ Quality of life (AUC: 0.83, Effect size: 0.46, Mean change score: 23.7)
	Upper extremity function	Cronbach's α coefficient (0.72–0.74) Test–retest stability: ICC (0.93) MDC or SDC (Distribution: 9.5, anchor: 6.1) MCID (Anchor: 2.5–13.0) Responsiveness:

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	JOACMEQ Upper extremity function (AUC: 0.74, Effect size: 0.17, Mean change score: 10.7)		
Neuromuscular function			
JOA	Cronbach's α coefficient (0.72) Forward-backward translation [Brazilian Portuguese]: Comprehension rate (>81.2%) Inter-observer reliability: ICC (0.81) MDC or SDC (Distribution: 1.0, anchor: 2.5) LOA (1.2 [–1.2, 3.6]) MCID (Anchor: 2.5) Criterion validity (Likert scale) AUC: 0.59–0.62 Construct validity: JOACMEQ QOL (Spearman's rank correlation: 0.41) mJOA (Spearman's rank correlation: 0.87) NDI (Spearman's rank correlation: –0.50 to –0.76) SF-12 MCS (Spearman's rank correlation: –0.05) SF-12 PCS (Spearman's rank correlation: 0.50) Responsiveness: JOA (Mean change score: 4.6, normalised change: 0.21) JOA Motor function of lower extremity (Mean change score: 0.60) mJOA (Spearman's rank correlation: 0.75)	–	B

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	Bladder function	Intra-observer reliability (κ = 0.64) Inter-observer reliability (κ = 0.47)		
	Motor function of fingers	Intra-observer reliability (κ = 0.68) Inter-observer reliability (κ = 0.53)		
	Motor function of shoulder and elbow	Intra-observer reliability (κ = 0.50) Inter-observer reliability (κ = 0.31)		
	Motor function of lower extremity	Intra-observer reliability (κ = 0.55) Inter-observer reliability (κ = 0.49)		
	Sensory function of lower extremity	Intra-observer reliability (κ = 0.54) Inter-observer reliability (κ = 0.58)		
	Sensory function of upper extremity	Intra-observer reliability (κ = 0.51) Inter-observer reliability (κ = 0.42)		
mJOA		Cronbach's α coefficient (0.60–0.63) Forward-backward translation [Brazilian Portuguese and Italian]: n/a Test–retest stability (Spearman's rank correlation: 0.91) Intra-observer reliability (ICC: 0.87) Inter-observer reliability (ICC: 0.97, κ = 0.80) MDC or SDC (Distribution: 2.1) MCID (Distribution: 1.2–1.4, anchor: 1.3–3.1) SCB (14) Criterion validity (Nurick scale) Spearman's rank correlation: –0.41, Pearson's correlation: –0.62 to –0.63	–	C

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		Construct validity: 30MWT (Pearson's correlation: –0.38) EQ-5D (Spearman's rank correlation: 0.42) JOACMEQ QOL (Spearman's rank correlation: 0.41) NDI (Spearman's rank correlation: –0.51, Pearson's correlation: –0.33 to –0.34) SF-12 MCS (Pearson's correlation: 0.03) SF-12 PCS (Pearson's correlation: 0.42) SF-36 MCS (Pearson's correlation: 0.25) SF-36 PCS (Pearson's correlation: 0.30) Responsiveness: mJOA (Effect size: 0.87–1.0, normalised change: 1.47)
	Motor dysfunction of lower extremities	Inter-observer reliability (ICC: 0.73) Criterion validity (Nurick scale) Pearson's correlation: –0.65 to –0.68 Construct validity: 30MWT (Pearson's correlation: –0.43) NDI (Pearson's correlation: –0.31) SF-36 MCS (Pearson's correlation: 0.21) SF-36 PCS (Pearson's correlation: 0.31–0.50)
	Motor dysfunction of upper extremities	Inter-observer reliability (ICC: 0.77) Criterion validity (Nurick scale) Pearson's correlation: –0.42 Construct validity:

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		30MWT (Pearson's correlation: –0.21) NDI (Pearson's correlation: –0.24) SF-36 MCS (Pearson's correlation: 0.20) SF-36 PCS (Pearson's correlation: 0.22)		
	Sensory dysfunction of sphincter dysfunction	Inter-observer reliability (ICC: 0.78) Criterion validity (Nurick scale) Pearson's correlation: –0.25 Construct validity: 30MWT (Pearson's correlation: –0.23) NDI (Pearson's correlation: –0.16) SF-36 MCS (Pearson's correlation: 0.08) SF-36 PCS (Pearson's correlation: 0.06)		
	Sensory dysfunction of upper extremities	Inter-observer reliability (ICC: 0.93) Criterion validity (Nurick scale) Pearson's correlation: –0.23 Construct validity: 30MWT (Pearson's correlation: –0.05) NDI (Pearson's correlation: –0.23) SF-36 MCS (Pearson's correlation: 0.19) SF-36 PCS (Pearson's correlation: 0.19)		
Pain				
NDI		MDC or SDC (Distribution: 6.2%, anchor: 5.2%) MCID (Anchor: 5–13) SCB (Anchor: 9.5–36)	+	B

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		Criterion validity (Likert scale) AUC: 0.66–0.75 Construct validity: Arm pain score (Pearson's correlation: 0.68) mJOA (Pearson's correlation: –0.36) Neck pain score (Pearson's correlation: 0.64) SF-12 MCS (Pearson's correlation: –0.40) SF-12 PCS (Pearson's correlation: –0.54) Responsiveness: Anchor (AUC: 0.66) NDI (Mean change score: –15.8)		
	Pain, "Numeric rating scale" (Arm pain score)	MCID (Anchor: 2.5) SCB (3.5) Construct validity: mJOA (Pearson's correlation: –0.19) Neck pain score (Pearson's correlation: 0.72)	–	B
	Pain, "Numeric rating scale" (Neck pain scores)	MCID (Anchor: 2.5) SCB (3.5) Construct validity: mJOA (Pearson's correlation: –0.07)	–	B
	VAS for pain	MDC or SDC (Distribution: 3.1) MCID (Distribution: 24.0–30.0, anchor: 0.4–2.7) SCB (1.1)	+	B

n/a = No info available

LOA: Limits of agreement

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Feasibility

++ = No barriers

+ = Minimal barriers

– = Barriers

Interpretability

+ = Interpretable

– = Uninterpretable, due to absence of anchor-based MCIDs (23-25)

Recommendation category

A = Measurement instruments with evidence for sufficient content validity (any level) AND at least low-quality evidence for sufficient internal consistency

B = Measurement instruments categorized not in A or C

C = Measurement instruments with high quality evidence for an insufficient measurement property

*Comparators shown as indented tools

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DISCUSSION

DCM is measured and reported inconsistently across clinical trials (1-4). In light of these inconsistencies, AO Spine launched RECODE-DCM (www.aospine.org/recode) with the aim of creating a 'research toolkit' that helps to accelerate knowledge discovery and improve outcomes in DCM. One of the objectives of the RECODE-DCM initiative was to develop a CMS (3, 6). This systematic review consists of an initial step towards building this CMS by identifying tools that have been used in DCM research and examining their quality, in accordance with the COSMIN standards (23-25).

Overall, we identified 29 instruments with at least one in 10 measurement properties evaluated (**Figure 2**); none, however, had evaluations for all 10 properties and <50% had more than one property evaluated per measurement domain (i.e., reliability, validity, and responsiveness) (**Table 2**). We also noted a paucity in the quantity and quality of studies evaluating DCM instruments; this is visible by the absence of category A recommendations and the classification of most tools in category B (**Table 4**). Acknowledging both the stringency of the COSMIN standards and that absence of category A evidence is not equivalent to presence of poor-quality evidence, we proposed nine instruments that seem interpretable to clinicians and appear to offer qualitative meaning to clinicians and people with lived experience of DCM. These instruments are the SF-12 and SF-36; JOA, mJOA, and JOACMEQ; NDI; and VAS for pain (**Table 5**).

The fact that most outcomes received B-category recommendations due to absence of high-quality evidence is not unexpected. In this review, the most common reasons for low-quality evidence, as per the COSMIN guidelines, were (a) important methodological flaws in study design or statistical methods, (b) uncertainty of approaches used to analyse the data, and (c) imprecision due to sample size below the recommended power and significance levels. The rigor (or stringency) of the COSMIN standards may have accentuated these limitations due to the highly specific nature of some standards and the expectation of psychometric expertise within the DCM context. For example, results for internal consistency must be rated 'indeterminate' if there is not at least low-quality evidence for structural validity. No such studies were available in this review, possibly because this is a more recent and complex criterion, or because of the search or selection criteria. Similarly, studies on

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content validity cannot score higher than 'inadequate' if there are no recordings/verbatim transcriptions of patient focus groups or interviews. Likewise, analyses of reliability cannot score higher than 'doubtful' if statistics other than the Pearson or Spearman correlation coefficients are used. These thresholds of acceptability may account for some of the lacking information and are an important entry challenge for instruments into DCM research—a field where the routine involvement of stakeholders with lived experience is at an early stage (3, 8), inconsistent study reporting is prevalent (2, 4), few studies have involved >100 patients, and where there is a bias in the availability of measurement literature (i.e., some tools, such as the SF-12, are used because they are the only tools available and, therefore, have available literature due to their routine use). From the application of these COSMIN criteria in other research fields, however, it appears that these methodological deficiencies are not exclusive to DCM instruments, including those in current use (43-45). The lack of high-quality assessments, thus, should not necessarily imply that (a) the identified outcome measures are generally inadequate, or (b) that the COSMIN standards are not fit for the DCM context.

Measurement rigor is universally important and, in DCM, particularly relevant as the development of new instruments is a top 10 research priority. This rank reinforces the decision of the steering committee to make the initial CMS recommendations based existing on tools, rather than on tools under development. This decision was taken recognising that the success of a CMS requires widespread adoption, and that the adoption of clinical recommendations can be challenging without stakeholder awareness, familiarity, and/or confidence (46-49). We hypothesised that asking the global field to align with new innovations would be more challenging, and premature, at this stage. Thus, for this first iteration of the DCM CMS, there is a focus on current instruments in academic usage. Whilst, currently, few have met the bar set by the COSMIN methodology, there are nine reasonable candidates using our post-hoc thresholds (**Table 5**). Ultimately, the CMS process will need to lean significantly on the expertise of those involved in the consensus phase in order to make final recommendations that are methodologically rigorous and representative of those with lived experience.

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Despite its conscientious design, this systematic review has limitations. In searching for existing instruments, we have neither identified nor assessed tools under development, or those currently being translated into clinical or research settings or published in languages other than English. To the extent that DCM instruments are currently in use, however, this review only identified tools in four of the six core domains from RECODE-DCM's minimum data set (36), and did not considered the construct of the disease as a factor in evaluating the outcomes. For those missing outcomes, focused scoping reviews (informed by a gap analysis that will be published separately) will be conducted in the future. Next, clinician-reported outcome measures and performance-based outcome measures were analysed with the exact same methods as PROMs. Whilst COSMIN explicitly allows this (23-25), methods may be differentially adapted to tailor to these distinct instrument types; we chose not to do so out of prudence and consistency, and results across these instrument groups should be interpreted accordingly. Feasibility and interpretability were also evaluated using bespoke criteria which, despite being adapted from the COSMIN methodology, may not weigh all criteria accurately. Importantly, our decision to shortlist the clinically interpretable instruments was made a posteriori due to the unexpected absence of category A recommendations. This decision was informed by our judgement that instruments in a CMS should be interpretable by clinicians and offer qualitative meaning to clinicians and people with lived experience. While the COSMIN taxonomy does indeed class interpretability as an important and stand-alone characteristic (23-25), the aforementioned shortlist may inevitably represent a placement bias. Notably, some nuances of different versions of measurement instruments (e.g., mJOA) were not extensively evaluated (50). Lastly, and as is frequently the case in this body of reviews (43-45), none of the authors is specifically trained in measurement theory and, therefore, this work represents our best attempt to implement the guidelines and standards set forward by the COSMIN methodology in the context of DCM.

CONCLUSIONS

Currently, none of the measurement instruments used in DCM holds sufficient evidence to meet the COSMIN criteria for a strong recommendation for use. However, there are leading contenders that appear to offer qualitative meaning to clinicians and people with lived experience of DCM; namely, the SF-12 and SF-36; JOA, mJOA, and JOACMEQ; NDI; and VAS for pain. The findings of this review will inform a consensus

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process to form a CMS for DCM. As the development of new assessments for DCM is an active research priority, greater awareness of the COSMIN framework is pertinent to DCM researchers.

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AUTHOR CONTRIBUTIONS

BMD was responsible for conceiving the article. KSL conducted the search and AYT, AB, ED, and FB conducted the screening. AYT and BMD extracted and analysed the data and wrote the manuscript. AYT, JCF, MGF, JH, CMZ, RRP, JM, ES, AC, VRM, BA, TFB, LT, RC, JDG, SKR, AGKM, MRN, and BMD provided critical appraisal of the manuscript. All authors critically revised and approved the manuscript.

ETHICS APPROVAL

Ethics approval for this study was not applicable because this study did not involve human participants.

DATA AVAILABILITY STATEMENT

All data relevant to the study are included in the article or uploaded as supplementary information.

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COMPETING INTERESTS

None declared.

REFERENCES

For peer review only

1. Fehlings MG, Tetreault LA, Riew KD, Middleton JW, Aarabi B, Arnold PM, et al. A Clinical Practice Guideline for the Management of Patients With Degenerative Cervical Myelopathy: Recommendations for Patients With Mild, Moderate, and Severe Disease and Nonmyelopathic Patients With Evidence of Cord Compression. Global Spine Journal. 2017;7(3_suppl):70S-83S.
2. Davies BM, McHugh M, Elgheriani A, Kolias AG, Tetreault LA, Hutchinson PJA, et al. Reported Outcome Measures in Degenerative Cervical Myelopathy: A Systematic Review. PLOS ONE. 2016;11(8):e0157263.
3. Davies BM, Khan DZ, Mowforth OD, McNair AGK, Gronlund T, Kolias AG, et al. RE-CODE DCM (REsearch Objectives and Common Data Elements for Degenerative Cervical Myelopathy): A Consensus Process to Improve Research Efficiency in DCM, Through Establishment of a Standardized Dataset for Clinical Research and the Definition of the Research Priorities. Global Spine Journal. 2019;9(1_suppl):65S-76S.
4. Davies BM, McHugh M, Elgheriani A, Kolias AG, Tetreault L, Hutchinson PJA, et al. The reporting of study and population characteristics in degenerative cervical myelopathy: A systematic review. PloS one. 2017;12(3):e0172564-e.
5. Davies BM, Mowforth OD, Smith EK, Kotter MR. Degenerative cervical myelopathy. BMJ. 2018;360:k186.
6. AO Spine. AO Spine RECODE-DCM: REsearch Objectives and COmmon Data Elements for Degenerative Cervical Myelopathy 2021 [Available from: www.aospine.org/recode.
7. Davies B, Mowforth O, Sadler I, Aarabi B, Kwon B, Kurpad S, et al. Recovery priorities in degenerative cervical myelopathy: a cross-sectional survey of an international, online community of patients. BMJ Open. 2019;9(10):e031486.
8. Boerger TF, Davies BM, Sadler I, Sarewitz E, Kotter MRN. Patient, sufferer, victim, casualty or person with cervical myelopathy: let us decide our identifier. Integrated Healthcare Journal. 2020;2(1):e000023.
9. Clarke M. Standardising outcomes for clinical trials and systematic reviews. Trials. 2007;8(1):39.
10. Kirkham JJ, Dwan KM, Altman DG, Gamble C, Dodd S, Smyth R, et al. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. BMJ. 2010;340:c365.
11. Kirkham JJ, Gargon E, Clarke M, Williamson PR. Can a core outcome set improve the quality of systematic reviews? – a survey of the Co-ordinating Editors of Cochrane review groups. Trials. 2013;14(1):21.
12. Prinsen CAC, Vohra S, Rose MR, Boers M, Tugwell P, Clarke M, et al. How to select outcome measurement instruments for outcomes included in a "Core Outcome Set" – a practical guideline. Trials. 2016;17(1):449.
13. Williamson PR, Altman DG, Bagley H, Barnes KL, Blazeby JM, Brookes ST, et al. The COMET Handbook: version 1.0. Trials. 2017;18(3):280.
14. Boers M, Kirwan JR, Wells G, Beaton D, Gossec L, d'Agostino M-A, et al. Developing Core Outcome Measurement Sets for Clinical Trials: OMERACT Filter 2.0. Journal of Clinical Epidemiology. 2014;67(7):745-53.
15. Schmitt J, Apfelbacher C, Spuls PI, Thomas KS, Simpson EL, Furue M, et al. The Harmonizing Outcome Measures for Eczema (HOME) roadmap: a methodological framework to develop core sets of outcome measurements in dermatology. J Invest Dermatol. 2015;135(1):24-30.

For peer review only

16. Kirkham JJ, Davis K, Altman DG, Blazeby JM, Clarke M, Tunis S, et al. Core Outcome Set-STAndards for Development: The COS-STAD recommendations. PLOS Medicine. 2017;14(11):e1002447.
17. Kirkham JJ, Gorst S, Altman DG, Blazeby JM, Clarke M, Devane D, et al. Core Outcome Set-STAndards for Reporting: The COS-STAR Statement. PLoS Med. 2016;13(10):e1002148.
18. Potter S, Davies C, Holcombe C, Weiler-Mithoff E, Skillman J, Vidya R, et al. International development and implementation of a core measurement set for research and audit studies in implant-based breast reconstruction: a study protocol. BMJ Open. 2020;10(1):e035505.
19. Grieve S, Perez R, Birklein F, Brunner F, Bruehl S, Harden RN, et al. Recommendations for a first Core Outcome Measurement set for complex regional PAin syndrome Clinical sTudies (COMPACT). Pain. 2017;158(6):1083-90.
20. Davies CF, Macefield R, Avery K, Blazeby JM, Potter S. Patient-Reported Outcome Measures for Post-mastectomy Breast Reconstruction: A Systematic Review of Development and Measurement Properties. Ann Surg Oncol. 2021;28(1):386-404.
21. Kalsi-Ryan S, Singh A, Massicotte EM, Arnold PM, Brodke DS, Norvell DC, et al. Ancillary outcome measures for assessment of individuals with cervical spondylotic myelopathy. Spine (Phila Pa 1976). 2013;38(22 Suppl 1):S111-22.
22. Mokkink LB, Terwee CB, Knol DL, Stratford PW, Alonso J, Patrick DL, et al. Protocol of the COSMIN study: COnsensus-based Standards for the selection of health Measurement INstruments. BMC Medical Research Methodology. 2006;6(1):2.
23. Prinsen CAC, Mokkink LB, Bouter LM, Alonso J, Patrick DL, de Vet HCW, et al. COSMIN guideline for systematic reviews of patient-reported outcome measures. Qual Life Res. 2018;27(5):1147-57.
24. Mokkink LB, de Vet HCW, Prinsen CAC, Patrick DL, Alonso J, Bouter LM, et al. COSMIN Risk of Bias checklist for systematic reviews of Patient-Reported Outcome Measures. Quality of Life Research. 2018;27(5):1171-9.
25. Terwee CB, Prinsen CAC, Chiarotto A, Westerman MJ, Patrick DL, Alonso J, et al. COSMIN methodology for evaluating the content validity of patient-reported outcome measures: a Delphi study. Qual Life Res. 2018;27(5):1159-70.
26. Davies BM, Goh S, Yi K, Kuhn I, Kotter MRN. Development and validation of a MEDLINE search filter/hedge for degenerative cervical myelopathy. BMC Medical Research Methodology. 2018;18(1):73.
27. Khan DZ, Khan MS, Kotter MR, Davies BM. Tackling Research Inefficiency in Degenerative Cervical Myelopathy: Illustrative Review. JMIR Res Protoc [Internet]. 2020 2020/06//; 9(6):[e15922 p.]. Available from: http://europepmc.org/abstract/MED/32525490 https://doi.org/10.2196/15922 https://europepmc.org/articles/PMC7317636.
28. Khan MA, Mowforth OD, Kuhn I, Kotter MRN, Davies BM. Development of a validated search filter for Ovid Embase for degenerative cervical myelopathy. Health Information & Libraries Journal.n/a(n/a).
29. Singh A, Tetreault L, Casey A, Laing R, Statham P, Fehlings MG. A summary of assessment tools for patients suffering from cervical spondylotic myelopathy: a systematic review on validity, reliability and responsiveness. Eur Spine J. 2015;24 Suppl 2:209-28.
30. Mokkink LB, Terwee CB, Patrick DL, Alonso J, Stratford PW, Knol DL, et al. The COSMIN study reached international consensus on taxonomy, terminology, and definitions of

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measurement properties for health-related patient-reported outcomes. J Clin Epidemiol. 2010;63(7):737-45.
31. van Kampen DA, Willems WJ, van Beers LWAH, Castelein RM, Scholtes VAB, Terwee CB. Determination and comparison of the smallest detectable change (SDC) and the minimal important change (MIC) of four-shoulder patient-reported outcome measures (PROMs). Journal of Orthopaedic Surgery and Research. 2013;8(1):40.
32. Nwachukwu BU, Beck EC, Kunze KN, Chahla J, Rasio J, Nho SJ. Defining the Clinically Meaningful Outcomes for Arthroscopic Treatment of Femoroacetabular Impingement Syndrome at Minimum 5-Year Follow-up. The American Journal of Sports Medicine. 2020;48(4):901-7.
33. Streiner DL, Norman GR, Cairney J. Health Measurement ScalesA practical guide to their development and use: A practical guide to their development and use: Oxford University Press; 2015 2015-01.
34. Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. BMJ. 2020;368:l6890.
35. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 2021;372:n71.
36. AO Spine. Minimum Dataset [Internet] 2021 [Available from: https://aospine.aofoundation.org/research/recode-dcm/minimum--dataset.
37. de Vet HCW, Terwee CB, Mokkink LB, Knol DL. Measurement in Medicine: A Practical Guide. Cambridge: Cambridge University Press; 2011.
38. Longo UG, Berton A, Denaro L, Salvatore G, Denaro V. Development of the Italian version of the modified Japanese orthopaedic association score (mJOA-IT): cross-cultural adaptation, reliability, validity and responsiveness. European Spine Journal. 2016;25(9):2952-7.
39. de Vet HC, Terwee CB. The minimal detectable change should not replace the minimal important difference. J Clin Epidemiol. 2010;63(7):804-5; author reply 6.
40. de Vet HC, Ostelo RW, Terwee CB, van der Roer N, Knol DL, Beckerman H, et al. Minimally important change determined by a visual method integrating an anchor-based and a distribution-based approach. Qual Life Res. 2007;16(1):131-42.
41. de Vet HC, Terwee CB, Ostelo RW, Beckerman H, Knol DL, Bouter LM. Minimal changes in health status questionnaires: distinction between minimally detectable change and minimally important change. Health Qual Life Outcomes. 2006;4:54-.
42. Crosby RD, Kolotkin RL, Williams GR. Defining clinically meaningful change in health-related quality of life. J Clin Epidemiol. 2003;56(5):395-407.
43. Francis DO, McPheeters ML, Noud M, Penson DF, Feurer ID. Checklist to operationalize measurement characteristics of patient-reported outcome measures. Syst Rev. 2016;5(1):129.
44. Francis DO, Daniero JJ, Hovis KL, Sathe N, Jacobson B, Penson DF, et al. Voice-Related Patient-Reported Outcome Measures: A Systematic Review of Instrument Development and Validation. Journal of Speech, Language, and Hearing Research. 2017;60(1):62-88.
45. Terwee CB, de Vet HC, Prinsen CAC, Mokkink LB. Comment on "Checklist to operationalize measurement characteristics of patient-reported outcome measures" [Internet] [Available from: https://www.cosmin.nl/wp-content/uploads/Letter-comment-on-Francis.pdf.

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46. Kirkham JJ, Clarke M, Williamson PR. A methodological approach for assessing the uptake of core outcome sets using ClinicalTrials.gov: findings from a review of randomised controlled trials of rheumatoid arthritis. BMJ. 2017;357:j2262.
47. Bauer MS, Kirchner J. Implementation science: What is it and why should I care? Psychiatry Research. 2020;283:112376.
48. Gupta DM, Boland RJ, Aron DC. The physician's experience of changing clinical practice: a struggle to unlearn. Implementation Science. 2017;12(1):28.
49. Braithwaite J, Churruca K, Long JC, Ellis LA, Herkes J. When complexity science meets implementation science: a theoretical and empirical analysis of systems change. BMC Medicine. 2018;16(1):63.
50. Furlan JC, Catharine Craven B. Psychometric analysis and critical appraisal of the original, revised, and modified versions of the Japanese Orthopaedic Association score in the assessment of patients with cervical spondylotic myelopathy. Neurosurg Focus. 2016;40(6):E6.

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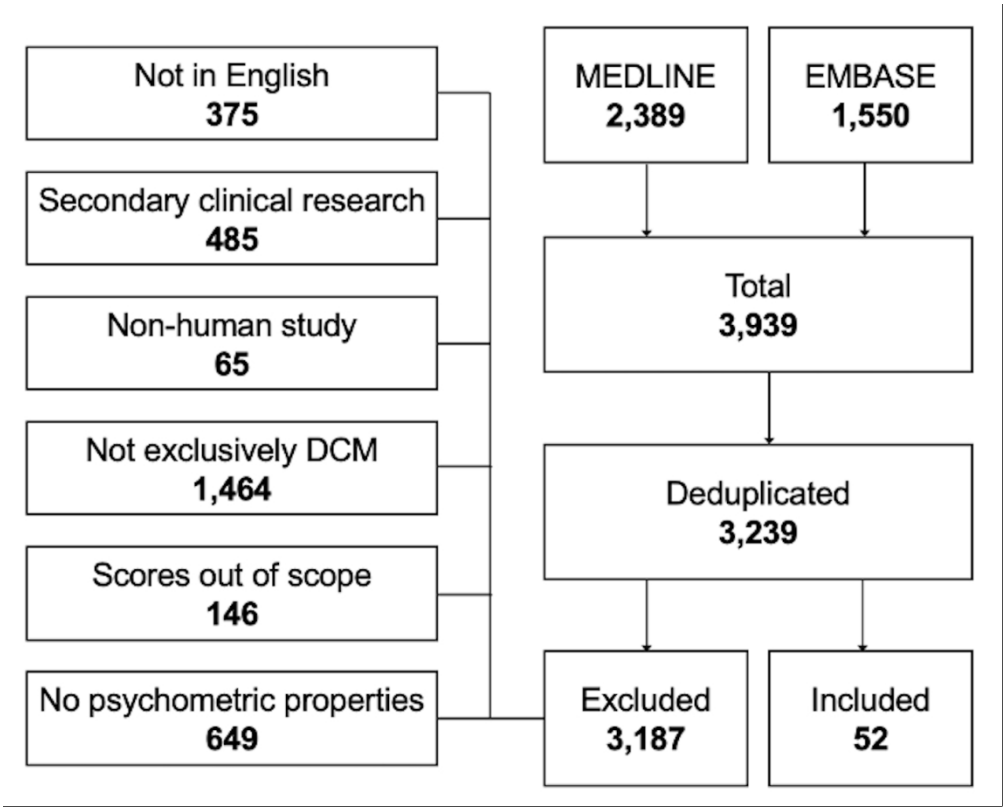


Figure 1. PRISMA flow chart. A systematic review of MEDLINE and EMBASE was conducted between August 2020 and January 2021 to identify original research on the measurement properties of instruments currently used in DCM research.

119x96mm (300 x 300 DPI)

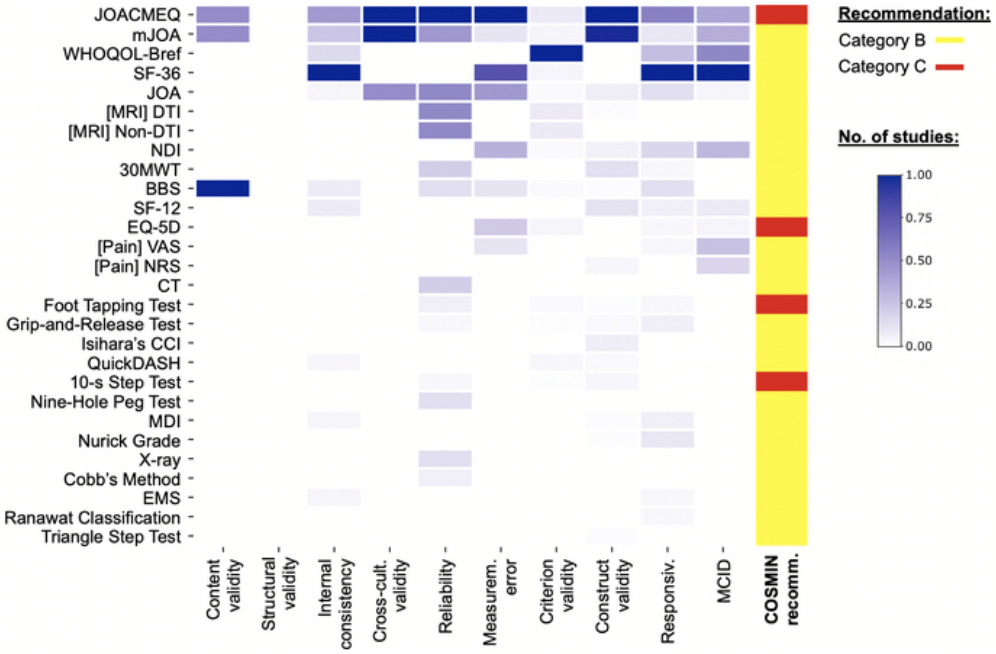


Figure 2. Number of studies for each outcome measure and property [normalised]. Included studies reported on at least one of the 10 COSMIN properties for all instruments. No instrument had evidence for all 10 properties and <50% (13/29) of instruments had evidence for at least one property per measurement domain (see Table 2 for definitions). Notably, no instruments were evaluated for structural validity, attained sufficient evidence for content validity, or obtained a Category A recommendation based on COSMIN criteria.

58x38mm (300 x 300 DPI)

SUPPLEMENTARY INFORMATION

Supplementary Table 1. Full search phrases used for MEDLINE and EMBASE on 4 August 2020

Ovid MEDLINE		**2389 articles**
Cervical spine concept		
1	exp Cervical Vertebrae/ or exp Cervical Cord/ or cervical.tw	
DCM concept		
2	Exp Spinal Cord Diseases/ or Exp Spinal Diseases/	
3	degenerat*.tw	
4	2 and 3	
5	Myelopath*.tw	
6	Myeloradiculopath*.tw	
7	Radiculopath*.tw	
8	Exp Spinal Cord Compression/	
9	Exp “Ossification of the Posterior Longitudinal Ligament”/	
10	Ossification of the Posterior Longitudinal Ligament.tw	
11	OPLL.tw	
12	Exp Spinal Stenosis/	
13	Cervical stenosis.tw	
14	Exp Spondylosis/	
15	Spondylosis.tw	
16	Spondylotic.tw	
17	Degenerative cervical myelopathy.tw	
18	DCM.tw	
19	Cervical spondylotic myelopathy.tw	
20	CSM.tw	
21	4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20	
22	1 and 21	
Tools for function concept (neurological/gait/mobility centric) concept		
23	Japanese Orthopaedic Association score.tw	
24	JOA.tw	
25	modified JOA.tw	
26	mJOA.tw	
27	Graded Redefined Assessment of Sensation Strength and Prehension.tw	
28	GRASSP.tw	
29	Quick Disability of the Arm Shoulder and Hand.tw	
30	QuickDASH.tw	

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31 Myelopathy Disability Index.tw
32 MDI.tw
33 Nurick score.tw
34 Neck functional disability scale.tw
35 NFDS.tw
36 Neck Disability Index.tw
37 NDI.tw
38 Cooper myelopathy scale.tw
39 CMS.tw
40 European myelopathy score.tw
41 EMS.tw
42 Bournemouth questionnaire.tw
43 BQ.tw
44 Cervical spine outcomes questionnaire.tw
45 CSOQ.tw
46 Patient specific functional scale.tw
47 PSFS.tw
48 World Health Organization Quality of Life Instruments.tw
49 WHOQOL.tw
50 Grip and release test.tw
51 GRT.tw
52 Grip Dynamometer.tw
53 Triangle step test.tw
54 Foot tapping test.tw
55 30 m walking test.tw
56 30MWT.tw
57 10 m walking test.tw
58 10MWT.tw
59 Berg Balance Scale.tw
60 BBS.tw
61 GAITRite.tw
62 10 second step test.tw
63 9 hole peg test.tw
64 Prolo.tw
65 Mental component score.tw
66 MCS.tw
67 Physical component score.tw
68 PCS.tw

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69	Hospital anxiety depression scale.tw
70	HADS.tw
71	Global rating of change.tw
72	GROC.tw
73	23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67 or 68 or 69 or 70 or 71 or 72
Tools for QOL concept (including pain) concept	
74	Exp “Quality of Life”/ or exp “Surveys and Questionnaires”/
75	Short Form Health Survey.tw
76	SF-36.tw
77	SF-12.tw
78	EQ-5D.tw
79	Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire.tw
80	JOACMEQ.tw
81	Visual Analogue Scale.tw
82	VAS.tw
83	Likert scale.tw
84	Numeric pain rating scale.tw
85	NPRS.tw
86	North American Spine Satisfaction.tw
87	NASS.tw
88	74 or 75 or 76 or 77 or 78 or 79 or 80 or 81 or 82 or 83 or 84 or 85 or 86 or 87
Psychometric concept	
89	Exp Psychometrics/
90	Pyschometr*.tw
91	(clinimetr* or clinometr*).tw.
92	Outcome assessment*.tw
93	exp Health Status Indicators/
94	Exp “Reproducibility of Results”/
95	Reproducib*.tw
96	Exp Validation Study/
97	exp Discriminant Analysis/
98	(reliab* or unreliab* or valid* or coefficient or homogeneity or homogeneous or internal consistency).tw
99	(cronbach* and (alpha or alphas)).tw.
100	(item and (correlation* or selection* or reduction*)).tw

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101 (agreement or precision or imprecision or precise values or test–retest).tw
102 (reliab* and (test or retest)).tw

103 (stability or interrater or inter-rater or intrarater or intra-rater or intertester or inter-tester or intratester or intra-tester or interobserver or inter-observer or intraobserver or intraobserver or intertechnician or inter-technician or intratechnician or intra-technician or interexaminer or inter-examiner or intraexaminer or intra-examiner or interassay or interassay or intraassay or intra-assay or interindividual or inter-individual or intraindividual or intra-individual or interparticipant or inter-participant or intraparticipant or intra-participant or kappa or kappas or repeatab*).tw
104 ((replicab* or repeated) and (measure or measures or findings or result or results or test or tests)).tw
105 (generaliza* or generalisa* or concordance).tw
106 (intraclass and correlation*).tw
107 Exp Observer Variation/
108 Observer variation.tw
109 (multitrait and scaling and (analysis or analyses)).tw
110 Measurement error*.tw
111 (item discriminant or interscale correlation* or error or errors or individual variability).tw
112 (variability and (analysis or values)).tw
113 (uncertainty and (measurement or measuring)).tw
114 Exp Diagnostic Errors/
115 Exp Data accuracy/
116 Exp Dimensional Measurement Accuracy/
117 Accuracy.tw
118 ((minimal or minimally or clinical or clinically) and (important or significant or detectable) and (change or difference)).tw
119 Minimally clinically important difference*.tw
120 MCID.tw
121 (small* and (real or detectable) and (change or difference)).tw
122 (meaningful change or ceiling effect or floor effect or Item response model or IRT or Rasch or Differential item functioning or DIF or computer adaptive testing or item bank or cross-cultural equivalence).tw
123 Exp Bias/ or exp Selection Bias/
124 Bias.tw
125 Exp “Predictive Value of Test”/

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126	89 or 90 or 91 or 92 or 93 or 94 or 95 or 96 or 97 or 98 or 99 or 100 or 101 or 102 or 103 or 104 or 105 or 106 or 107 or 108 or 109 or 110 or 111 or 112 or 113 or 114 or 115 or 116 or 117 or 118 or 119 or 120 or 121 or 122 or 123 or 124 or 125
Combined concepts	
127	73 or 88
128	22 and 126 and 127

EMBASE **1550 articles**

Cervical spine concept	
1	exp Cervical Vertebra/ or cervical spine/ or exp Cervical spinal cord/ or cervical.tw
DCM concept	
2	Exp Spinal Cord Disease/ or Exp Spine Disease/
3	Exp degeneration/
4	degenerat*.tw
5	3 or 4
6	2 and 5
7	Myelopath*.tw
8	Myeloradiculopath*.tw
9	Exp radiculopathy/
10	Radiculopath*.tw
11	Exp Spinal Cord Compression/
12	Exp Posterior Longitudinal Ligament/ and exp ossification/
13	Ossification of the Posterior Longitudinal Ligament.tw
14	OPLL.tw
15	Exp vertebral canal stenosis/
16	Cervical stenosis.tw
17	Exp Cervical Spondylosis/
18	Exp Spondylosis/
19	Spondylosis.tw
20	Spondylotic.tw
21	Exp Cervical myelopathy/
22	Degenerative cervical myelopathy.tw
23	DCM.tw
24	Exp Cervical spondylotic myelopathy/
25	Cervical spondylotic myelopathy.tw
26	CSM.tw
27	6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26

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28	1 and 27
Tools for function concept (neurological/gait/mobility centric) concept	
29	Exp Japanese Orthopaedic Association score/
30	Japanese Orthopaedic Association score.tw
31	JOA.tw
32	modified JOA.tw
33	mJOA.tw
34	Exp “Disabilities of the Arm, Shoulder and Hand (score)”/
35	Graded Redefined Assessment of Sensation Strength and Prehension.tw
36	GRASSP.tw
37	Quick Disability of the Arm Shoulder and Hand.tw
38	QuickDASH.tw
39	Myelopathy Disability Index.tw
40	MDI.tw
41	Exp “Nurick (grade)”/
42	Nurick score.tw
43	Neck functional disability scale.tw
44	NFDS.tw
45	Exp Neck Disability Index/
46	Neck Disability Index.tw
47	NDI.tw
48	Cooper myelopathy scale.tw
49	CMS.tw
50	European myelopathy score.tw
51	EMS.tw
52	Bournemouth questionnaire.tw
53	BQ.tw
54	Cervical spine outcomes questionnaire.tw
55	CSOQ.tw
56	Patient specific functional scale.tw
57	PSFS.tw
58	World Health Organization Quality of Life Instruments.tw
59	WHOQOL.tw
60	Grip and release test.tw
61	GRT.tw
62	Grip Dynamometer.tw
63	Triangle step test.tw
64	Foot tapping test.tw

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65	30 m walking test.tw
66	30MWT.tw
67	10 m walking test.tw
68	10MWT.tw
69	Berg Balance Scale.tw
70	BBS.tw
71	GAITRite.tw
72	10 second step test.tw
73	9 hole peg test.tw
74	Prolo.tw
75	Mental component score.tw
76	MCS.tw
77	Physical component score.tw
78	PCS.tw
79	Hospital anxiety depression scale.tw
80	HADS.tw
81	Global rating of change.tw
82	GROC.tw
83	29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67 or 68 or 69 or 70 or 71 or 72 or 73 or 74 or 75 or 76 or 77 or 78 or 79 or 80 or 81 or 82
Tools for QOL concept (including pain) concept	
84	Short Form Health Survey.tw
85	Exp Short Form 36/
86	SF-36.tw
87	Exp Short Form 12/
88	SF-12.tw
89	Exp “European Quality of Life 5 Dimensions questionnaire”/
90	EQ-5D.tw
91	Exp Japanese Orthopaedic Association Cervical Myelopathy Evaluation/
92	Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire.tw
93	JOACMEQ.tw
94	Exp visual analog scale/
95	Visual Analogue Scale.tw
96	VAS.tw
97	Exp Likert scale/
98	Likert scale.tw

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99	Numeric pain rating scale.tw
100	NPRS.tw
101	North American Spine Satisfaction.tw
102	NASS.tw
103	84 or 85 or 86 or 87 or 88 or 89 or 90 or 91 or 92 or 93 or 94 or 95 or 96 or 97 or 98 or 99 or 100 or 101 or 102
Psychometric concept	
104	Exp Psychometry/
105	Pyschometr*.tw
106	(clinimetr* or clinometr*).tw.
107	Outcome assessment*.tw
108	exp Health Status Indicator/
109	Exp Reproducibility/
110	Reproducib*.tw
111	Exp Validation Study/
112	exp Discriminant Analysis/
113	(reliab* or unreliab* or valid* or coefficient or homogeneity or homogeneous or internal consistency).tw
114	(cronbach* and (alpha or alphas)).tw.
115	(item and (correlation* or selection* or reduction*)).tw
116	(agreement or precision or imprecision or precise values or test–retest).tw
117	(reliab* and (test or retest)).tw
118	(stability or interrater or inter-rater or intrarater or intra-rater or intertester or inter-tester or intratester or intra-tester or interobserver or inter-observer or intraobserver or intraobserver or intertechnician or inter-technician or intratechnician or intra-technician or interexaminer or inter-examiner or intraexaminer or intra-examiner or interassay or interassay or intraassay or intra-assay or interindividual or inter-individual or intraindividual or intra-individual or interparticipant or inter-participant or intraparticipant or intra-participant or kappa or kappas or repeatab*).tw
119	((replicab* or repeated) and (measure or measures or findings or result or results or test or tests)).tw
120	(generaliza* or generalisa* or concordance).tw
121	(intraclass and correlation*).tw
122	Exp Observer Variation/
123	Observer variation.tw
124	(multitrait and scaling and (analysis or analyses)).tw
125	Measurement error*.tw
126	(item discriminant or interscale correlation* or error or errors or individual variability).tw

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127	(variability and (analysis or values)).tw
128	(uncertainty and (measurement or measuring)).tw
129	Exp Diagnostic Error/
130	Exp Data accuracy/
131	Exp Dimensional Measurement Accuracy/
132	((minimal or minimally or clinical or clinically) and (important or significant or detectable) and (change or difference)).tw
133	Minimally clinically important difference*.tw
134	MCID.tw
135	(small* and (real or detectable) and (change or difference)).tw
136	(meaningful change or ceiling effect or floor effect or Item response model or IRT or Rasch or Differential item functioning or DIF or computer adaptive testing or item bank or cross-cultural equivalence).tw
137	Exp Bias/ or exp Selection Bias/
138	Bias.tw
139	104 or 105 or 106 or 107 or 108 or 109 or 110 or 111 or 112 or 113 or 114 or 115 or 116 or 117 or 118 or 119 or 120 or 121 or 122 or 123 or 124 or 125 or 126 or 127 or 128 or 129 or 130 or 131 or 132 or 133 or 134 or 135 or 136 or 137 or 138
Combined concepts	
140	83 or 103
141	28 and 139 and 140

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Supplementary Table 2. Study characteristics

Study	Country	Sample size	Psychometric properties	Outcome measures
Auffinger, Lall (1)	United States	30	MCID/SCB Measurement error	NDI VAS for pain SF-36
Augusto, Diniz (2)	Brazil	30	Cross-cultural validity/Measurement invariance Reliability Responsiveness	JOA NDI
Azimi, Rezaei (3)	Iran	87	Cross-cultural validity/Measurement invariance Responsiveness	JOACMEQ
Badhiwala, Witiw (4)	Canada	606	MCID/SCB	SF-36 mJOA Nurick Scale NDI
Bohm, Fehlings (5)	Multicenter/ Global	601	Reliability Hypotheses testing for construct validity Responsiveness	Walking tests (timed or steps) mJOA Nurick Scale NDI SF-36
Carreon, Glassman (6)	United States	505	MCID/SCB	NDI SF-36 "Numeric rating scale" for pain
Chang, Kong (7)	Korea	108	Reliability	CT / CTM
Chiba, Kato (8)	Japan		Reliability	X-rays
Chien, Lai (9)	Taiwan	45	Responsiveness MCID/SCB	JOACMEQ NDI
Chiu and Pang (10)	Hong Kong	72	Internal consistency Reliability Content validity Hypotheses testing for construct validity Criterion validity Measurement error	BBS mJOA

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			Responsiveness	
Fukui, Chiba (11)	Japan	368	Content validity	JOACMEQ
Fukui, Chiba (12)	Japan	201	Reliability	JOACMEQ
Goyal, Murphy (13)	United States	118	Responsiveness	NDI SF-12
Gwinn, Iannotti (14)	United States	20	Reliability	X-rays Cobb's method
Hosono, Sakaura (15)	Japan	30	Reliability Criterion validity	Grip-and-release test JOA
Hosono, Takenaka (16)	Japan	48	Responsiveness	Grip-and-release test JOA
Kang, Lee (17)	Korea	82	Reliability	MRI (not DTI)
Kato, Oshima (18)	Japan	92	Measurement error Hypotheses testing for construct validity Responsiveness	JOA mJOA JOACMEQ NDI SF-12
Kato, Oshima (19)	Japan	101	Measurement error Criterion validity MCID/SCB	JOA Likert scale
Kato, Oshima (20)	Japan	101	Measurement error Criterion validity MCID/SCB	JOACMEQ NDI EQ-5D SF-36 Likert scale
King and Roberts (21)	United States	88	Internal consistency	SF-36
Ko, Choi (22)	Korea	357	Reliability	MRI (not DTI)
Kopjar, Tetreault (23)	USA	277	Responsiveness Hypotheses testing for construct validity Internal consistency	mJOA Nurick Scale NDI SF-36 Walking tests (timed or steps)
Latimer, Haden (24)	England	70	Responsiveness	SF-36 NDI

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				VAS for pain MDI
Longo, Berton (25)	Italy	75	Cross-cultural validity/Measurement invariance Reliability Internal consistency Hypotheses testing for construct validity Responsiveness Criterion validity	mJOA Nurick Scale NDI SF-36
Lubelski, Alvin (26)	United States	119	Hypotheses testing for construct validity Responsiveness Criterion validity	mJOA Nurick Scale EQ-5D
Mihara, Kondo (27)	Japan	270	Hypotheses testing for construct validity	Grip-and-release test Triangle step test
Nakamoto, Oshima (28)	Japan	94	Internal consistency Hypotheses testing for construct validity Criterion validity	QuickDASH JOA NDI SF-36 "Numeric rating scale" for pain
Nakashima, Yukawa (29)	Japan	101	Hypotheses testing for construct validity	
Nicholson, Millhouse (30)	United States	235	Hypotheses testing for construct validity	MRI (not DTI) mJOA NDI SF-12 "Numeric rating scale" for pain Isihara's Cervical Curvature Index
Nikaido, Kikuchi (31)	Japan	87	Hypotheses testing for construct validity)	JOACMEQ SF-36
Numasawa, Ono (32)	Japan	126	Hypotheses testing for construct validity Responsiveness Reliability	JOA Foot tapping test Grip-and-release test
Olindo, Signate (33)	France	40	Reliability	9-Hole peg test MRI (not DTI) Nurick Scale mJOA Walking tests (timed or steps)

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Park, Kim (34)	Korea	100	Reliability	MRI (not DTI)
Pratali, Smith (35)	Brazil		Cross-cultural validity	mJOA
Pratali, Smith (36)	Brazil	55	Reliability	mJOA
Rhee, Shi (37)	United States	100	Criterion validity Reliability Content validity	mJOA
Sato, Horikoshi (38)	Japan	66	Hypotheses testing for construct validity	MRI (DTI) JOA
Shim, Lee (39)	Korea	79	Reliability Criterion validity	MRI (not DTI)
Singh and Crockard (40)	England	100	Internal consistency Responsiveness	Odom's Criteria Nurick Scale Ranawat classification of disease severity MDI JOA EMS SF-36
Singh and Crockard (41)	United Kingdom	41	Hypotheses testing for construct validity	Walking tests (timed or steps) MDI Nurick Scale
Singh, Gnanalingham (42)	England	105	Internal consistency Criterion validity Responsiveness	SF-12 SF-36
Spurgas, Abbas (43)	USA	35	MCID/SCB	VAS for pain NDI SF-12 mJOA
Tetreault, Nouri (44)	Canada	755	MCID/SCB	mJOA NDI
Thakar and Rajshekhar (45)	India	51	MCID/SCB Responsiveness	VAS for pain Nurick Scale SF-36
Thakar, Christopher (46)	India	70	Internal consistency Criterion validity	WHOQOL-Bref SF-36

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			Responsiveness MCID/SCB	Nurick Scale
Wada, Fukui (47)	Japan	137	Responsiveness	JOACMEQ JOA 10-s step test
Witayakom, Paholpak (48)	Thailand	70	Cross-cultural validity/Measurement invariance Reliability Internal consistency Hypotheses testing for construct validity	JOACMEQ SF-36
Yonenobu, Abumi (49)	Japan	29	Reliability	JOA
Yukawa, Kato (50)	Japan	163	Hypotheses testing for construct validity Reliability Criterion validity	10-s step test JOA Grip-and-release test
Zhang, Zhou (51)	China	142	Internal consistency Responsiveness MCID/SCB	SF-36 mJOA
Zhou, Zhang (52)	China	113	MCID/SCB Measurement error	mJOA SF-36

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Supplementary Table 3. Interpretability (i.e., MCID and SCB).

Instrument	Result summary	Overall rating
EQ-5D	MCID: 0.05; total sample size: 101	Sufficient
JOA	MCID: 2.5; total sample size: 101	Sufficient
JOACMEQ		
Bladder function	MCID: 6.0; total sample size: 78	Sufficient
Cervical spine function	MCID: 2.5; total sample size: 179	Sufficient
Lower extremity function	MCID range 2.5–9.4; total sample size: 179	Sufficient
QOL	MCID range 8.5–9.5; total sample size: 179	Sufficient
Upper extremity function	MCID range 2.5–13.0; total sample size: 179	Sufficient
mJOA	MCID range 1.3–3.1; total sample size: 868	Sufficient
	SCB: 14; total sample size: 35	Indeterminate
NDI	MCID range 5–13; total sample size: 108	Sufficient
	SCB range 9.5–36; total sample size: 65	Indeterminate
Pain, "Numeric rating scale" (Arm pain)	MCID: 2.5; total sample size: 30 SCB: 3.5; total sample size: 30	Indeterminate
Pain, "Numeric rating scale" (Neck pain)	MCID: 2.5; total sample size: 30 SCB: 3.5; total sample size: 30	Indeterminate
SF-12		

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	MCS	SCB: 51.5; total sample size: 35	Indeterminate
	PCS	SCB: 30.1; total sample size: 35	Indeterminate
SF-36			
	MCS	MCID range 3.0–7.4; total sample size: 749	Sufficient
	PCS	"MCID range 3.9–9.6; total sample size: 890 SCB: 16; total sample size: 30"	Sufficient
VAS for pain		MCID range 0.4–2.7; total sample size: 30	Sufficient
		SCB: 1.1; total sample size: 30	Indeterminate

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Supplementary Table 4. Feasibility assessment.

Tool	Time (min)	Equipment	Training	License	Money	Ease of administration	Overall assessment
10-s step test	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
30MWT	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
9-Hole peg test	5	Yes	No	No	No	Barriers	Barriers
Berg Balance Scale	>15	Yes	Yes	No	No	Barriers	Barriers
Cobb's method (C2-C7)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
CT (Tsuyama's classification, 2D & 3D)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
CT (Tsuyama's classification, lateral + axial)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
EQ-5D	5	Minimal	No	Yes	Yes	Minimal barriers	Minimal barriers
European Myelopathy Scale	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
Foot tapping test	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
Grip-and-release test	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
Isihara's Cervical Curvature Index	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
JOA	5	No	No	No	No	No barriers	No barriers

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JOACMEQ	5–15	Minimal	No	No	No	Minimal barriers	Minimal barriers
MDI	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
mJOA	5	No	No	No	No	No barriers	No barriers
MRI (Depiction of intramedullary hyperintensity at eight cervical disc levels, T2W, 1.5-T or 3-T)	5–15	Minimal	No	No	No	Minimal barriers	Minimal barriers
MRI (Kang's classification, 1.5-T or 3-T)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
MRI (Muhle's classification, 1.5-T)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
MRI (Vaccaro's classification, 1.5-T)	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
NDI	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
Nurick scale	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
P-mJOA	5	Minimal	No	No	No	Minimal barriers	Minimal barriers

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Arm pain score	5	No	No	No	No	No barriers	No barriers
Neck pain score	5	No	No	No	No	No barriers	No barriers
QuickDASH	5	Minimal	No	Yes	Yes	Minimal barriers	Barriers
Ranawat classification of disease severity	5	No	No	No	No	No barriers	No barriers
SF-12	5	Minimal	No	Yes	Yes	Minimal barriers	Barriers
SF-36	5–15	Minimal	No	Yes	Yes	Minimal barriers	Barriers
Triangle step test	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
VAS for pain	5	Minimal	No	No	No	Minimal barriers	Minimal barriers
WHOQOL-Bref	5–15	Minimal	No	No	No	Minimal barriers	Minimal barriers
X-rays (Computer-assisted measurement of length & thickness)	5–15	Minimal	No	No	No	Minimal barriers	Minimal barriers

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Supplementary Table 5. Content validity.

Instrument	Result summary	Overall rating	Quality of evidence
BBS	Patient comprehensibility: Item discrimination index >0.589	Indeterminate	Very low
JOACMEQ	Patient comprehensibility: "No questions elicited no answer or “I am not sure” in more than 5% of patients"	Indeterminate	Very low
P-mJOA	Patient comprehensibility: "In patients preferring to complete the mJOA them- selves, the most popular answers were: “ease of answering the questions” (n = 33), “understanding of the questions” (n = 17)"	Indeterminate	Very low

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Supplementary Table 6. Internal consistency.

Instrument	Result summary	Overall rating	Quality of evidence
BBS	Cronbach's alpha range 0.95–0.98; consistent; total sample size: 72	Indeterminate	Moderate
European Myelopathy Scale	Cronbach's alpha: 0.68; consistent; total sample size: 100	Indeterminate	Low
JOA	Cronbach's alpha: 0.72; consistent; total sample size: 100	Indeterminate	Low
JOACMEQ	Cronbach's alpha: 0.91; total sample size: 70	Indeterminate	Moderate
Bladder function	Cronbach's alpha range 0.77–0.78; consistent; total sample size: 157	Indeterminate	High
Cervical spine function	Cronbach's alpha range 0.75–0.88; consistent; total sample size: 157	Indeterminate	High
QOL	Cronbach's alpha range 0.80–0.86; consistent; total sample size: 157	Indeterminate	High
Upper extremity function	Cronbach's alpha range 0.72–0.74; consistent; total sample size: 157	Indeterminate	High
MDI	Cronbach's alpha: 0.92; consistent; total sample size: 100	Indeterminate	Low
mJOA	Cronbach's alpha range 0.60–0.63; consistent; total sample size: 352	Indeterminate	High
QuickDASH	Cronbach's alpha: 0.94; consistent; total sample size: 94	Indeterminate	Very low
SF-12	Cronbach's alpha: 0.77; consistent; total sample size: 105	Indeterminate	n/a

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SF-36	Cronbach's alpha range 0.79–0.93; consistent; total sample size: 473	Indeterminate	n/a
WHOQOL-Bref	Cronbach's alpha range 0.86–0.87; consistent; total sample size: 38	Indeterminate	n/a

n/a = No info available

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Supplementary Table 7. Cross-cultural validity.

Instrument	Result summary	Overall rating	Quality of evidence
JOA	Forward-backward translation [Brazilian Portuguese] Comprehension rate: >81.2%	Indeterminate	Very low
JOACMEQ	Forward-backward translation [Persian and Thai] No info available	Indeterminate	Very low
mJOA	Forward-backward translation [Brazilian Portuguese and Italian] No info available	Indeterminate	Very low

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Supplementary Table 8. Reliability.

Instrument	Result summary	Overall rating	Quality of evidence
10-s step test	Test–retest stability: Spearman's rank correlation: 0.89; total sample size: 163	Indeterminate	Low
30MWT	Test–retest stability: Pearson's correlation range 0.89–1.00; total sample size: 16	Indeterminate	Very low
9-Hole peg test	Intra-observer reliability: ICC range 0.97–0.98; consistent; total sample size: 41 Inter-observer reliability: ICC range 0.97–0.99; consistent; total sample size: 41	Sufficient	Very low
BBS*	Test–retest stability: ICC: 0.99; total sample size: 32 Inter-observer reliability: ICC: 0.99; total sample size: 32	Sufficient	Very low
	Test–retest stability: Kappa: 0.67; total sample size: 32 Inter-observer reliability: Kappa: 0.43; total sample size: 32	Insufficient	Very low
Cobb's method	Intra-observer reliability: ICC: 0.84; total sample size: 20 Inter-observer reliability: ICC: 0.77; total sample size: 20	Sufficient	Very low
CT	Intra-observer reliability:	Sufficient	Moderate

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(Tsuyama's classification, 2D & 3D)	Kappa range 0.85–0.86; consistent; total sample size: 108 Inter-observer reliability: Kappa range 0.71–0.76; consistent; total sample size: 108		
CT (Tsuyama's classification, lateral + axial)	Intra-observer reliability: Kappa: 0.67; total sample size: 108 Inter-observer reliability: Kappa: 0.51; total sample size: 108	Insufficient	Moderate
Foot tapping test	Test–retest stability: Pearson's correlation range 0.90–0.93; total sample size: 126	Indeterminate	Low
Grip-and-release test	Inter-observer reliability: ICC: 0.99; total sample size: 30	Sufficient	Very low
JOA	Inter-observer reliability: ICC: 0.81; total sample size: 29	Sufficient	Very low
Bladder function	Intra-observer reliability: Kappa: 0.64; total sample size: 29 Inter-observer reliability: Kappa: 0.47; total sample size: 29	Insufficient	Very low
Motor function of fingers	Intra-observer reliability: Kappa: 0.68; total sample size: 29 Inter-observer reliability: Kappa: 0.53; total sample size: 29	Insufficient	Very low
Motor function of	Intra-observer reliability: Kappa: 0.50; total sample size: 29	Insufficient	Very low

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	shoulder and elbow	Inter-observer reliability: Kappa: 0.31; total sample size: 29		
	Motor function of lower extremity	Intra-observer reliability: Kappa: 0.55; total sample size: 29 Inter-observer reliability: Kappa: 0.49; total sample size: 29	Insufficient	Very low
	Sensory function of lower extremity	Intra-observer reliability: Kappa: 0.44; total sample size: 29 Inter-observer reliability: Kappa: 0.34; total sample size: 29	Insufficient	Very low
	Sensory function of trunk	Intra-observer reliability: Kappa: 0.54; total sample size: 29 Inter-observer reliability: Kappa: 0.58; total sample size: 29	Insufficient	Very low
	Sensory function of upper extremity	Intra-observer reliability: Kappa: 0.51; total sample size: 29 Inter-observer reliability: Kappa: 0.42; total sample size: 29	Insufficient	Very low
JOACMEQ				
	Bladder function	Test–retest stability: ICC: 0.62; total sample size: 70	Insufficient	Very low
	Cervical spine function	Test–retest stability: ICC: 0.63; total sample size: 70	Insufficient	Very low
	Lower extremity function	Test–retest stability: ICC: 0.93; total sample size: 70	Sufficient	Very low
	QOL	Test–retest stability:	Sufficient	Very low

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		ICC: 0.83; total sample size: 70		
	Upper extremity function	Test–retest stability: ICC: 0.93; total sample size: 70	Sufficient	Very low
mJOA		Test–retest stability: Spearman's rank correlation: 0.91; total sample size: 75	Indeterminate	Very low
		Intra-observer reliability: ICC: 0.87; total sample size: 55	Sufficient	Very low
		Inter-observer reliability: ICC: 0.97; total sample size: 55 Kappa: 0.80; total sample size: 75	Sufficient	Low
	Motor dysfunction of lower extremities	Inter-observer reliability: ICC: 0.73; total sample size: 75	Sufficient	Low
	Motor dysfunction of upper extremities	Inter-observer reliability: ICC: 0.77; total sample size: 75	Sufficient	Low
	Sensory dysfunction of sphincter dysfunction	Inter-observer reliability: ICC: 0.78; total sample size: 75	Sufficient	Low
	Sensory dysfunction of upper extremities	Inter-observer reliability: ICC: 0.93; total sample size: 75	Sufficient	Low
MRI (Depiction of intramedullary hyperintensity at eight cervical disc		Inter-observer reliability: Kendall's W range 0.72–0.78; total sample size: 79	Indeterminate	Very low

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levels, T2W, 1.5-T or 3-T)			
MRI (Kang's classification, 1.5-T or 3-T)	Intra-observer reliability: Kappa: 0.67; total sample size: 439 ICC: 0.77, total sample size: 82 Inter-observer reliability: Kappa range 0.60–0.93; total sample size: 539 ICC range 0.74–0.75; total sample size: 82	Inconsistent	n/a
MRI (Muhle's classification, 1.5-T)	Intra-observer reliability: Kappa: 0.72; total sample size: 357 Inter-observer reliability: Kappa range 0.61; total sample size: 357	Inconsistent	n/a
MRI (Vaccaro's classification, 1.5-T)	Intra-observer reliability: Kappa: 0.71; total sample size: 357 Inter-observer reliability: Kappa range 0.69; total sample size: 357	Sufficient	Moderate
P-mJOA			
Motor dysfunction of lower extremities	Inter-observer reliability: Kappa: 0.61; total sample size: 755	Insufficient	Moderate
Motor dysfunction of upper extremities	Inter-observer reliability: Kappa: 0.66; total sample size: 755	Insufficient	Moderate
Sensory dysfunction of sphincter dysfunction	Inter-observer reliability: Kappa: 0.55; total sample size: 755	Insufficient	Moderate

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Sensory dysfunction of upper extremities	Inter-observer reliability: Kappa: 0.55; total sample size: 755	Insufficient	Moderate
X-rays (Computer-assisted measurement of length)	Intra-observer reliability: ICC: 0.94; total sample size: 9 Inter-observer reliability: ICC: 0.93; total sample size: 9	Sufficient	Very low
X-rays (Computer-assisted measurement of thickness)	Intra-observer reliability: ICC: 0.96; total sample size: 9 Inter-observer reliability: ICC: 0.97; total sample size: 9	Sufficient	Very low

*Result ratings for BBS were split by statistic used due to their associated differences in sufficiency.

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Supplementary Table 9. Measurement error.

Instrument	Result summary	Overall rating	Quality of evidence
BBS	MDC or SDC Distribution: 1.5; total sample size: 32	Indeterminate	n/a
EQ-5D	MDC or SDC Distribution: 0.13; total sample size: 101 Anchor: 0.04; total sample size: 101	Inconsistent	n/a
JOA	MDC or SDC Distribution: 1.0; total sample size: 101 Anchor: 2.5; total sample size: 101 LOA 1.2 (–1.2, 3.6); total sample size: 92	Sufficient	Very low
JOACMEQ			
Bladder function	MDC or SDC Distribution: 7.7; total sample size: 101	Insufficient	Very low
Cervical spine function	MDC or SDC Distribution: 12.9; total sample size: 101 Anchor: 12.5; total sample size: 101	Insufficient	Very low
Lower extremity function	MDC or SDC Distribution: 7.3; total sample size: 101 Anchor: 9.4; total sample size: 101	Inconsistent	n/a
QOL	MDC or SDC Distribution: 6.6; total sample size: 101 Anchor: 8.5; total sample size: 101	Sufficient	Very low

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	Upper extremity function	MDC or SDC Distribution: 9.5; total sample size: 101 Anchor: 6.1; total sample size: 101	Sufficient	Very low
mJOA		MDC or SDC Distribution: 2.1; total sample size: 113	Inconsistent	Very low
		MCID range; total sample size: 868 Distribution: 1.2–1.4	Sufficient	High
NDI		MDC or SDC Distribution: 6.2%; total sample size: 101 Anchor: 5.2%; total sample size: 101	Insufficient	Very low
SF-36				
	MCS	MDC or SDC Distribution: 3.3–5.7; total sample size: 244	Inconsistent	n/a
		MCID; total sample size: 748 Distribution: 3.4–6.8	Inconsistent	n/a
	PCS	MDC or SDC Distribution: 5.2–5.7; total sample size: 214 Anchor: 4.9; total sample size: 101	Inconsistent	n/a
		MCID range; total sample size: 861 Distribution: 2.9–5.5	Inconsistent	n/a
		MCID; total sample size: 51 Distribution: 10		
VAS for pain		MDC or SDC Distribution: 3.1; total sample size: 30	Insufficient	Very low

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	MCID range 24.0–30.0; total sample size: 51	Insufficient	Very low
WHOQOL-Bref			
PH	MCID Distribution: 8.2; total sample size: 38	Indeterminate	n/a
PS	MCID Distribution: 7.9; total sample size: 38	Indeterminate	n/a
SR	MCID Distribution: 8.0; total sample size: 38	Indeterminate	n/a
EN	MCID Distribution: 5.6; total sample size: 38	Indeterminate	n/a
PF	MCID Distribution: 10.5; total sample size: 38	Indeterminate	n/a
RP	MCID Distribution: 17.2; total sample size: 38	Indeterminate	n/a
BP	MCID Distribution: 13.2; total sample size: 38	Indeterminate	n/a
GH	MCID Distribution: 12.3; total sample size: 38	Indeterminate	n/a
VT	MCID Distribution: 10.8; total sample size: 38	Indeterminate	n/a
SF	MCID Distribution: 13.6; total sample size: 38	Indeterminate	n/a
RE	MCID Distribution: 18.0; total sample size: 38	Indeterminate	n/a
MH	MCID	Indeterminate	n/a

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	Distribution: 11.2; total sample size: 38

n/a = No info available

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Supplementary Table 10. Criterion validity.

Instrument	Result summary*	Overall rating	Quality of evidence
10-s step test	JOA Spearman's rank correlation: 0.66; total sample size: 163	Insufficient	High
BBS	mJOA AUC range 0.88–0.94; total sample size: 31	Sufficient	Low
Foot tapping test	JOA Pearson's correlation: 0.66; total sample size: 126 JOA MFLE Pearson's correlation: 0.70; total sample size: 126	Insufficient	High
Grip-and-release test	JOA Pearson's correlation: 0.72; total sample size: 30	Sufficient	Low
JOA	Likert scale, "Health transition question" AUC: 0.59; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.62; total sample size: 101	Insufficient	Very low
JOACMEQ			
Cervical spine function	Likert scale, "Health transition question" AUC: 0.58; total sample size: 101 Likert scale, "Patient satisfaction question"	Insufficient	Very low

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		AUC: 0.58; total sample size: 101		
	Upper extremity function	Likert scale, "Health transition question" AUC: 0.66; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.65; total sample size: 101	Insufficient	Very low
	Lower extremity function	Likert scale, "Health transition question" AUC: 0.61; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.66; total sample size: 101	Insufficient	Very low
	QOL	Likert scale, "Health transition question" AUC: 0.70; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.66; total sample size: 101	Insufficient	Very low
mJOA		Nurick scale [convergent] Spearman's rank correlation: –0.41; total sample size: 119 Pearson's correlation range: –0.62 to –0.63; total sample size: 352	Sufficient	High
	Motor dysfunction of upper extremities	Nurick scale [convergent] Pearson's correlation range –0.42 to –0.42; total sample size: 352	Insufficient	High

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	Motor dysfunction of lower extremities	Nurick scale [convergent] Pearson's correlation: –0.65 to –0.68; total sample size: 352	Sufficient	High
	Sensory dysfunction of upper extremities	Nurick scale [convergent] Pearson's correlation: –0.23; total sample size: 277	Insufficient	High
	Sensory dysfunction of sphincter dysfunction	Nurick scale [convergent] Pearson's correlation: –0.25; total sample size: 277	Insufficient	High
NDI		Likert scale, "Health transition question" AUC: 0.66; total sample size: 101 Likert scale, "Patient satisfaction question" AUC: 0.75; total sample size: 101	Inconsistent	n/a
P-mJOA		mJOA Spearman's rank correlation: 0.83; total sample size: 755	Sufficient	High
QuickDASH		JOA MFSE Spearman's rank correlation: –0.50; total sample size: 94 JOA SFUE Spearman's rank correlation: –0.32; total sample size: 94	Insufficient	Moderate
SF-36				
	PCS	Likert scale, "Health transition question" AUC: 0.67; total sample size: 101	Insufficient	Very low

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	Likert scale, "Patient satisfaction question" AUC: 0.69; total sample size: 101		
WHOQOL-Bref			
PH	SF-36 PCS Pearson's correlation: 0.51; total sample size: 38 SF-36 MCS Pearson's correlation: 0.30; total sample size: 38	Inconsistent	n/a
PS	SF-36 PCS Pearson's correlation: 0.34; total sample size: 38 SF-36 MCS Pearson's correlation: 0.23; total sample size: 38	Insufficient	Low
SR	SF-36 PCS Pearson's correlation: 0.35; total sample size: 38 SF-36 MCS Pearson's correlation: 0.28; total sample size: 38	Insufficient	Low
EN	SF-36 PCS Pearson's correlation: 0.05; total sample size: 38 SF-36 MCS Pearson's correlation: 0.03; total sample size: 38	Insufficient	Low

n/a = No info available

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*Instruments listed are comparators

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Supplementary Table 11. Construct validity.

Instrument	Result summary*	Overall rating	Quality of evidence
10-s step test	Grip-and-release test [convergent] Spearman's rank correlation: 0.53; total sample size: 163	Sufficient	Moderate
30MWT	mJOA [convergent] Pearson's correlation: –0.44; total sample size: 16 MDI [convergent] Spearman's rank correlation: 0.65; total sample size: 41 Nurick scale [convergent] Pearson's correlation: 0.50; total sample size: 16 Spearman's rank correlation: 0.61; total sample size: 41	Sufficient	Moderate
	NDI Pearson's correlation: 0.21; total sample size: 16	Sufficient	Low
	SF-36 PCS Pearson's correlation: –0.35; total sample size: 16	Sufficient	Low
	SF-36 MCS Pearson's correlation: –0.20; total sample size: 16	Sufficient	Low
BBS	mJOA [convergent] Spearman's rank correlation: 0.81; total sample size: 72	Sufficient	Moderate

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EQ-5D	mJOA AUC: 0.68; total sample size: 119 Nurick scale AUC: 0.61; total sample size: 119	Insufficient	High
Foot tapping test	Grip-and-release test [convergent] Pearson's correlation: 0.58; total sample size: 126	Sufficient	High
Isihara's Cervical Curvature Index	mJOA Pearson's correlation: 0.04; total sample size: 235	Sufficient	High
	SF-12 PCS Pearson's correlation: 0.06; total sample size: 235 SF-12 MCS Pearson's correlation: 0.11; total sample size: 235	Sufficient	High
	Pain, "Numeric rating scale" (Arm pain score) Pearson's correlation: –0.28; total sample size: 235 Pain, "Numeric rating scale" (Neck pain scores) Pearson's correlation: –0.27; total sample size: 235 NDI Pearson's correlation: –0.10; total sample size: 235	Sufficient	High
JOA	mJOA [convergent]	Sufficient	Low

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		Spearman's rank correlation: 0.87; total sample size: 92		
		JOACMEQ QOL [convergent] Spearman's rank correlation: 0.41; total sample size: 92	Sufficient	Low
		SF-12 PCS Spearman's rank correlation: 0.50; total sample size: 92	Sufficient	Low
		SF-12 MCS Spearman's rank correlation: –0.05; total sample size: 92	Sufficient	Low
		NDI Spearman's rank correlation range –0.50 to –0.76; total sample size: 122	Sufficient	Moderate
JOACMEQ				
	QOL	NDI Spearman's rank correlation: –0.66; total sample size: 92	Sufficient	Low
	QOL	SF-12 PCS Spearman's rank correlation: 0.29; total sample size: 92 SF-12 MCS Spearman's rank correlation: 0.40; total sample size: 92	Insufficient	Low
MDI		Nurick scale [convergent] Spearman's rank correlation: 0.66; total sample size: 41	Sufficient	Low

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mJOA		30MWT [convergent] Pearson's correlation: –0.38; total sample size: 193	Insufficient	High
		JOACMEQ QOL [convergent] Spearman's rank correlation: 0.41; total sample size: 92	Insufficient	Low
		EQ-5D Spearman's rank correlation: 0.42; total sample size: 119	Insufficient	High
		SF-36 PCS Pearson's correlation range: 0.30–0.30; total sample size: 338 SF-12 PCS Spearman's rank correlation: 0.47; total sample size: 92	Sufficient	High
		SF-36 MCS Pearson's correlation: 0.25–0.25; total sample size: 338 SF-12 MCS Spearman's rank correlation: 0.03; total sample size: 92	Sufficient	High
		NDI Spearman's rank correlation: –0.51; total sample size: 92 Pearson's correlation rage –0.33 to –0.34; total sample size: 336	Sufficient	High
	Motor dysfunction	30MWT [convergent]	Insufficient	High

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	of lower extremities	Pearson's correlation: –0.43; total sample size: 193		
		SF-36 PCS Pearson's correlation range: 0.31–0.50; total sample size: 338	Sufficient	High
		SF-36 MCS Pearson's correlation: 0.21; total sample size: 268	Sufficient	High
		NDI Pearson's correlation: –0.31; total sample size: 261	Sufficient	High
	Motor dysfunction of upper extremities	30MWT [convergent] Pearson's correlation: –0.21; total sample size: 193	Insufficient	High
		SF-36 PCS Pearson's correlation: 0.22; total sample size: 268	Insufficient	High
		SF-36 MCS Pearson's correlation: 0.20; total sample size: 268	Sufficient	High
		NDI Pearson's correlation: –0.24; total sample size: 261	Sufficient	High
	Sensory dysfunction of sphincter dysfunction	30MWT [convergent] Pearson's correlation: –0.23; total sample size: 193	Insufficient	High
		SF-36 PCS	Sufficient	High

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		Pearson's correlation: 0.06; total sample size: 268 SF-36 MCS Pearson's correlation: 0.08; total sample size: 268		
		NDI Pearson's correlation: –0.16; total sample size: 261	Sufficient	High
	Sensory dysfunction of upper extremities	30MWT [convergent] Pearson's correlation: –0.05; total sample size: 193	Insufficient	High
		SF-36 PCS Pearson's correlation: 0.19; total sample size: 268 SF-36 MCS Pearson's correlation: 0.19; total sample size: 268	Sufficient	High
		NDI Pearson's correlation: –0.23; total sample size: 261	Insufficient	High
NDI		mJOA Pearson's correlation: –0.36; total sample size: 235	Sufficient	High
		SF-12 PCS Pearson's correlation: –0.54; total sample size: 235 SF-12 MCS	Sufficient	High

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	Pearson's correlation: –0.40; total sample size: 235		
	Pain, "Numeric rating scale" (Arm pain score) Pearson's correlation: 0.68; total sample size: 235 Pain, "Numeric rating scale" (Neck pain scores) Pearson's correlation: 0.64; total sample size: 235	Sufficient	High
Nurick scale	EQ-5D Spearman's rank correlation: –0.28; total sample size: 119	Sufficient	High
Pain, "Numeric rating scale" (Arm pain scores)	mJOA Pearson's correlation: –0.19; total sample size: 235	Sufficient	High
	Pain, “Numeric rating scale” (Neck pain score) [convergent] Pearson’s correlation: 0.72; total sample size: 235	Sufficient	High
Pain, "Numeric rating scale" (Neck pain scores)	mJOA Pearson's correlation: –0.07; total sample size: 235	Sufficient	High
QuickDASH	SF-36 Spearman's rank correlation: –0.75; total sample size: 94	Sufficient	Moderate
	NDI and Pain, "Numeric rating scale" [convergent]	Sufficient	Moderate

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		Spearman's rank correlation range 0.69–0.83; total sample size: 94		
SF-12				
	MCS	mJOA Pearson's correlation: 0.19; total sample size: 235	Sufficient	High
		Pain, "Numeric rating scale" (Arm pain score) Pearson's correlation: –0.23; total sample size: 235 Pain, "Numeric rating scale" (Neck pain score) Pearson's correlation: –0.28; total sample size: 235	Sufficient	High
		NDI Spearman's rank correlation: –0.17; total sample size: 92	Sufficient	Moderate
		SF-12 PCS Pearson's correlation: 0.01; total sample size: 235	Sufficient	High
	PCS	mJOA Pearson's correlation: 0.43; total sample size: 235	Sufficient	High
		Pain, "Numeric rating scale" (Arm pain score) Pearson's correlation: –0.44; total sample size: 235 Pain, "Numeric rating scale" (Neck pain score)	Sufficient	High

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	Pearson's correlation: –0.41; total sample size: 235		
	NDI Spearman's rank correlation: –0.49; total sample size: 92	Sufficient	Moderate
	SF-12 MCS Spearman's rank correlation: –0.29; total sample size: 92	Sufficient	Low
Triangle step test	Grip-and-release test [convergent] Spearman's rank correlation: 0.55; total sample size: 270	Sufficient	High

*Instruments listed are comparators

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Supplementary Table 12. Responsiveness.

Instrument	Result summary*	Overall rating	Quality of evidence
30MWT	30MWT SRM: 0.3; total sample size: 484	Insufficient	High
BBS	mJOA Sensitivity range 77.4-80.0; total sample size: 31 Specificity range 87.8–92.9; total sample size: 31	Sufficient	Low
EQ-5D	EQ-5D Mean change score: 0.06; total sample size: 108	Indeterminate	High
European Myelopathy Scale	EMS Normalised change: 0.18; total sample size: 99	Indeterminate	Very low
Foot tapping test	Foot tapping test Mean change score: 6; total sample size: 6	Indeterminate	Very low
Grip-and-release test	Grip-and-release test Spearman's rank correlation: 0.69; total sample size: 48	Sufficient	Very low
	JOA Spearman's rank correlation: 0.32; total sample size: 48	Insufficient	Low
JOA	mJOA Spearman's rank correlation: 0.75; total sample size: 92	Sufficient	Very low
	JOA	Indeterminate	Very low

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		Mean change score range 4.6; total sample size: 126 Normalised change: 0.21; total sample size: 99 JOA MFLE Mean change score range 0.6; total sample size: 126		
JOACMEQ				
	Bladder function	JOACMEQ BF AUC: 0.82; total sample size: 78	Sufficient	Moderate
		JOACMEQ BF Mean change score: 18.0; total sample size: 87	Indeterminate	Very low
		JOACMEQ BF Effect size: 0.33; total sample size: 78	Insufficient	Moderate
	Cervical spine function	JOACMEQ CF AUC: 0.72; total sample size: 78	Sufficient	Moderate
		JOACMEQ CF Mean change score: 25.8; total sample size: 87	Indeterminate	Very low
		JOACMEQ CF Effect size: 0.28; total sample size: 78	Insufficient	Moderate
	Lower extremity function	JOACMEQ LEF AUC: 0.75; total sample size: 78	Sufficient	Moderate
		JOACMEQ LEF Mean change score: 28.4; total sample size: 87	Indeterminate	Very low

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		JOACMEQ LEF Effect size: 0.02; total sample size: 78	Insufficient	Moderate
	Upper extremity function	JOACMEQ UEF AUC: 0.74; total sample size: 78	Sufficient	Moderate
		JOACMEQ UEF Mean change score: 10.7; total sample size: 87	Indeterminate	Very low
		JOACMEQ UEF Effect size: 0.17; total sample size: 78	Insufficient	Moderate
	QOL	JOACMEQ QOL AUC: 0.83; total sample size: 78	Sufficient	Moderate
		JOACMEQ QOL Mean change score: 23.7; total sample size: 87	Indeterminate	Very low
		JOACMEQ QOL Effect size: 0.46; total sample size: 78	Insufficient	Moderate
MDI		MDI Normalised change: 0.52; total sample size: 99	Indeterminate	Very low
mJOA		mJOA Effect size range 0.87–1.0; total sample size: 352	Sufficient	High
		mJOA Normalised change: 1.47; total sample size: 42	Indeterminate	Very low
NDI		Anchor-based approach AUC: 0.66; total sample size: 78	Insufficient	Moderate

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	Effect size: 0.44; total sample size: 78		
	NDI Mean change score: –15.8; total sample size: 118	Indeterminate	Very low
Nurick scale	Nurick scale Normalised change: 0.42; total sample size: 99 Mean change score range –0.76 to –1.3; total sample size: 93	Indeterminate	Very low
Ranawat classification of disease severity	Ranawat classification of disease severity Normalised change: 0.34; total sample size: 99	Indeterminate	Very low
SF-12			
PCS	SF-12 PCS Mean change score: 8.17; total sample size: 118	Indeterminate	Very low
SF-36	SF-36 Normalised change: 0.32; total sample size: 99	Indeterminate	Very low
PCS	SF-36 PCS Effect size range: 0.84; total sample size: 142	Sufficient	Low
	SF-36 PCS Sensitivity: 0.85; total sample size: 105	Sufficient	Moderate
MCS	SF-36 MCS Effect size range: 0.81; total sample size: 142	Sufficient	Low
	SF-36 MCS	Sufficient	Moderate

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	Sensitivity: 0.67; total sample size: 105		
WHOQOL-Bref			
PH	WHOQOL-Bref PH Effect size: 0.68; total sample size: 38	Insufficient	Low
PS	WHOQOL-Bref PS Effect size: 0.39; total sample size: 38	Insufficient	Low
SR	WHOQOL-Bref SR Effect size: 0.03; total sample size: 38	Insufficient	Low
EN	WHOQOL-Bref EN Effect size: 0.45; total sample size: 38	Insufficient	Low

*Instruments listed are comparators

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REFERENCES

1. Auffinger BM, Lall RR, Dahdaleh NS, Wong AP, Lam SK, Koski T, et al. Measuring surgical outcomes in cervical spondylotic myelopathy patients undergoing anterior cervical discectomy and fusion: assessment of minimum clinically important difference. PLoS One. 2013;8(6):e67408.
2. Augusto MT, Diniz JM, Dantas FLR, Oliveira MF, Rotta JM, Botelho RV. Development of the Portuguese Version of the Modified Japanese Orthopaedic Association Score: Cross-Cultural Adaptation, Reliability, Validity, and Responsiveness. World Neurosurg. 2018;116:e1092-e7.
3. Azimi P, Rezaei O, Montazeri A. An outcome measure of functionality and quality of life in patients with cervical myelopathy. Iran Red Crescent Med J. 2014;16(6):e8102.
4. Badhiwala JH, Witiw CD, Nassiri F, Akbar MA, Jaja B, Wilson JR, et al. Minimum Clinically Important Difference in SF-36 Scores for Use in Degenerative Cervical Myelopathy. Spine (Phila Pa 1976). 2018;43(21):E1260-e6.
5. Bohm PE, Fehlings MG, Kopjar B, Tetreault LA, Vaccaro AR, Anderson KK, et al. Psychometric properties of the 30-m walking test in patients with degenerative cervical myelopathy: results from two prospective multicenter cohort studies. Spine J. 2017;17(2):211-7.
6. Carreon LY, Glassman SD, Campbell MJ, Anderson PA. Neck Disability Index, short form-36 physical component summary, and pain scales for neck and arm pain: the minimum clinically important difference and substantial clinical benefit after cervical spine fusion. Spine J. 2010;10(6):469-74.
7. Chang H, Kong CG, Won HY, Kim JH, Park JB. Inter- and intra-observer variability of a cervical OPLL classification using reconstructed CT images. Clin Orthop Surg. 2010;2(1):8-12.
8. Chiba K, Kato Y, Tsuzuki N, Nagata K, Toyama Y, Iwasaki M, et al. Computer-assisted measurement of the size of ossification in patients with ossification of the posterior longitudinal ligament in the cervical spine. J Orthop Sci. 2005;10(5):451-6.
9. Chien A, Lai DM, Cheng CH, Wang SF, Hsu WL, Wang JL. Responsiveness of the Chinese versions of the Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire and Neck Disability Index in postoperative patients with cervical spondylotic myelopathy. Spine (Phila Pa 1976). 2015;40(17):1315-21.
10. Chiu AY, Pang MY. Assessment of psychometric properties of various balance assessment tools in persons with cervical spondylotic myelopathy. journal of orthopaedic & sports physical therapy. 2017;47(9):673-82.
11. Fukui M, Chiba K, Kawakami M, Kikuchi S, Konno S, Miyamoto M, et al. Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire (JOACMEQ): Part 2. Endorsement of the alternative item. J Orthop Sci. 2007;12(3):241-8.
12. Fukui M, Chiba K, Kawakami M, Kikuchi S, Konno S, Miyamoto M, et al. Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire: part 3. Determination of reliability. J Orthop Sci. 2007;12(4):321-6.
13. Goyal DKC, Murphy HA, Hollern DA, Divi SN, Nicholson K, Stawicki C, et al. Is the Neck Disability Index an Appropriate Measure for Changes in Physical Function After Surgery for Cervical Spondylotic Myelopathy? Int J Spine Surg. 2020;14(1):53-8.
14. Gwinn DE, Iannotti CA, Benzel EC, Steinmetz MP. Effective lordosis: analysis of sagittal spinal canal alignment in cervical spondylotic myelopathy. J Neurosurg Spine. 2009;11(6):667-72.

For peer review only

15. Hosono N, Sakaura H, Mukai Y, Kaito T, Makino T, Yoshikawa H. A simple performance test for quantifying the severity of cervical myelopathy. J Bone Joint Surg Br. 2008;90(9):1210-3.
16. Hosono N, Takenaka S, Mukai Y, Makino T, Sakaura H, Miwa T, et al. Postoperative 24-hour result of 15-second grip-and-release test correlates with surgical outcome of cervical compression myelopathy. Spine (Phila Pa 1976). 2012;37(15):1283-7.
17. Kang Y, Lee JW, Koh YH, Hur S, Kim SJ, Chai JW, et al. New MRI grading system for the cervical canal stenosis. AJR Am J Roentgenol. 2011;197(1):W134-40.
18. Kato S, Oshima Y, Oka H, Chikuda H, Takeshita Y, Miyoshi K, et al. Comparison of the Japanese Orthopaedic Association (JOA) score and modified JOA (mJOA) score for the assessment of cervical myelopathy: a multicenter observational study. PloS one. 2015;10(4):e0123022.
19. Kato S, Oshima Y, Matsubayashi Y, Taniguchi Y, Tanaka S, Takeshita K. Minimum Clinically Important Difference and Patient Acceptable Symptom State of Japanese Orthopaedic Association Score in Degenerative Cervical Myelopathy Patients. Spine (Phila Pa 1976). 2019;44(10):691-7.
20. Kato S, Oshima Y, Matsubayashi Y, Taniguchi Y, Tanaka S, Takeshita K. Minimum clinically important difference in outcome scores among patients undergoing cervical laminoplasty. Eur Spine J. 2019;28(5):1234-41.
21. King JT, Jr., Roberts MS. Validity and reliability of the Short Form-36 in cervical spondylotic myelopathy. J Neurosurg. 2002;97(2 Suppl):180-5.
22. Ko S, Choi W, Chae S. Comparison of inter- and intra-observer reliability among the three classification systems for cervical spinal canal stenosis. Eur Spine J. 2017;26(9):2290-6.
23. Kopjar B, Tetreault L, Kalsi-Ryan S, Fehlings M. Psychometric properties of the modified Japanese Orthopaedic Association scale in patients with cervical spondylotic myelopathy. Spine (Phila Pa 1976). 2015;40(1):E23-8.
24. Latimer M, Haden N, Seeley HM, Laing RJ. Measurement of outcome in patients with cervical spondylotic myelopathy treated surgically. Br J Neurosurg. 2002;16(6):545-9.
25. Longo UG, Berton A, Denaro L, Salvatore G, Denaro V. Development of the Italian version of the modified Japanese orthopaedic association score (mJOA-IT): cross-cultural adaptation, reliability, validity and responsiveness. Eur Spine J. 2016;25(9):2952-7.
26. Lubelski D, Alvin MD, Nesterenko S, Sundar SJ, Thompson NR, Benzel EC, et al. Correlation of quality of life and functional outcome measures for cervical spondylotic myelopathy. Journal of Neurosurgery: Spine. 2016;24(3):483-9.
27. Mihara H, Kondo S, Murata A, Ishida K, Niimura T, Hachiya M. A new performance test for cervical myelopathy: the triangle step test. Spine (Phila Pa 1976). 2010;35(1):32-5.
28. Nakamoto H, Oshima Y, Takeshita K, Chikuda H, Ono T, Taniguchi Y, et al. Usefulness of QuickDASH in patients with cervical laminoplasty. J Orthop Sci. 2014;19(2):218-22.
29. Nakashima H, Yukawa Y, Ito K, Machino M, Kanbara S, Morita D, et al. Validity of the 10-s step test: prospective study comparing it with the 10-s grip and release test and the 30-m walking test. Eur Spine J. 2011;20(8):1318-22.
30. Nicholson KJ, Millhouse PW, Pflug E, Woods B, Schroeder GD, Anderson DG, et al. Cervical Sagittal Range of Motion as a Predictor of Symptom Severity in Cervical Spondylotic Myelopathy. Spine (Phila Pa 1976). 2018;43(13):883-9.
31. Nikaido T, Kikuchi S, Yabuki S, Otani K, Konno S. Surgical treatment assessment using the Japanese orthopedic association cervical myelopathy evaluation questionnaire in patients

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with cervical myelopathy: a new outcome measure for cervical myelopathy. Spine (Phila Pa 1976). 2009;34(23):2568-72.
32. Numasawa T, Ono A, Wada K, Yamasaki Y, Yokoyama T, Aburakawa S, et al. Simple foot tapping test as a quantitative objective assessment of cervical myelopathy. Spine (Phila Pa 1976). 2012;37(2):108-13.
33. Olindo S, Signate A, Richech A, Cabre P, Catonne Y, Smadja D, et al. Quantitative assessment of hand disability by the Nine-Hole-Peg test (9-HPT) in cervical spondylotic myelopathy. Journal of Neurology, Neurosurgery & Psychiatry. 2008;79(8):965-7.
34. Park H-J, Kim SS, Chung E-C, Lee S-Y, Park N-H, Rho M-H, et al. Clinical correlation of a new practical MRI method for assessing cervical spinal canal compression. American Journal of Roentgenology. 2012;199(2):W197-W201.
35. Pratali RR, Smith JS, Motta RL, Martins SM, Motta MM, Rocha RD, et al. A Brazilian Portuguese cross-cultural adaptation of the modified JOA scale for myelopathy. Clinics. 2017;72:103-5.
36. Pratali RR, Smith JS, Rocha RD, Matos TD, Defino HLA, Herrero C. RELIABILITY OF A BRAZILIAN PORTUGUESE TRANSLATED AND CROSS-CULTURALLY ADAPTED VERSION OF THE MJOA SCALE. Acta Ortop Bras. 2018;26(5):335-7.
37. Rhee JM, Shi WJ, Cyriac M, Kim JY, Zhou F, Easley KA, et al. The P-mJOA: A Patient-derived, Self-reported Outcome Instrument for Evaluating Cervical Myelopathy: Comparison with the mJOA. Clin Spine Surg. 2018;31(2):E115-e20.
38. Sato T, Horikoshi T, Watanabe A, Uchida M, Ishigame K, Araki T, et al. Evaluation of cervical myelopathy using apparent diffusion coefficient measured by diffusion-weighted imaging. AJNR Am J Neuroradiol. 2012;33(2):388-92.
39. Shim E, Lee JW, Lee E, Kang Y, Kang HS, Kang WY, et al. Cervical spondylotic myelopathy: diagnostic performance of radiologists with varying levels of experience in comparing MR images acquired using field strengths of 1.5 and 3 Tesla. Acta Radiol. 2019;60(10):1314-20.
40. Singh A, Crockard HA. Comparison of seven different scales used to quantify severity of cervical spondylotic myelopathy and post-operative improvement. J Outcome Meas. 2001;5(1):798-818.
41. Singh A, Crockard HA. Quantitative assessment of cervical spondylotic myelopathy by a simple walking test. Lancet. 1999;354(9176):370-3.
42. Singh A, Gnanalingham K, Casey A, Crockard A. Quality of life assessment using the Short Form-12 (SF-12) questionnaire in patients with cervical spondylotic myelopathy: comparison with SF-36. Spine (Phila Pa 1976). 2006;31(6):639-43.
43. Spurgas MP, Abbas SF, Szewczyk BS, Yim B, Ata A, German JW. The effect of length of follow-up on substantial clinical benefit thresholds in patients undergoing surgery for cervical degenerative myelopathy. Journal of Clinical Neuroscience. 2019;62:88-93.
44. Tetreault L, Nouri A, Kopjar B, Côté P, Fehlings MG. The Minimum Clinically Important Difference of the Modified Japanese Orthopaedic Association Scale in Patients with Degenerative Cervical Myelopathy. Spine (Phila Pa 1976). 2015;40(21):1653-9.
45. Thakar S, Rajshekhar V. Evaluation of pain as a preference-based health status measure in patients with cervical spondylotic myelopathy undergoing central corpectomy. Acta Neurochir (Wien). 2012;154(2):335-40.
46. Thakar S, Christopher S, Rajshekhar V. Quality of life assessment after central corpectomy for cervical spondylotic myelopathy: comparative evaluation of the 36-Item Short

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Form Health Survey and the World Health Organization Quality of Life-Bref. J Neurosurg Spine. 2009;11(4):402-12.
47. Wada E, Fukui M, Takahashi K, Takeuchi D, Hashizume H, Kanamori M, et al. Japanese orthopaedic association cervical myelopathy evaluation questionnaire (JOACMEQ): Part 5. Determination of responsiveness. Journal of Orthopaedic Science. 2019;24(1):57-61.
48. Witayakom W, Paholpak P, Jirarattanaphochai K, Kosuwon W, Sirichativapee W, Wisanuyotin T, et al. Validation of the reliability of the Thai version of the Japanese Orthopaedic Association Cervical Myelopathy Evaluation Questionnaire (JOACMEQ). Journal of Orthopaedic Science. 2016;21(2):124-7.
49. Yonenobu K, Abumi K, Nagata K, Taketomi E, Ueyama K. Interobserver and intraobserver reliability of the japanese orthopaedic association scoring system for evaluation of cervical compression myelopathy. Spine (Phila Pa 1976). 2001;26(17):1890-4; discussion 5.
50. Yukawa Y, Kato F, Ito K, Horie Y, Nakashima H, Masaaki M, et al. "Ten second step test" as a new quantifiable parameter of cervical myelopathy. Spine (Phila Pa 1976). 2009;34(1):82-6.
51. Zhang Y, Zhou F, Sun Y. Assessment of health-related quality of life using the SF-36 in Chinese cervical spondylotic myelopathy patients after surgery and its consistency with neurological function assessment: a cohort study. Health Qual Life Outcomes. 2015;13(1):1-7.
52. Zhou F, Zhang Y, Sun Y, Zhang F, Pan S, Liu Z. Assessment of the minimum clinically important difference in neurological function and quality of life after surgery in cervical spondylotic myelopathy patients: a prospective cohort study. Eur Spine J. 2015;24(12):2918-23.

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			
Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	3
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	7
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	7-8
Methods			

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Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	8-9
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	8
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	8, S1-S9
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	9
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	9-10
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	9-13
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	9-13
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	9-13
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	9-13
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention	13

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		characteristics and comparing against the planned groups for each synthesis (item #5))	
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	13
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	13
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	13
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	NA
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	NA
Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	9-13
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	NA
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	13
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	13-14
Study characteristics	#17	Cite each included study and present its characteristics	15, S10-S12
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	S13-S50
Results of individual	#19	For all outcomes, present for each study (a) summary statistics for	S13-S50

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studies		each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	16-18, 23-32
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	16-18, 23-32
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	NA
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	NA
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	NA
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	NA
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	33-34
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	34-35
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	34-35
Implications	#23d	Discuss implications of the results for practice, policy, and future research	34
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	2
Registration and	#24b	Indicate where the review protocol can be accessed, or state that a	2

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protocol		protocol was not prepared	
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	2
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	36
Competing interests	#26	Declare any competing interests of review authors	36
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	

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